Gene Set Analysis

HUGEN 2073

Jon Chernus

What can you do with a list of genes?

- We can do “variant → gene” inference at each locus, using
  - LD/position
  - Known trait associations
  - eQTLS (in a later lecture)
  - Etc.
- Put all the loci together → a list of genes

How can you interpret a list of genes?

Suppose your study (e.g., GWAS) yielded a list of dozens (or more) genes

- Why not just publish a list of genes and stop there?
  - Are all those genes necessarily relevant?
  - What scientific/statistical question(s) can you ask about the list?

Gene set analysis

- “Are certain categories of genes “enriched” in my results, compared to what would be expected by chance?”

- Originally used for differential expression studies

Includes or can be sometimes synonymous with

- “enrichment analysis”
- “over-representation analysis”
- Etc.

Some considerations

- What is the null hypothesis of a given method?
- Some methods can include effect sizes and directions (instead of simply using gene names)

Over-representation/enrichment

When asking what categories of genes are over-represented/enriched among a prioritized list of genes, what kinds of categories exist?

- What kinds of categories might you be interested in?

Over-representation

Example: a GWAS of telomere length

https://www.medrxiv.org/content/10.1101/2021.03.23.21253516v1

Key components of telomere regulatory complexes found within genome-wide significant loci.

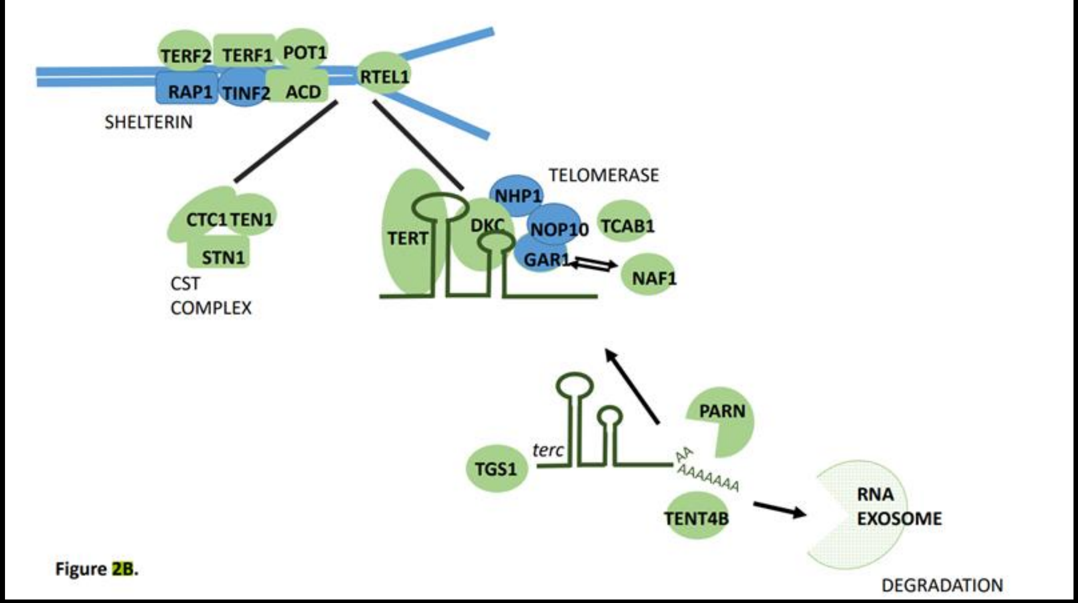


Figure 2B.

Proteins depicted in green are found within GWAS loci, those not found within GWAS loci are depicted in blue.

We find the majority of components of core telomere binding complexes alongside many proteins involved in the formation and activity of telomerase. Note that not all components of the RNA exosome are shown.

Over-representation

Eric Fauman @Eric_Fauman · Mar 26

Here I colored each gene by its closest proximity to a GWAS SNP.
Genes in red represent a closest gene to a GWAS SNP.
Orange means 1 other gene is closer.
TEN1 (yellow) is the 3rd closest gene to its SNP.
And 2 genes are further than 5th closest gene.
Pretty amazing I think!

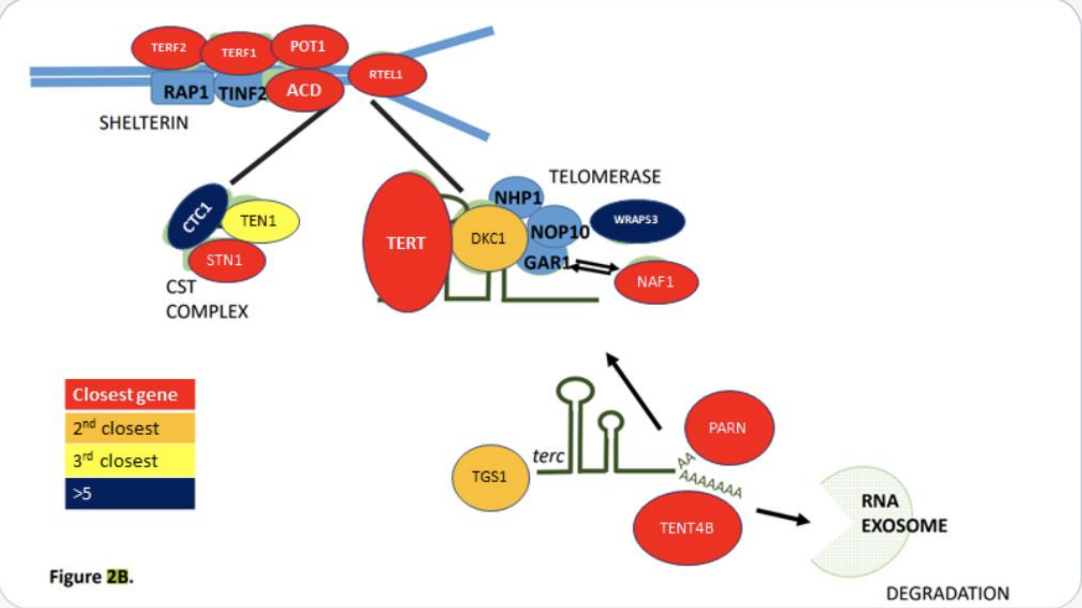


Figure 2B.

6

It looks like “telomerase-maintenance” genes are probably over-represented in these GWAS results.

(In a GWAS of telomere length, this is not surprising.)

But how to quantify the over-representation?

Some examples of gene sets/databases/tools

Some databases

- Gene Ontologies (GO) (http://geneontology.org)
- KEGG (https://www.genome.jp/kegg/)
- Reactome (https://reactome.org)
- MSigDB
- GTEx (tissue-specific gene sets)
- Pathway Commons (https://www.pathwaycommons.org)
- Many others

Some methods and implementations

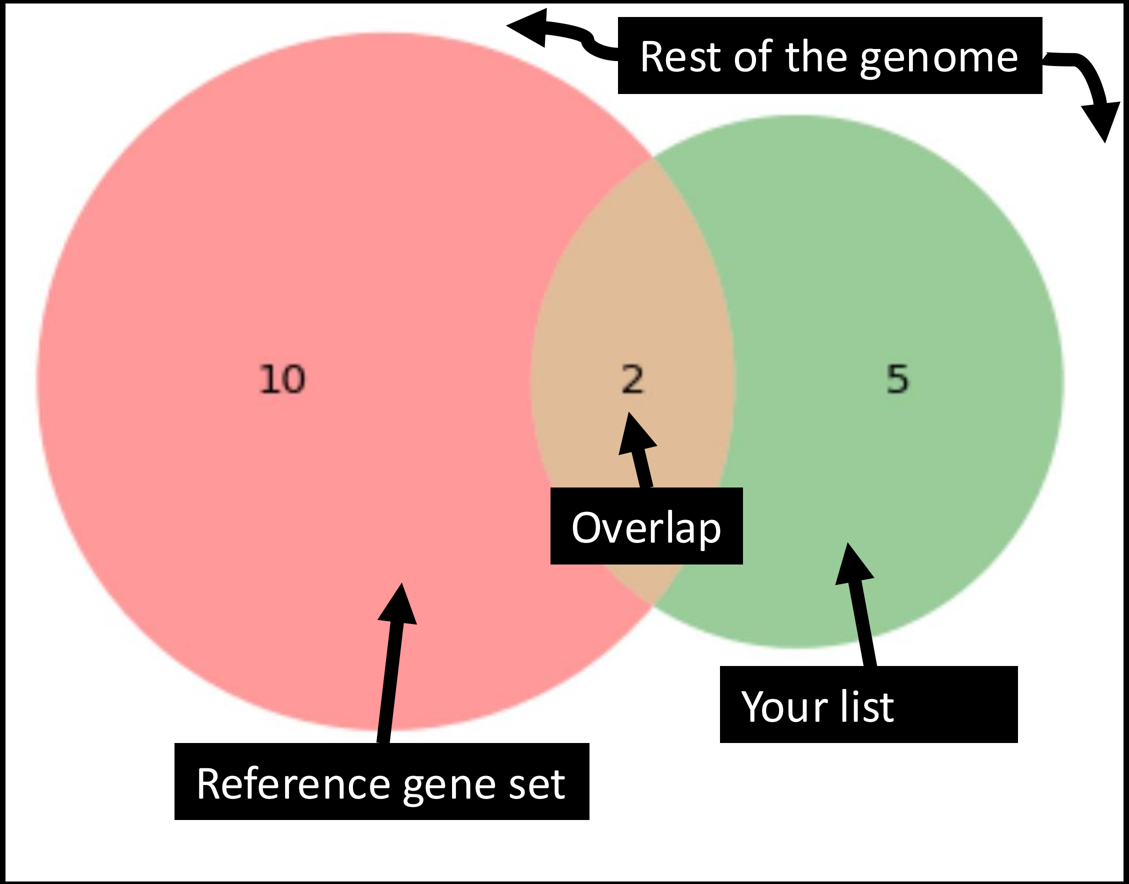
- PANTHER (http://www.pantherdb.org)
- IPA (https://digitalinsights.qiagen.com/products-overview/discovery-insights-portfolio/analysis-and-visualization/qiagen-ipa/; not free)
- GREAT (http://great.stanford.edu/public/html/)
- GSEA (https://www.gsea-msigdb.org/gsea/index.jsp)
- DAVID (https://david.ncifcrf.gov)
- MAGMA (https://ctg.cncr.nl/software/magma)
- FUMA (GENE2FUNC function; MAGMA) (https://fuma.ctglab.nl)
- Enrichr (https://maayanlab.cloud/Enrichr/)
- STRING
- R packages: fgsea, clusterProfiler

Types of gene set analysis

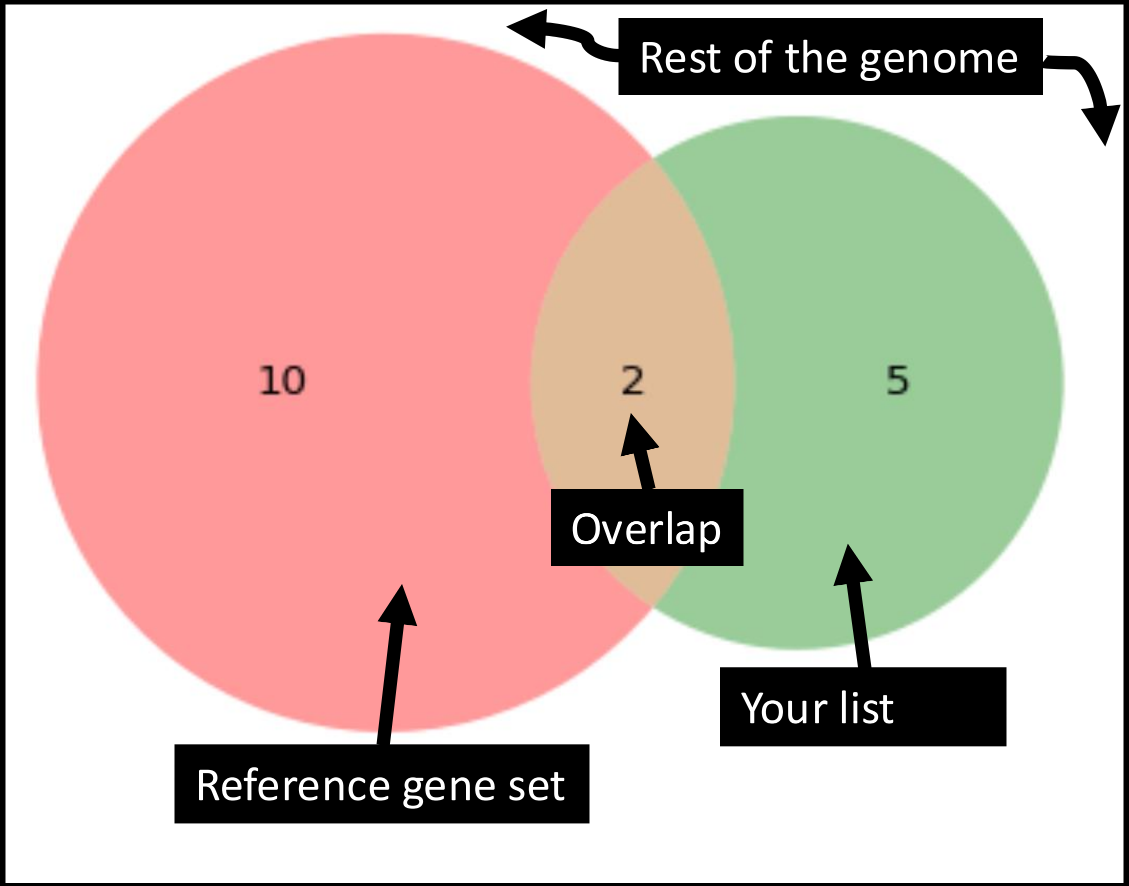
- Over-representation analysis (think Fisher’s exact test)
  - Simplest
  - You have a thresholded gene list
  - Compare overlap with a gene set
  - Interpretation: “More hits in this pathway than expected by chance”
- Functional class scoring (think GSEA)
  - Uses all genes you have results for, ranked (not thresholded)
  - Interpretation: “This pathway has *stronger* hits than expected by chance”
- Pathway topology-based (e.g., SPIA)
  - Use pathway structure
  - Weigh genes by position/connectivity
  - Interpretation: “Important parts of this pathway show signal”

The simplest over-representation test

Venn diagram to show enrichment



Hypergeometric test (Fisher's exact test)



What's the probability of an overlap at least this big?

- You picked n=7 GWAS genes out of the genome (N≈20,000)
- The gene set had K=10 genes, overlapping k=2 of your genes

Number of ways to pick *b* balls from an urn with *a* total balls:

$$ {}_{a}C_b = \binom{a}{b} = \frac{a!}{(a-b)!(b!)} $$

How many ways could you pick 7 genes out of the genome?

$$ {}_{N}C_n = \binom{20000}{7} = \frac{20000!}{(20000-7)!(7!)} = ? $$

How many ways could you get an overlap of exactly 2?

$$ {}_{K}C_k * {}_{N-K}C_{n-k} = \binom{10}{2} * \binom{20{,}000-10}{5} = \frac{10!}{(10-2)!(2!)} * \frac{(20{,}000-10)!}{((20{,}000-10)-5)!(5!)} = 😱 $$

So P[overlap exactly k=2] = 😱 /?

Finally, $P_{hypergeometric} = \mathrm{P}[\text{see overlap } \textit{at least} \text{ this big})] = \sum_{i=k}^{min(K,n)} P(\text{overlap } exactly\ i)$

Gene Ontologies

http://geneontology.org

Gene ontology (GO) is a formalized system for categorizing/annotating genes according to their biological functions.

This makes it easier to systematically compare gene sets and also can help keep track of similar genes across different organisms.

Every gene has assigned three “aspects”

- Molecular function
- Cellular component
- Biological process

Gene Ontologies - aspects

Molecular Function	Molecular-level activities performed by gene products. Molecular function terms describe activities that occur at the molecular level, such as “catalysis” or “transport”. GO molecular function terms represent activities rather than the entities (molecules or complexes) that perform the actions, and do not specify where, when, or in what context the action takes place. Molecular functions generally correspond to activities that can be performed by individual gene products (*i.e.* a protein or RNA), but some activities are performed by molecular complexes composed of multiple gene products. Examples of broad functional terms are *catalytic activity* and *transporter activity*; examples of narrower functional terms are *adenylate cyclase activity* or *Toll-like receptor binding*. To avoid confusion between gene product names and their molecular functions, GO molecular functions are often appended with the word “activity” (a *protein kinase* would have the GO molecular function *protein kinase activity*).
Cellular Component	The locations relative to cellular structures in which a gene product performs a function, either cellular compartments (*e.g.*, *mitochondrion*), or stable macromolecular complexes of which they are parts (*e.g.*, the *ribosome*). Unlike the other aspects of GO, cellular component classes refer not to processes but rather a cellular anatomy.
Biological Process	The larger processes, or ‘biological programs’ accomplished by multiple molecular activities. Examples of broad biological process terms are *DNA repair* or *signal transduction*. Examples of more specific terms are *pyrimidine nucleobase biosynthetic process* or *glucose transmembrane transport*. Note that a biological process is not equivalent to a pathway. At present, the GO does not try to represent the dynamics or dependencies that would be required to fully describe a pathway.

In an example of GO annotation, the gene product “cytochrome c” can be described by the **molecular function** *oxidoreductase activity*, the **biological process** *oxidative phosphorylation*, and the **cellular component** *mitochondrial matrix*.

The GO vocabulary is designed to be species-agnostic, and includes terms applicable to prokaryotes and eukaryotes, as well as single and multicellular organisms.

Gene Ontologies – GO terms

Each possible aspect is arranged in a hierarchy or “tree” ranging from the most general to the most specific.

If a gene belong to a specific class, it also belongs to call of the “parent” classes in the hierarchy containing that class.

The entire system of classes can be described with a formalized logical vocabulary.

Gene Ontologies – GO terms

General principles of GO annotations

- Annotations represent the normal functions of gene products.
- A gene product can be annotated to zero or more terms from each ontology.
- Each annotation is supported by an GO Evidence Codes from the Evidence and Conclusions Ontology and a reference.
- Gene products are annotated to the most granular term in the ontology that is supported by the available evidence.
- By the transitivity principle, an annotation to a GO term implies annotation to all its parents.
- GO annotations are meant to reflect the most up-to-date view of a gene product's role in biology.
- Because biological knowledge changes, annotations for a given gene product may change to reflect changes in knowledge and/or changes in the ontology.
- There is an open-world assumption, that is, if a gene product is unannotated then its role is still unknown.

Gene Ontologies – GO terms

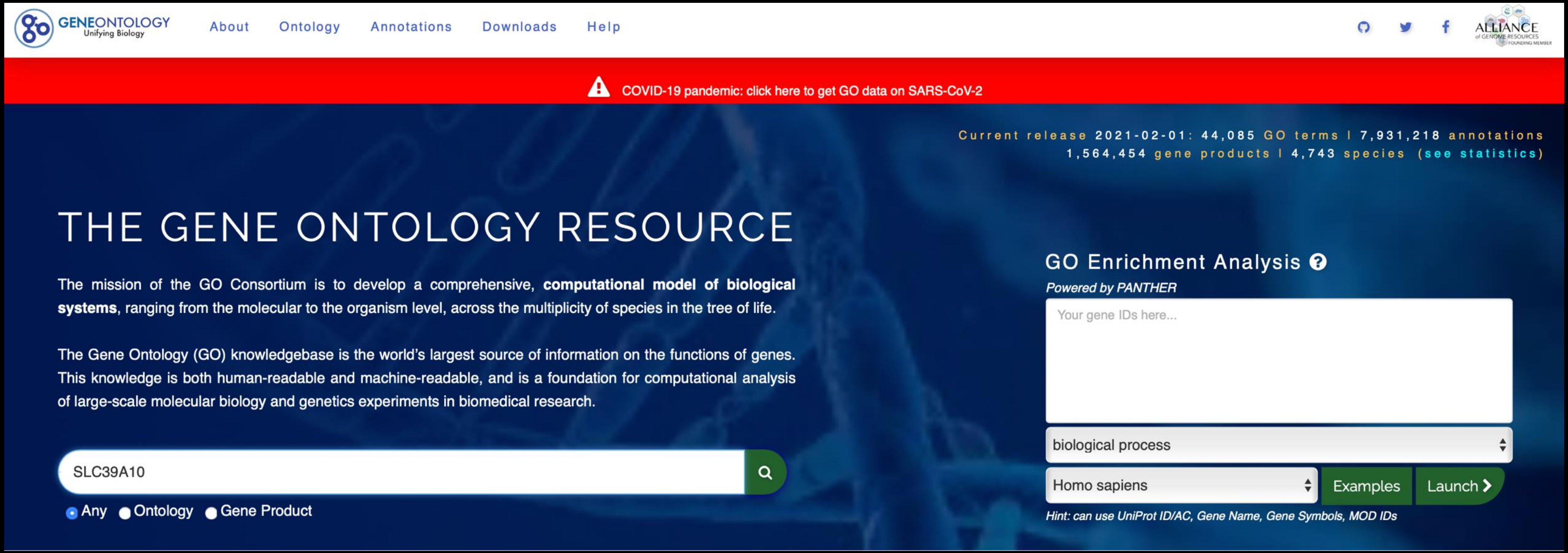
- The possible values for a given aspect are called GO terms or classes and have unique identifiers and descriptions:

Unique identifier and term name

Every term has a human-readable term name — e.g. mitochondrion, glucose transmembrane transport, or amino acid binding — and a GO ID, a unique seven digit identifier prefixed by GO:, e.g. GO:0005739, GO:1904659, or GO:0016597.

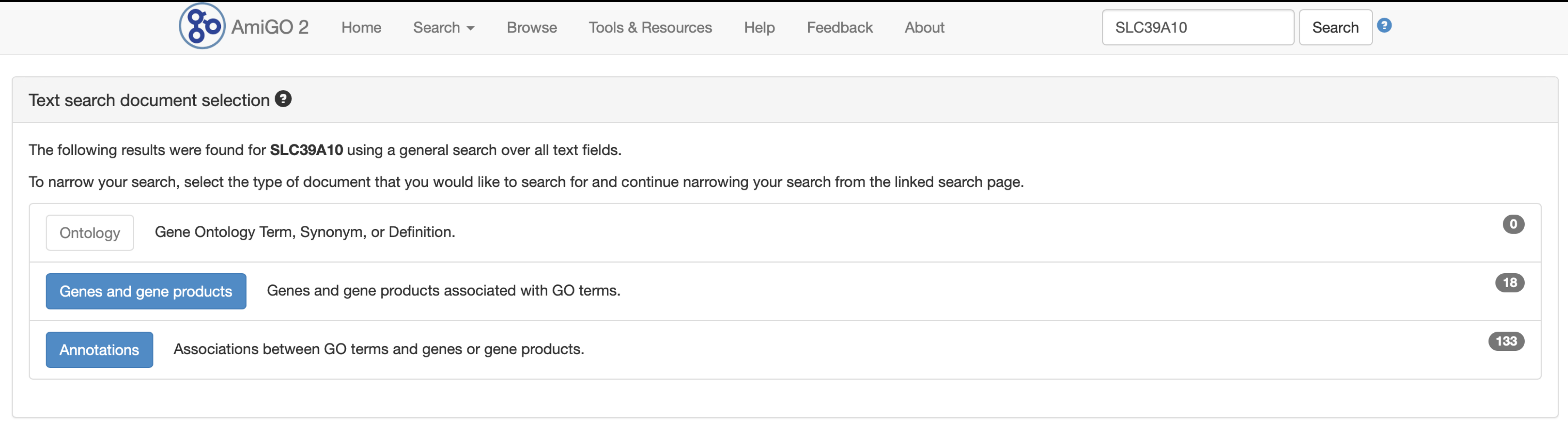
Gene Ontologies

Searching a gene



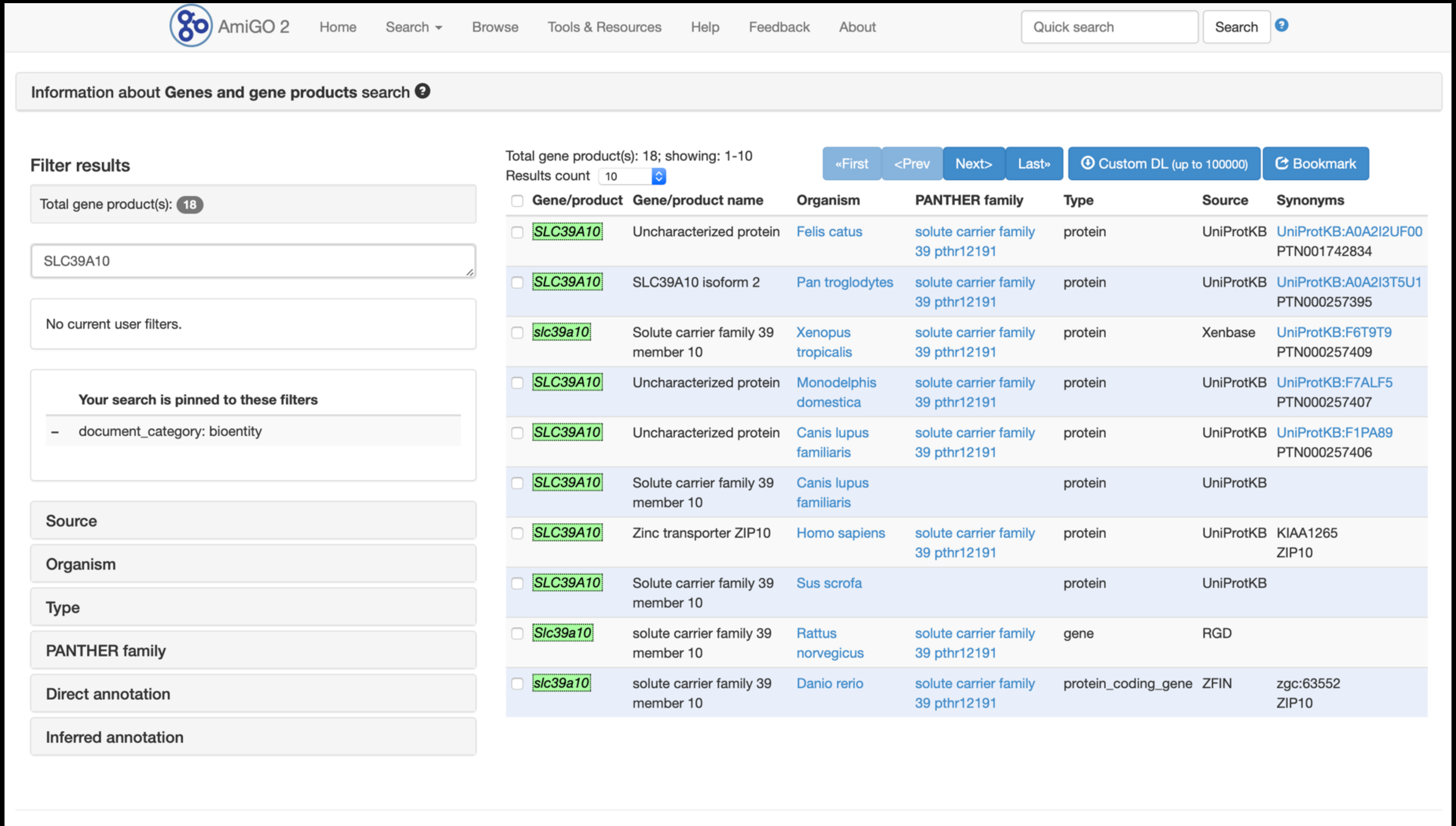
Gene Ontologies

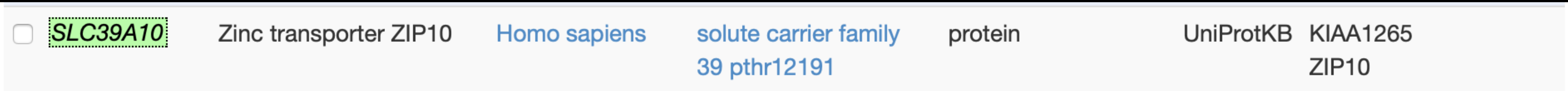
Search results. Let’s click on “genes and gene products.”



Gene Ontologies

Let’s select Homo sapiens.





Gene Ontologies

Results for *SLC39A10*

Zinc transporter ZIP10

Gene Product Information

Data health

Symbol SLC39A10
Name(s) Zinc transporter ZIP10
Type protein
Taxon Homo sapiens
Synonyms KIAA1265
ZIP10
Database UniProtKB, Q9ULF5
Related Link to **annotations page** to SLC39A10.
Link to annotations **download** for SLC39A10.

Gene Product Associations

Filter results

Total annotations: 8

No current user filters.

Your search is pinned to these filters

- document_category: annotation
- bioentity: UniProtKB:Q9ULF5

Ontology (aspect)

+ –	(6)	P
+ –	(1)	C
+ –	(1)	F

Organism
Type
Evidence
GO class
GO class (direct)
Annotation qualifier
Annotation extension
Contributor
PANTHER family

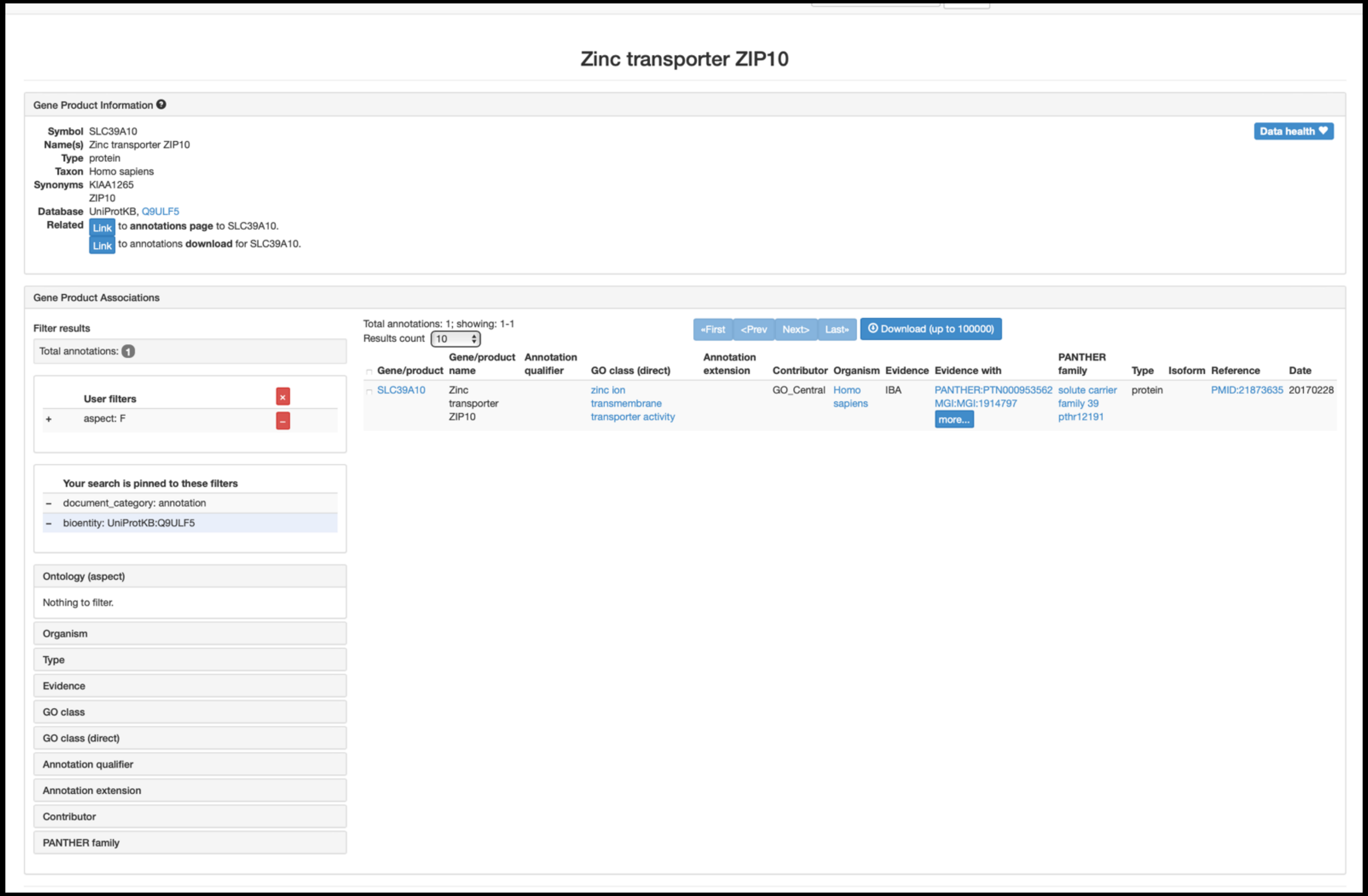
Total annotations: 8; showing: 1-8
Results count 10

«First <Prev Next> Last» Download (up to 100000)

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
SLC39A10	Zinc transporter ZIP10		negative regulation of B cell apoptotic process		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell proliferation		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of protein tyrosine phosphatase activity		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		zinc ion import across plasma membrane		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Zinc transporter ZIP10		cellular zinc ion homeostasis		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 UniProtKB:Q13433 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		zinc ion transmembrane transporter activity		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell receptor signaling pathway		GO_Central	Homo sapiens	IBA	PANTHER:PTN002638848 MGI:MGI:1914515	solute carrier family 39 pthr12191	protein		PMID:21873635	20200317
SLC39A10	Zinc transporter ZIP10		integral component of plasma membrane		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 FB:FBgn0024236 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228

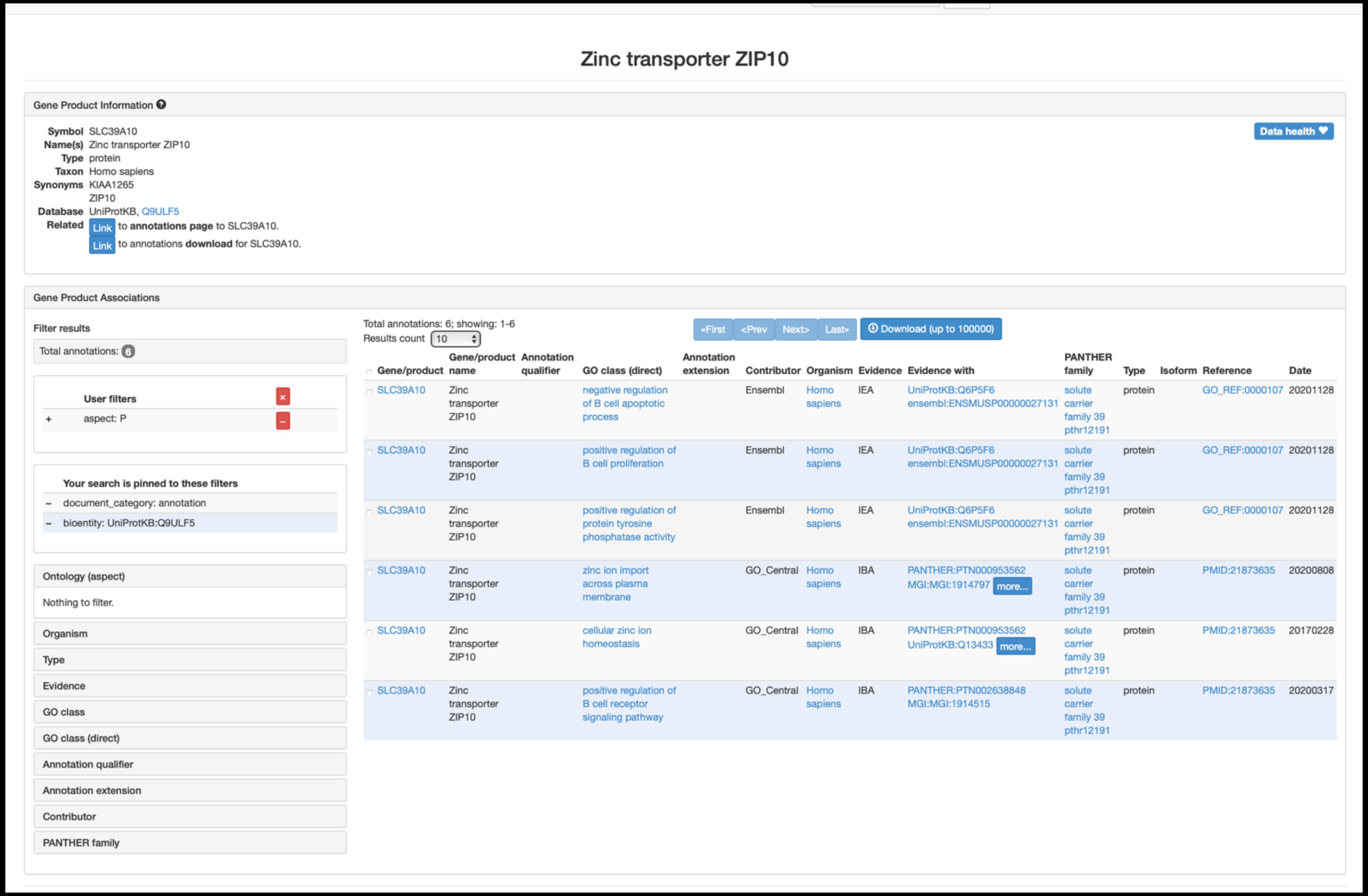
Gene Ontologies

Results for *SLC39A10*. Filter to show only aspect F (function)



Gene Ontologies

Results for *SLC39A10*. Filter to show only aspect P (process)



Zinc transporter ZIP10

Gene Product Information

Symbol SLC39A10
Name(s) Zinc transporter ZIP10
Type protein
Taxon Homo sapiens
Synonyms KIAA1265
ZIP10
Database UniProtKB, Q9ULF5
Related Link to **annotations page** to SLC39A10.
Link to annotations **download** for SLC39A10.

Data health

Gene Product Associations

Filter results

Total annotations: 6

User filters
+ aspect: P

Your search is pinned to these filters
– document_category: annotation
– bioentity: UniProtKB:Q9ULF5

Ontology (aspect)
Nothing to filter.

Organism
Type
Evidence
GO class
GO class (direct)
Annotation qualifier
Annotation extension
Contributor
PANTHER family

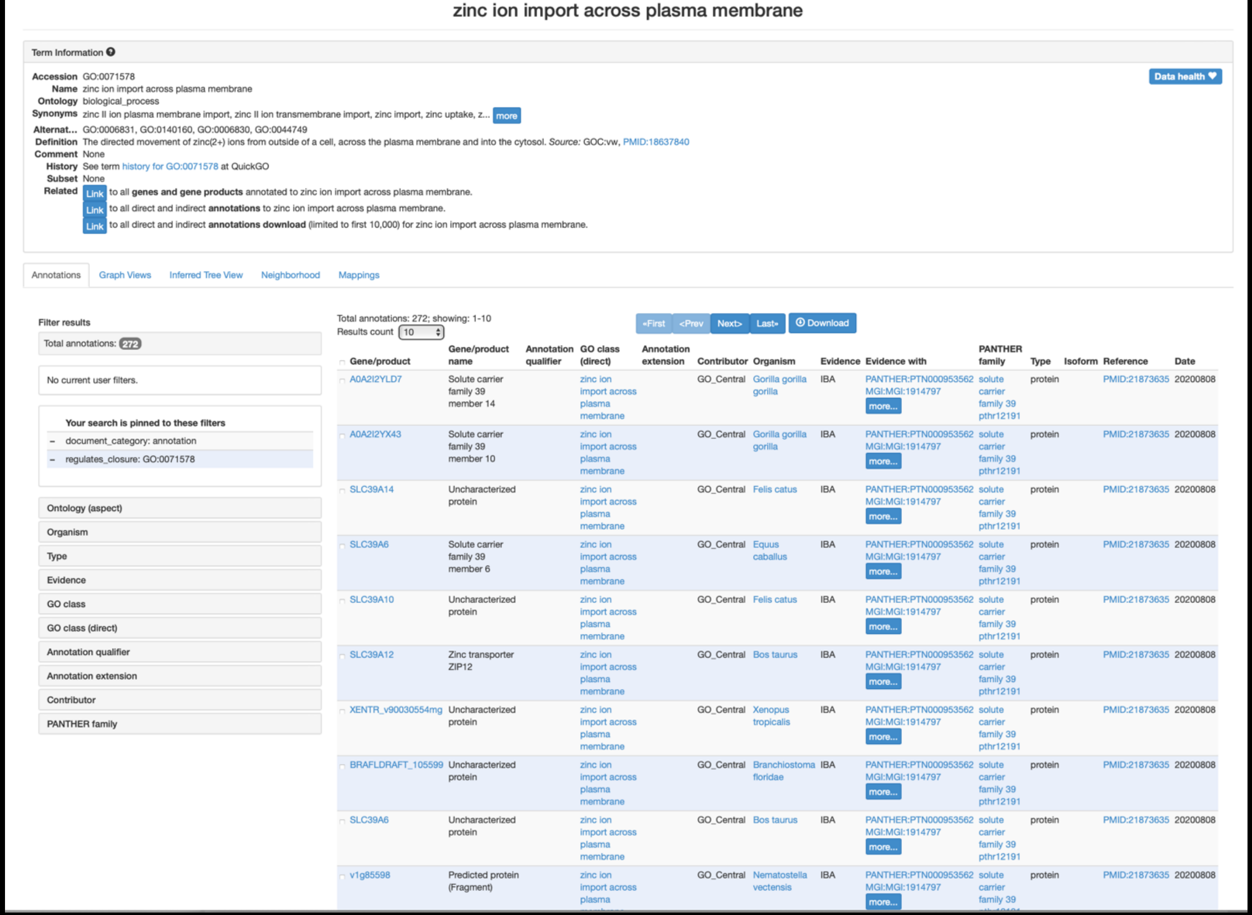
Total annotations: 6; showing: 1-6
Results count 10

«First <Prev Next> Last» Download (up to 100000)

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
SLC39A10	Zinc transporter ZIP10		negative regulation of B cell apoptotic process		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell proliferation		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		positive regulation of protein tyrosine phosphatase activity		Ensembl	Homo sapiens	IEA	UniProtKB:Q6P5F6 ensembl:ENSMUSP00000027131	solute carrier family 39 pthr12191	protein		GO_REF:0000107	20201128
SLC39A10	Zinc transporter ZIP10		zinc ion import across plasma membrane		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Zinc transporter ZIP10		cellular zinc ion homeostasis		GO_Central	Homo sapiens	IBA	PANTHER:PTN000953562 UniProtKB:Q13433 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20170228
SLC39A10	Zinc transporter ZIP10		positive regulation of B cell receptor signaling pathway		GO_Central	Homo sapiens	IBA	PANTHER:PTN002638848 MGI:MGI:1914515	solute carrier family 39 pthr12191	protein		PMID:21873635	20200317

Gene Ontologies

Result of clicking on one of the GO terms: “zinc ion import across plasma membrane.”



zinc ion import across plasma membrane

Term Information

Accession GO:0071578
Name zinc ion import across plasma membrane
Ontology biological_process
Synonyms zinc II ion plasma membrane import, zinc II ion transmembrane import, zinc import, zinc uptake, z... more
Alternat... GO:0006831, GO:0140160, GO:0006830, GO:0044749
Definition The directed movement of zinc(2+) ions from outside of a cell, across the plasma membrane and into the cytosol. *Source:* GOC:vw, PMID:18637840
Comment None
History See term history for GO:0071578 at QuickGO
Subset None
Related Link to all **genes and gene products** annotated to zinc ion import across plasma membrane.
Link to all direct and indirect **annotations** to zinc ion import across plasma membrane.
Link to all direct and indirect **annotations download** (limited to first 10,000) for zinc ion import across plasma membrane.

Data health

Annotations | Graph Views | Inferred Tree View | Neighborhood | Mappings

Filter results

Total annotations: 272

No current user filters.

Your search is pinned to these filters
- document_category: annotation
- regulates_closure: GO:0071578

Ontology (aspect)
Organism
Type
Evidence
GO class
GO class (direct)
Annotation qualifier
Annotation extension
Contributor
PANTHER family

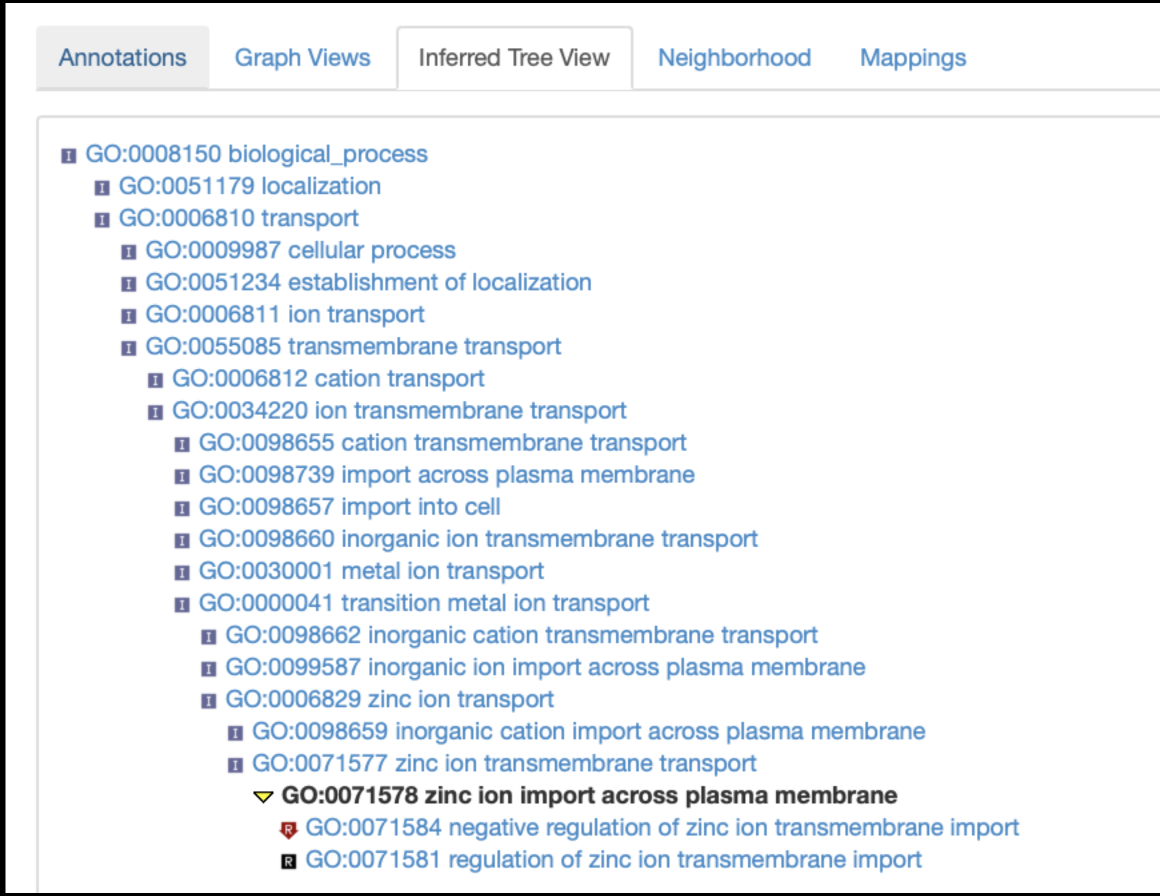
Total annotations: 272; showing: 1-10
Results count 10

«First <Prev Next> Last» Download

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
A0A2I2YLD7	Solute carrier family 39 member 14		zinc ion import across plasma membrane		GO_Central	Gorilla gorilla gorilla	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
A0A2I2YX43	Solute carrier family 39 member 10		zinc ion import across plasma membrane		GO_Central	Gorilla gorilla gorilla	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A14	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Felis catus	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A6	Solute carrier family 39 member 6		zinc ion import across plasma membrane		GO_Central	Equus caballus	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A10	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Felis catus	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A12	Zinc transporter ZIP12		zinc ion import across plasma membrane		GO_Central	Bos taurus	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
XENTR_v90030554mg	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Xenopus tropicalis	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
BRAFLDRAFT_105599	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Branchiostoma floridae	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
SLC39A6	Uncharacterized protein		zinc ion import across plasma membrane		GO_Central	Bos taurus	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39 pthr12191	protein		PMID:21873635	20200808
v1g85598	Predicted protein (Fragment)		zinc ion import across plasma		GO_Central	Nematostella vectensis	IBA	PANTHER:PTN000953562 MGI:MGI:1914797 more...	solute carrier family 39	protein		PMID:21873635	20200808

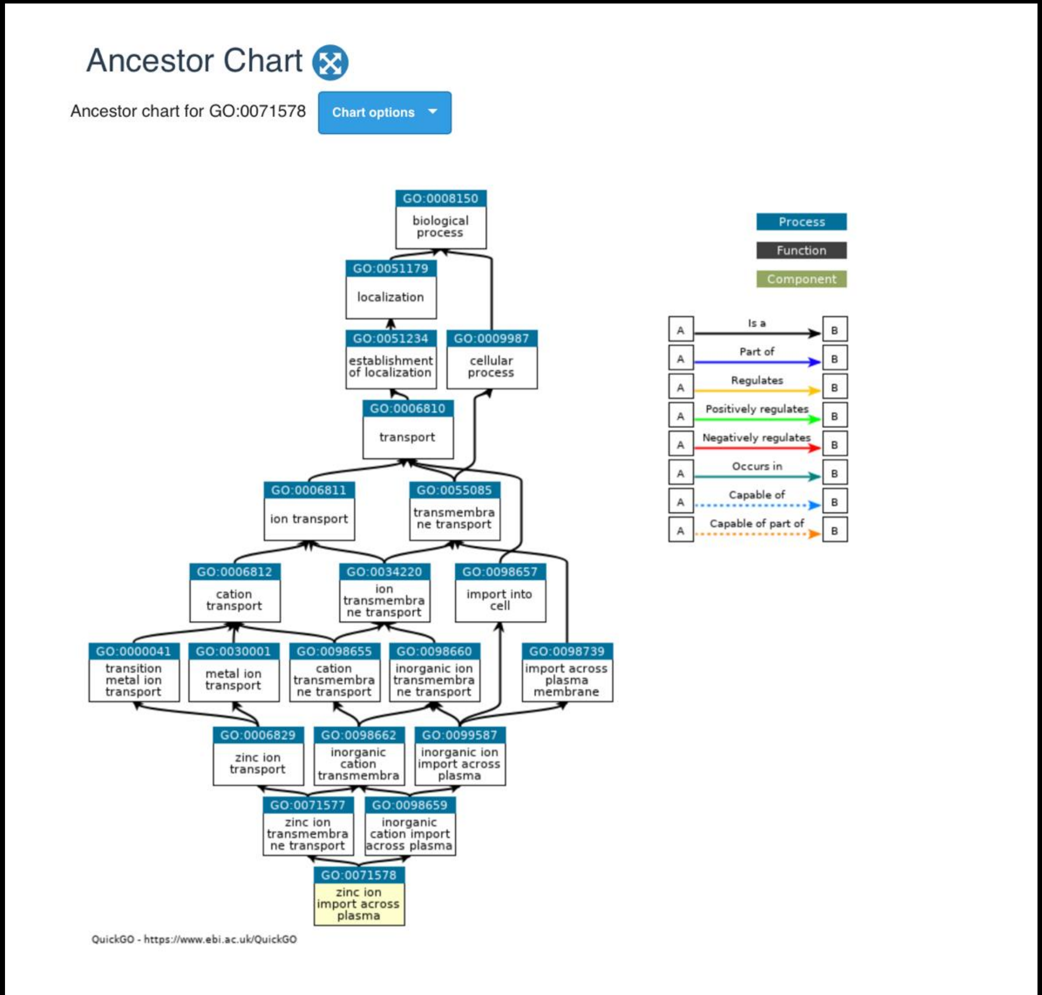
Gene Ontologies

Result of clicking on “Inferred Tree View”



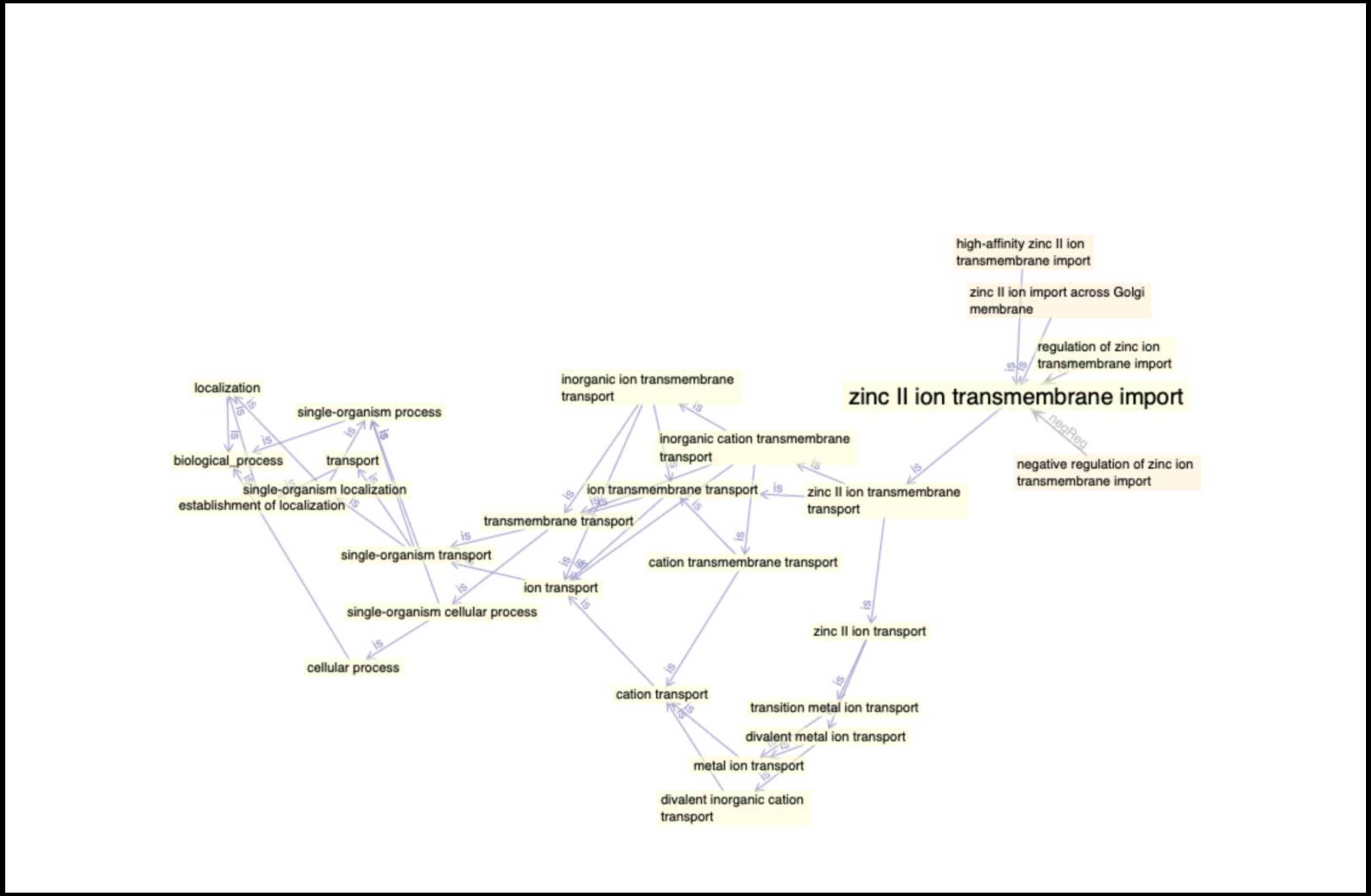
Gene Ontologies

An option from “Graph Views”



Gene Ontologies

Another option from "Graph Views"



Gene Ontologies – sources and evidence

Total annotations: 258; showing: 1-10

Results count 10

«First <Prev Next> Last» Download (up to 100000)

Gene/product	Gene/product name	Annotation qualifier	GO class (direct)	Annotation extension	Contributor	Organism	Evidence	Evidence with	PANTHER family	Type	Isoform	Reference	Date
BRCA1	Breast cancer type 1 susceptibility protein		ubiquitin ligase complex		UniProt	Homo sapiens	NAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:14976165	20050603
BRCA1	Breast cancer type 1 susceptibility protein		double-strand break repair via homologous recombination		HGNC-UCL	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:17349954	20070426
BRCA1	Breast cancer type 1 susceptibility protein		DNA double-strand break processing		Reactome	Homo sapiens	TAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		Reactome:R-HSA-5693608	20181122
BRCA1	Breast cancer type 1 susceptibility protein		lateral element		MGI	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9774970	20160830
BRCA1	Breast cancer type 1 susceptibility protein		transcription regulatory region sequence-specific DNA binding		BHF-UCL	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:20820192	20110418
BRCA1	Breast cancer type 1 susceptibility protein		DNA binding		PINC	Homo sapiens	TAS		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9662397	20030904
BRCA1	Breast cancer type 1 susceptibility protein		damaged DNA binding		Ensembl	Homo sapiens	IEA	UniProtKB:P48754 ensembl:ENSMUSP00000017290	breast cancer type 1 susceptibility protein brca1 pthr13763	protein		GO_REF:0000107	20201128
BRCA1	Breast cancer type 1 susceptibility protein		transcription coactivator activity		UniProt	Homo sapiens	IMP		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:9662397	20170706
BRCA1	Breast cancer type 1 susceptibility protein		RNA binding		MGI	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:12419249	20091111
BRCA1	Breast cancer type 1 susceptibility protein		ubiquitin-protein transferase activity		UniProt	Homo sapiens	IDA		breast cancer type 1 susceptibility protein brca1 pthr13763	protein		PMID:12890688	20100602

Experimental Evidence evidence codes:

- Inferred from Experiment (EXP)
- Inferred from Direct Assay (IDA)
- Inferred from Physical Interaction (IPI)
- Inferred from Mutant Phenotype (IMP)
- Inferred from Genetic Interaction (IGI)
- Inferred from Expression Pattern (IEP)

Computational Analysis evidence codes:

- Inferred from Sequence or structural Similarity (ISS)
- Inferred from Sequence Orthology (ISO)
- Inferred from Sequence Alignment (ISA)
- Inferred from Sequence Model (ISM)
- Inferred from Genomic Context (IGC)
- Inferred from Biological aspect of Ancestor (IBA)
- Inferred from Biological aspect of Descendant (IBD)
- Inferred from Key Residues (IKR)
- Inferred from Rapid Divergence (IRD)
- Inferred from Reviewed Computational Analysis (RCA)

Author Statement evidence codes:

- Traceable Author Statement (TAS)
- Non-traceable Author Statement (NAS)

Curatorial Statement codes:

- Inferred by Curator (IC)
- No biological Data available (ND) evidence code

Automatically-Assigned evidence code:

- Inferred from Electronic Annotation (IEA)

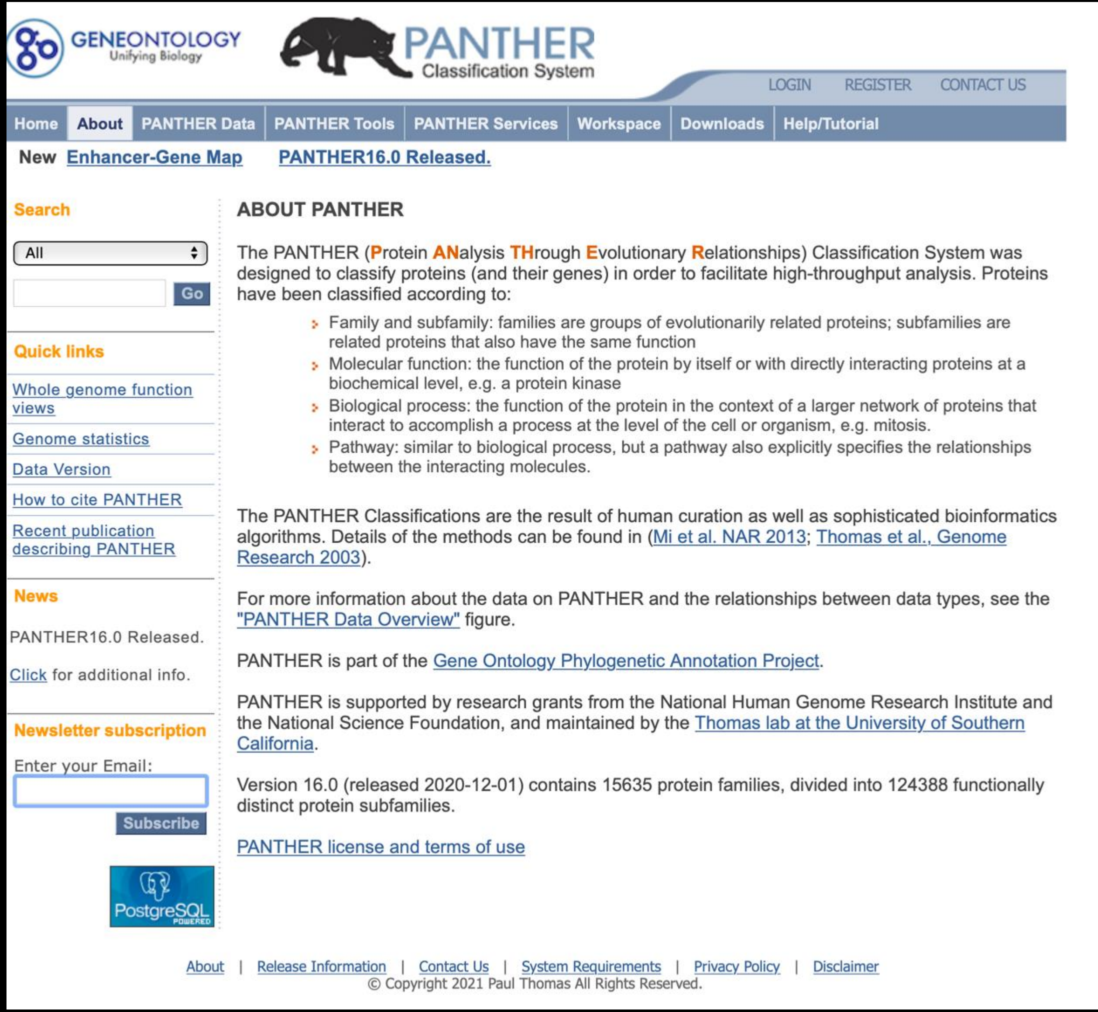
PANTHER

http://pantherdb.org

ABOUT PANTHER

The PANTHER (**P**rotein **AN**alysis **TH**rough **E**volutionary **R**elationships) Classification System was designed to classify proteins (and their genes) in order to facilitate high-throughput analysis. The core of PANTHER is a comprehensive, annotated "library" of gene family phylogenetic trees. All nodes in the tree have persistent identifiers that are maintained between versions of PANTHER, providing a stable substrate for annotations of protein properties like subfamily and function. Each phylogenetic tree is used to annotate each protein member of the family by its:

1. Family and Protein Class (supergrouping of protein families)
2. Subfamily (subgroup within the family phylogenetic tree)
3. Orthologs (genes in other organisms that derive from the same gene in the MRCA)
4. Paralogs (genes in the same organism that are related by gene duplication)
5. Function (using GO terms annotated on the trees by the GO Phylogenetic Annotation Project)
6. Pathways (curated by PANTHER and by Reactome)



GENEONTOLOGY Unifying Biology — PANTHER Classification System

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New Enhancer-Gene Map PANTHER16.0 Released.

Search: All — Go

Quick links: Whole genome function views; Genome statistics; Data Version; How to cite PANTHER; Recent publication describing PANTHER

News: PANTHER16.0 Released. Click for additional info.

Newsletter subscription: Enter your Email: — Subscribe

ABOUT PANTHER

The PANTHER (**P**rotein **AN**alysis **TH**rough **E**volutionary **R**elationships) Classification System was designed to classify proteins (and their genes) in order to facilitate high-throughput analysis. Proteins have been classified according to:

- Family and subfamily: families are groups of evolutionarily related proteins; subfamilies are related proteins that also have the same function
- Molecular function: the function of the protein by itself or with directly interacting proteins at a biochemical level, e.g. a protein kinase
- Biological process: the function of the protein in the context of a larger network of proteins that interact to accomplish a process at the level of the cell or organism, e.g. mitosis.
- Pathway: similar to biological process, but a pathway also explicitly specifies the relationships between the interacting molecules.

The PANTHER Classifications are the result of human curation as well as sophisticated bioinformatics algorithms. Details of the methods can be found in (Mi et al. NAR 2013; Thomas et al., Genome Research 2003).

For more information about the data on PANTHER and the relationships between data types, see the "PANTHER Data Overview" figure.

PANTHER is part of the Gene Ontology Phylogenetic Annotation Project.

PANTHER is supported by research grants from the National Human Genome Research Institute and the National Science Foundation, and maintained by the Thomas lab at the University of Southern California.

Version 16.0 (released 2020-12-01) contains 15635 protein families, divided into 124388 functionally distinct protein subfamilies.

PANTHER license and terms of use

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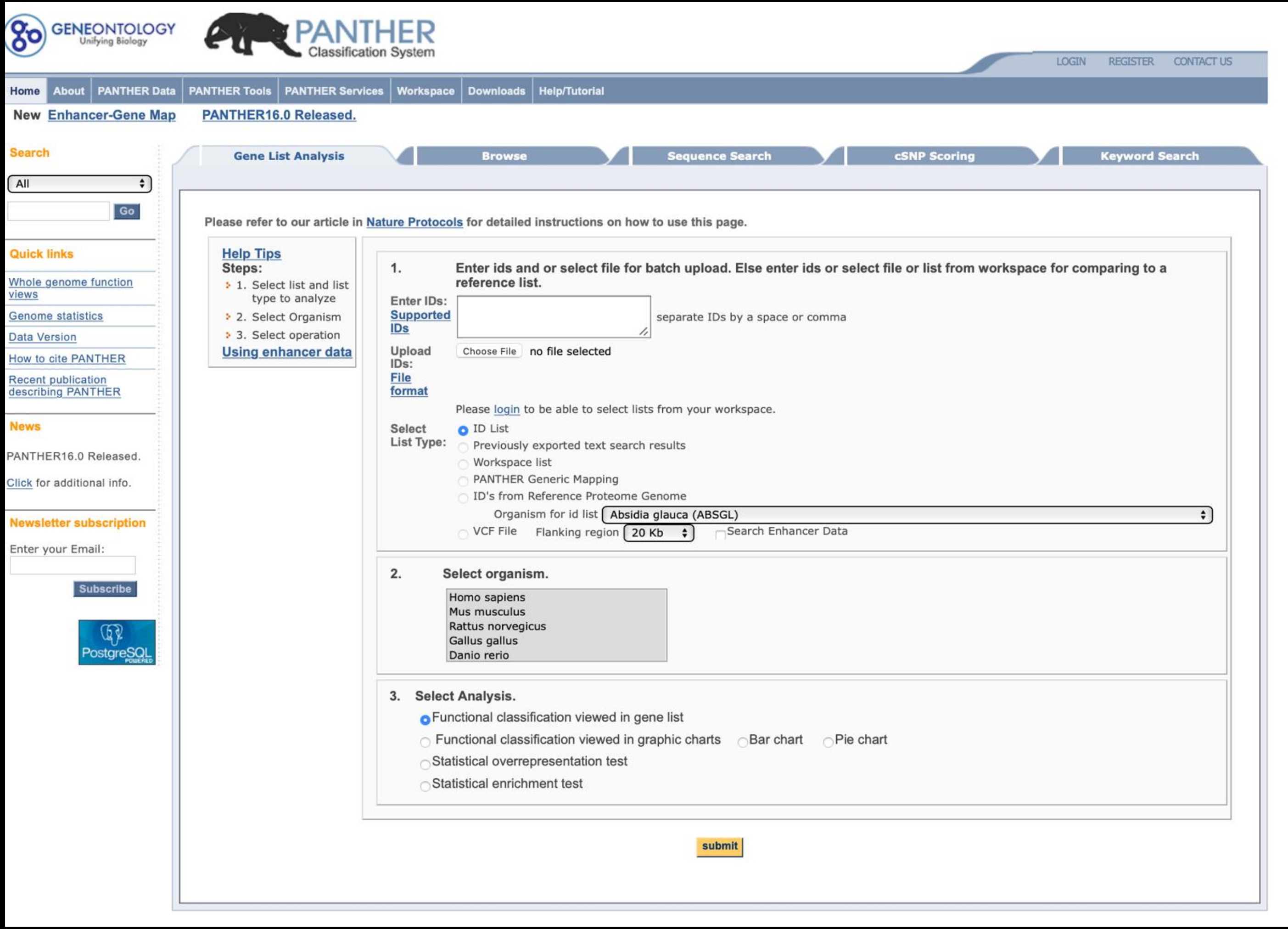
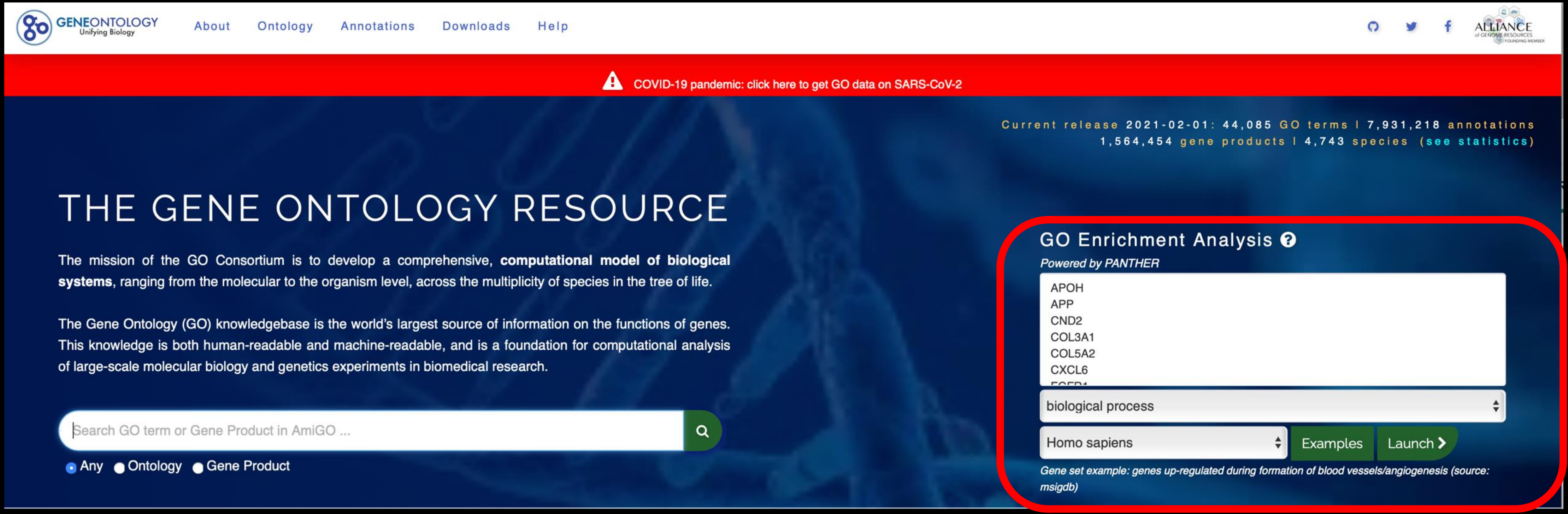


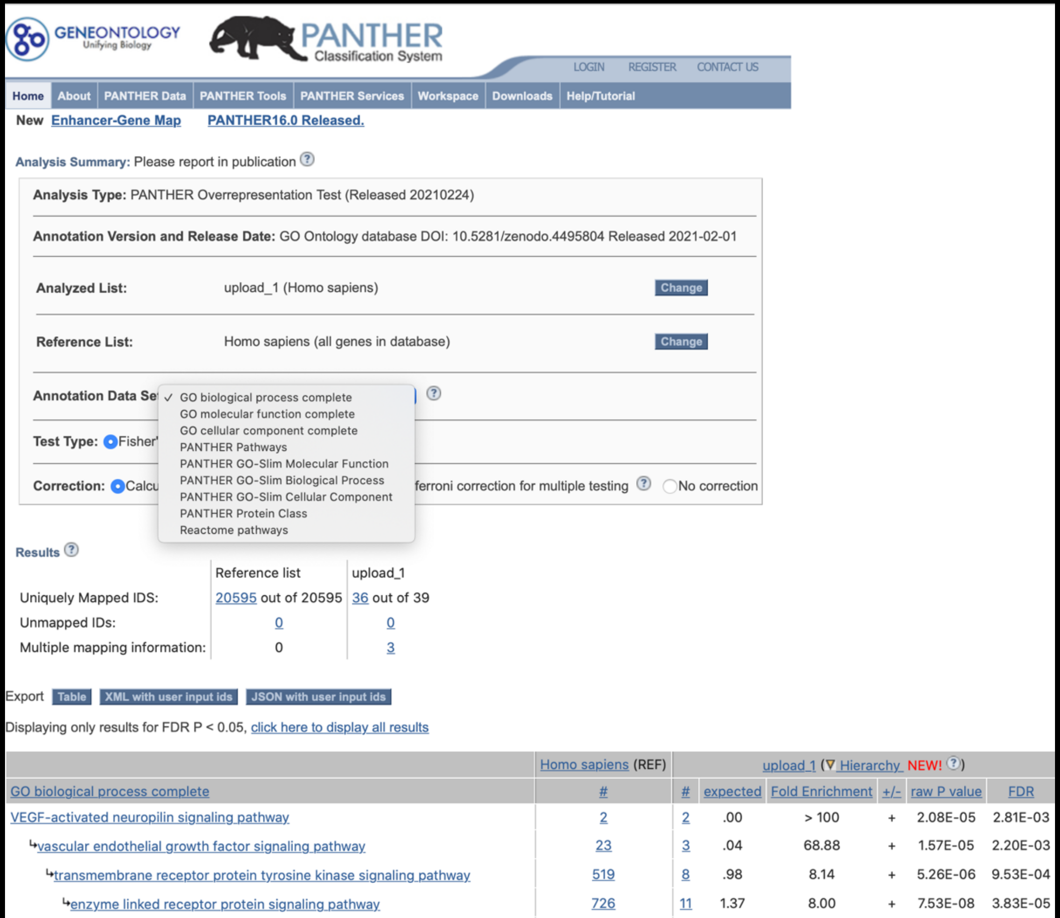
GO enrichment analysis (PANTHER)

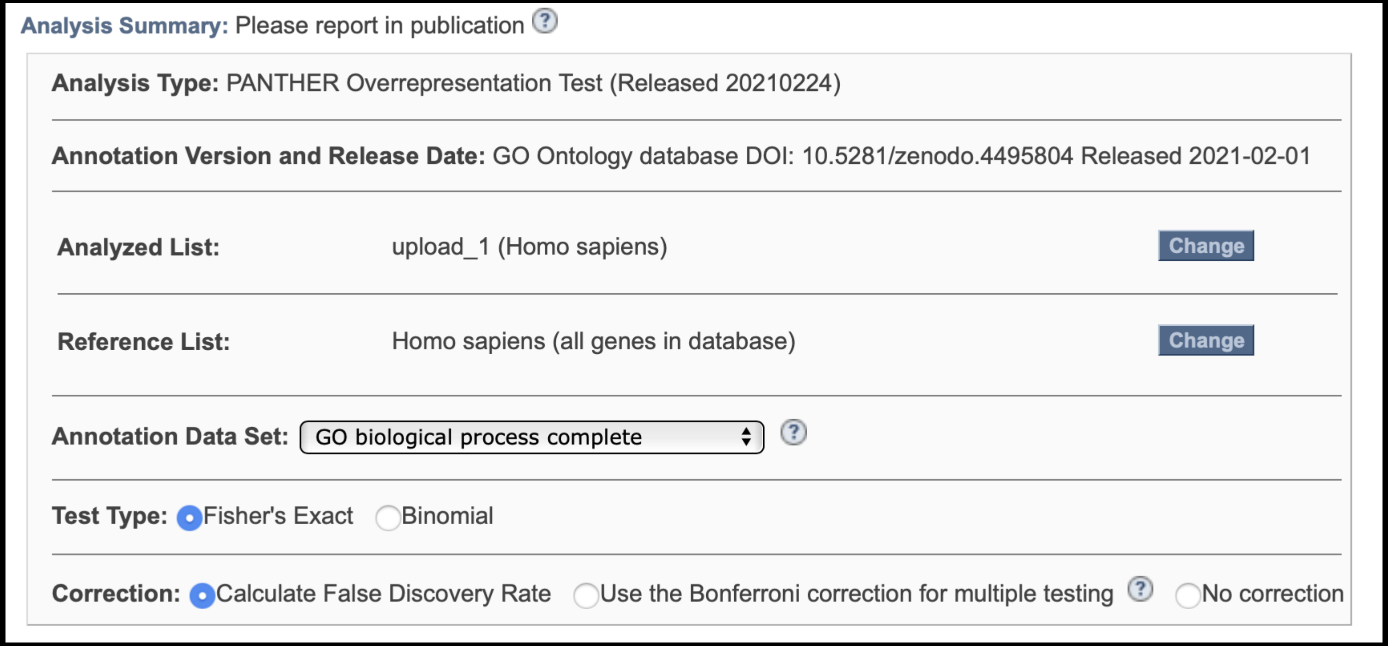
Enter a list of genes and click Launch >



GO enrichment analysis (PANTHER)

PANTHER results page. Notice you can select different gene sets/ontology sets and different statistical options.





GO enrichment analysis (PANTHER)

PANTHER results

- Test each annotation for enrichment
- Grouped by narrowest annotation category
- Sorted within group from most specific to most general

Total number of human genes mapping to a given annotation

Number of genes in your gene set mapping to a given annotation (vs expected)

/(expected #)

Enriched vs. depleted

P-value and false discovery rate

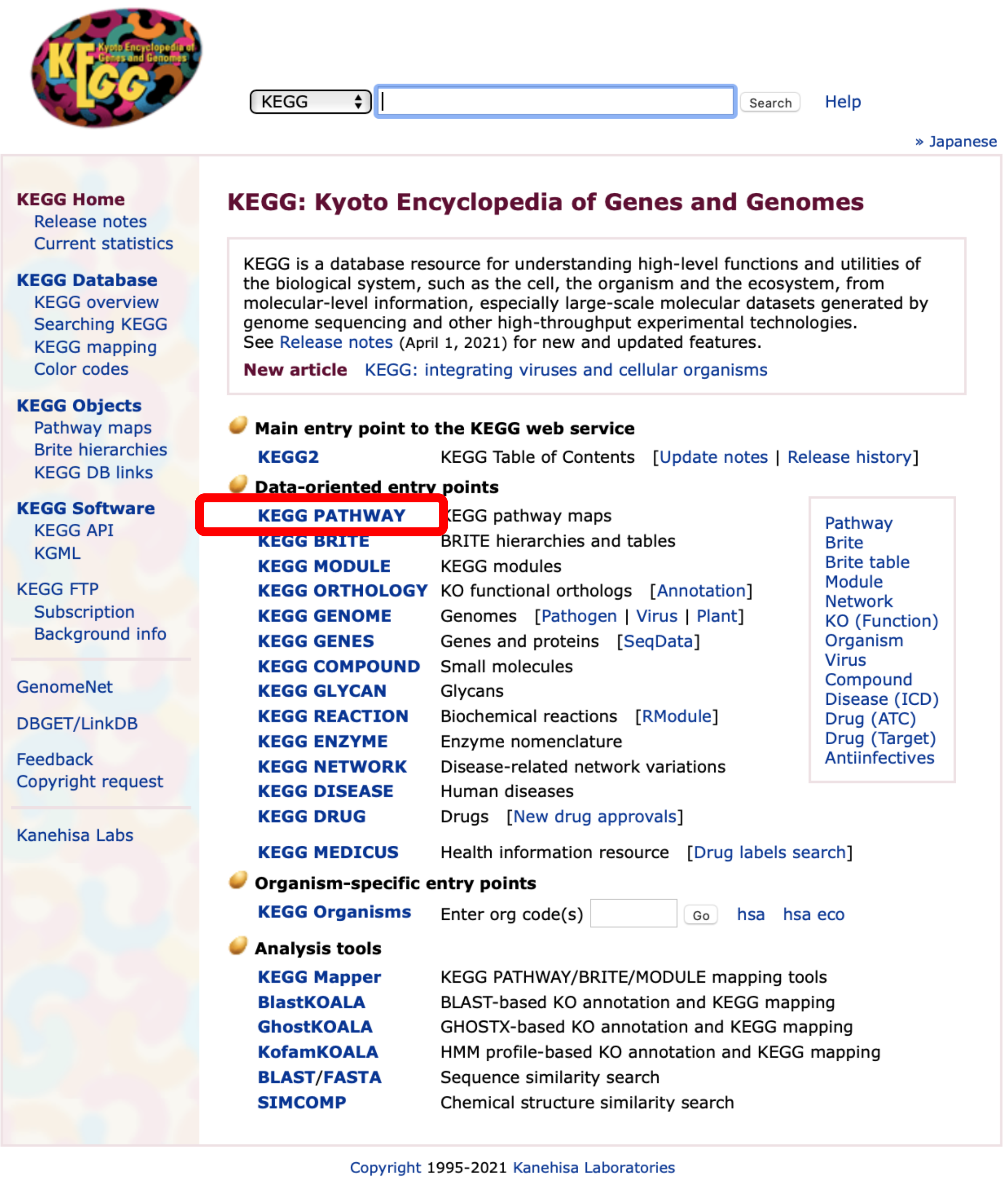
	Homo sapiens (REF)	upload_1					
GO biological process complete	#	#	expected	Fold Enrichment	+/-	raw P value	FDR
VEGF-activated neuropilin signaling pathway	2	2	.00	> 100	+	2.08E-05	2.81E-03
↳vascular endothelial growth factor signaling pathway	23	3	.04	68.88	+	1.57E-05	2.20E-03
↳transmembrane receptor protein tyrosine kinase signaling pathway	519	8	.98	8.14	+	5.26E-06	9.53E-04
↳enzyme linked receptor protein signaling pathway	726	11	1.37	8.00	+	7.53E-08	3.83E-05
↳cell surface receptor signaling pathway	2486	19	4.71	4.04	+	2.29E-08	1.44E-05
↳signal transduction	5138	23	9.73	2.36	+	6.76E-06	1.14E-03
↳signaling	5485	23	10.39	2.21	+	2.70E-05	3.31E-03
↳cellular process	15487	39	29.33	1.33	+	2.07E-05	2.81E-03
↳cell communication	5572	24	10.55	2.27	+	1.12E-05	1.69E-03
↳cellular response to stimulus	6770	26	12.82	2.03	+	1.99E-05	2.73E-03
↳response to stimulus	8496	35	16.09	2.18	+	3.82E-10	1.21E-06
↳regulation of cellular process	11302	33	21.40	1.54	+	1.39E-04	1.21E-02
↳regulation of biological process	11856	33	22.45	1.47	+	4.95E-04	3.15E-02
↳biological regulation	12571	34	23.81	1.43	+	4.61E-04	3.04E-02
↳cellular response to vascular endothelial growth factor stimulus	44	3	.08	36.01	+	9.56E-05	9.03E-03
↳cellular response to growth factor stimulus	503	10	.95	10.50	+	2.75E-08	1.67E-05
↳cellular response to organic substance	2331	18	4.41	4.08	+	5.67E-08	3.08E-05
↳cellular response to chemical stimulus	2898	19	5.49	3.46	+	2.71E-07	9.29E-05
↳response to chemical	4369	25	8.27	3.02	+	9.54E-09	7.53E-06
↳response to organic substance	2995	22	5.67	3.88	+	1.61E-09	2.31E-06
↳response to growth factor	532	11	1.01	10.92	+	3.26E-09	3.67E-06
↳neuropilin signaling pathway	4	2	.01	> 100	+	5.20E-05	5.50E-03
positive regulation of retinal ganglion cell axon guidance	2	2	.00	> 100	+	2.08E-05	2.79E-03
↳positive regulation of axon guidance	3	2	.01	> 100	+	3.47E-05	4.05E-03
↳regulation of axon guidance	14	2	.03	75.44	+	4.11E-04	2.78E-02
↳regulation of cellular component movement	1026	14	1.94	7.21	+	2.89E-09	3.51E-06
↳regulation of localization	2734	18	5.18	3.48	+	6.39E-07	1.87E-04
↳regulation of chemotaxis	222	5	.42	11.89	+	6.54E-05	6.53E-03
↳regulation of locomotion	990	14	1.87	7.47	+	1.84E-09	2.42E-06
↳regulation of response to external stimulus	1182	12	2.24	5.36	+	1.21E-06	2.81E-04
↳regulation of response to stimulus	4240	26	8.03	3.24	+	6.93E-10	1.37E-06
↳positive regulation of cellular component movement	556	9	1.05	8.55	+	8.27E-07	2.14E-04
↳positive regulation of cellular process	5656	21	10.71	1.96	+	5.11E-04	3.22E-02
↳positive regulation of biological process	6244	26	11.82	2.20	+	3.91E-06	7.82E-04
↳regulation of retinal ganglion cell axon guidance	5	2	.01	> 100	+	7.27E-05	7.12E-03
motor neuron migration	6	2	.01	> 100	+	9.68E-05	9.09E-03
↳cell migration	965	13	1.83	7.11	+	1.45E-08	9.94E-06
↳cell motility	1103	14	2.09	6.70	+	7.22E-09	6.33E-06
↳movement of cell or subcellular component	1591	16	3.01	5.31	+	1.14E-08	8.20E-06
↳localization of cell	1103	14	2.09	6.70	+	7.22E-09	6.70E-06
↳localization	5807	26	11.00	2.36	+	7.01E-07	1.91E-04
↳locomotion	1339	16	2.54	6.31	+	9.76E-10	1.71E-06
↳generation of neurons	1249	11	2.37	4.65	+	1.44E-05	2.07E-03
↳neurogenesis	1369	12	2.59	4.63	+	5.53E-06	9.79E-04
↳cell differentiation	3509	18	6.64	2.71	+	2.38E-05	3.02E-03
↳cellular developmental process	3562	18	6.75	2.67	+	2.93E-05	3.50E-03
↳developmental process	5729	26	10.85	2.40	+	5.26E-07	1.57E-04
↳nervous system development	2189	14	4.15	3.38	+	2.72E-05	3.30E-03
↳system development	4279	24	8.10	2.96	+	3.94E-08	2.22E-05
↳anatomical structure development	5266	26	9.97	2.61	+	8.61E-08	4.24E-05
↳multicellular organism development	4877	26	9.24	2.82	+	1.60E-08	1.05E-05
↳multicellular organismal process	6909	32	13.08	2.45	+	6.78E-10	1.53E-06

KEGG

https://www.genome.jp/kegg/

KEGG is a set of related databases.

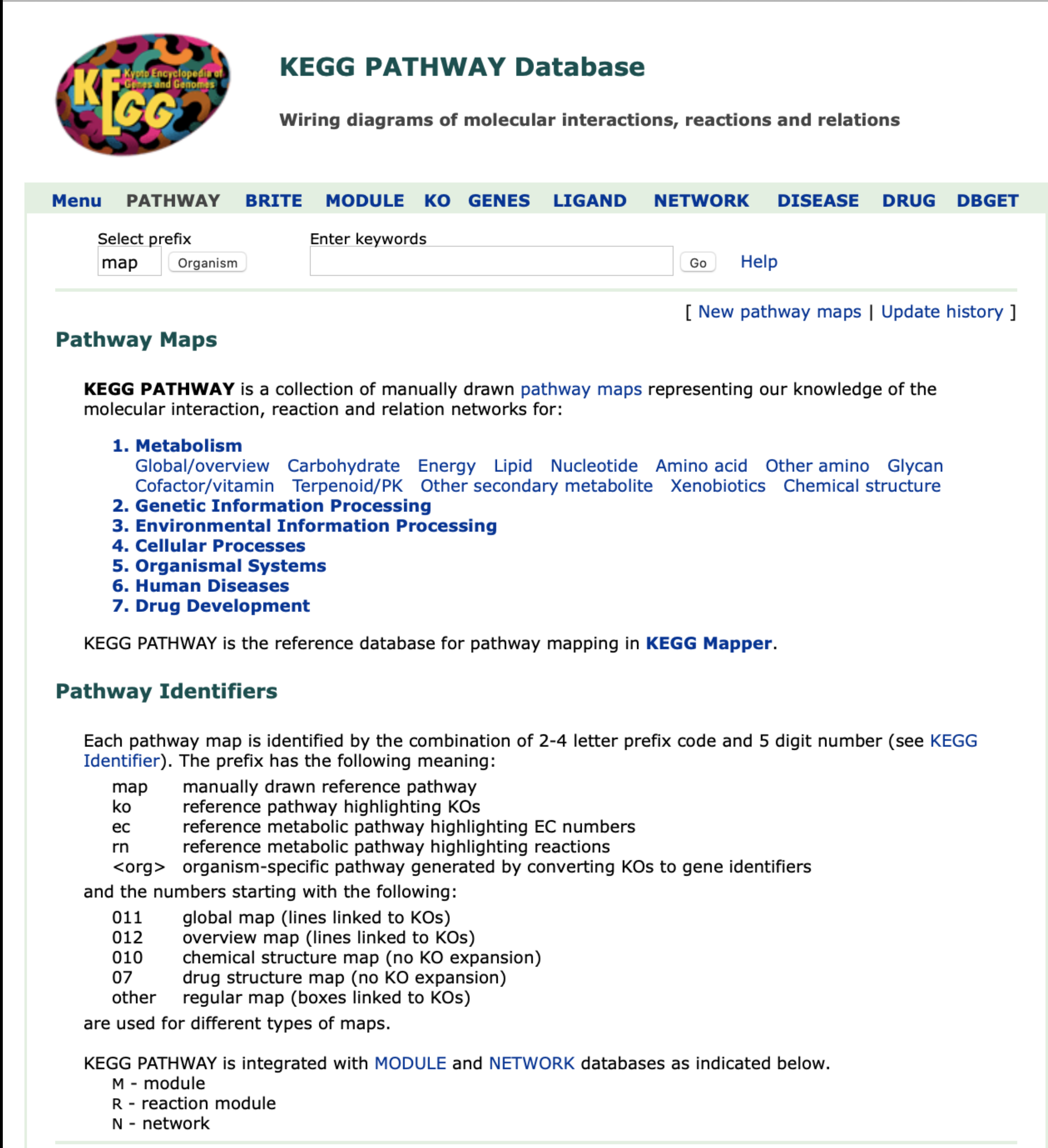
KEGG PATHWAY is the most important part for our purposes.



KEGG PATHWAY

A collection of manually drawn pathway maps.

Pathways fall into 7 categories.



KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations

Menu PATHWAY BRITE MODULE KO GENES LIGAND NETWORK DISEASE DRUG DBGET

Select prefix map Organism Enter keywords Go Help

[New pathway maps | Update history]

Pathway Maps

KEGG PATHWAY is a collection of manually drawn pathway maps representing our knowledge of the molecular interaction, reaction and relation networks for:

1. **Metabolism**
   Global/overview Carbohydrate Energy Lipid Nucleotide Amino acid Other amino Glycan
   Cofactor/vitamin Terpenoid/PK Other secondary metabolite Xenobiotics Chemical structure
2. **Genetic Information Processing**
3. **Environmental Information Processing**
4. **Cellular Processes**
5. **Organismal Systems**
6. **Human Diseases**
7. **Drug Development**

KEGG PATHWAY is the reference database for pathway mapping in **KEGG Mapper**.

Pathway Identifiers

Each pathway map is identified by the combination of 2-4 letter prefix code and 5 digit number (see KEGG Identifier). The prefix has the following meaning:

map	manually drawn reference pathway
ko	reference pathway highlighting KOs
ec	reference metabolic pathway highlighting EC numbers
rn	reference metabolic pathway highlighting reactions
<org>	organism-specific pathway generated by converting KOs to gene identifiers

and the numbers starting with the following:

011	global map (lines linked to KOs)
012	overview map (lines linked to KOs)
010	chemical structure map (no KO expansion)
07	drug structure map (no KO expansion)
other	regular map (boxes linked to KOs)

are used for different types of maps.

KEGG PATHWAY is integrated with MODULE and NETWORK databases as indicated below.

M - module
R - reaction module
N - network

KEGG PATHWAY

A few of the KEGG pathways.

Let's select "Metabolic pathways" (the entire category).

1. Metabolism

1.0 Global and overview maps

01100 M Metabolic pathways
01110 M Biosynthesis of secondary metabolites
01120 M Microbial metabolism in diverse environments
01200 M R Carbon metabolism
01210 M R 2-Oxocarboxylic acid metabolism
01212 M R Fatty acid metabolism
01230 M R Biosynthesis of amino acids
01240 M R Biosynthesis of cofactors *New!*
01220 M R Degradation of aromatic compounds

1.1 Carbohydrate metabolism

00010 M N Glycolysis / Gluconeogenesis
00020 M Citrate cycle (TCA cycle)
00030 M Pentose phosphate pathway
00040 M Pentose and glucuronate interconversions
00051 M Fructose and mannose metabolism
00052 M N Galactose metabolism
00053 M Ascorbate and aldarate metabolism
00500 M N Starch and sucrose metabolism
00520 M N Amino sugar and nucleotide sugar metabolism
00620 M Pyruvate metabolism
00630 M Glyoxylate and dicarboxylate metabolism
00640 M Propanoate metabolism
00650 M Butanoate metabolism
00660 M C5-Branched dibasic acid metabolism
00562 M Inositol phosphate metabolism

1.2 Energy metabolism

00190 M Oxidative phosphorylation
00195 M Photosynthesis
00196 Photosynthesis - antenna proteins
00710 M Carbon fixation in photosynthetic organisms
00720 M Carbon fixation pathways in prokaryotes
00680 M Methane metabolism
00910 M Nitrogen metabolism
00920 M Sulfur metabolism

1.3 Lipid metabolism

00061 M Fatty acid biosynthesis
00062 M Fatty acid elongation
00071 M N Fatty acid degradation
00072 M Synthesis and degradation of ketone bodies
00073 Cutin, suberine and wax biosynthesis
00100 M Steroid biosynthesis
00120 M N Primary bile acid biosynthesis
00121 Secondary bile acid biosynthesis
00140 M N Steroid hormone biosynthesis
00561 M Glycerolipid metabolism
00564 M Glycerophospholipid metabolism
00565 M Ether lipid metabolism
00600 M N Sphingolipid metabolism
00590 Arachidonic acid metabolism
00591 Linoleic acid metabolism
00592 M alpha-Linolenic acid metabolism
01040 M N Biosynthesis of unsaturated fatty acids

1.4 Nucleotide metabolism

00230 M N Purine metabolism
00240 M Pyrimidine metabolism

1.5 Amino acid metabolism

00250 M Alanine, aspartate and glutamate metabolism
00260 M Glycine, serine and threonine metabolism
00270 M Cysteine and methionine metabolism
00280 M N Valine, leucine and isoleucine degradation
00290 M Valine, leucine and isoleucine biosynthesis
00300 M Lysine biosynthesis
00310 M Lysine degradation
00220 M N Arginine biosynthesis
00330 M Arginine and proline metabolism

Metabolic pathways - Reference pathway

[Pathway menu | Pathway entry | Image (png) file | Help]

Change pathway type

▼ **Option**

Scale: 30%

Link: Normal

▼ **Search**

Go

▼ **User data** +

▼ **Module**

Pathway modules

Carbohydrate metabolism

Central carbohydrate metabo

M00001 Glycolysis (Embde

M00002 Glycolysis, core m

M00003 Gluconeogenesis

M00307 Pyruvate oxidation

M00009 Citrate cycle (TCA

M00010 Citrate cycle, first

M00011 Citrate cycle, seco

M00004 Pentose phosphate

M00006 Pentose phosphate

M00007 Pentose phosphate

M00580 Pentose phosphate

M00005 PRPP biosynthesis

M00008 Entner-Doudoroff p

M00308 Semi-phosphorylat

M00633 Semi-phosphorylat

M00309 Non-phosphorylati

Other carbohydrate metabolis

M00014 Glucuronate pathw

M00630 D-Galacturonate d

M00631 D-Galacturonate d

M00061 D-Glucuronate deg

M00081 Pectin degradation

M00632 Galactose degrada

M00552 D-galactonate deg

M00129 Ascorbate biosynth

M00114 Ascorbate biosynth

M00550 Ascorbate degrada

M00854 Glycogen biosynth

M00855 Glycogen degradat

M00565 Trehalose biosynth

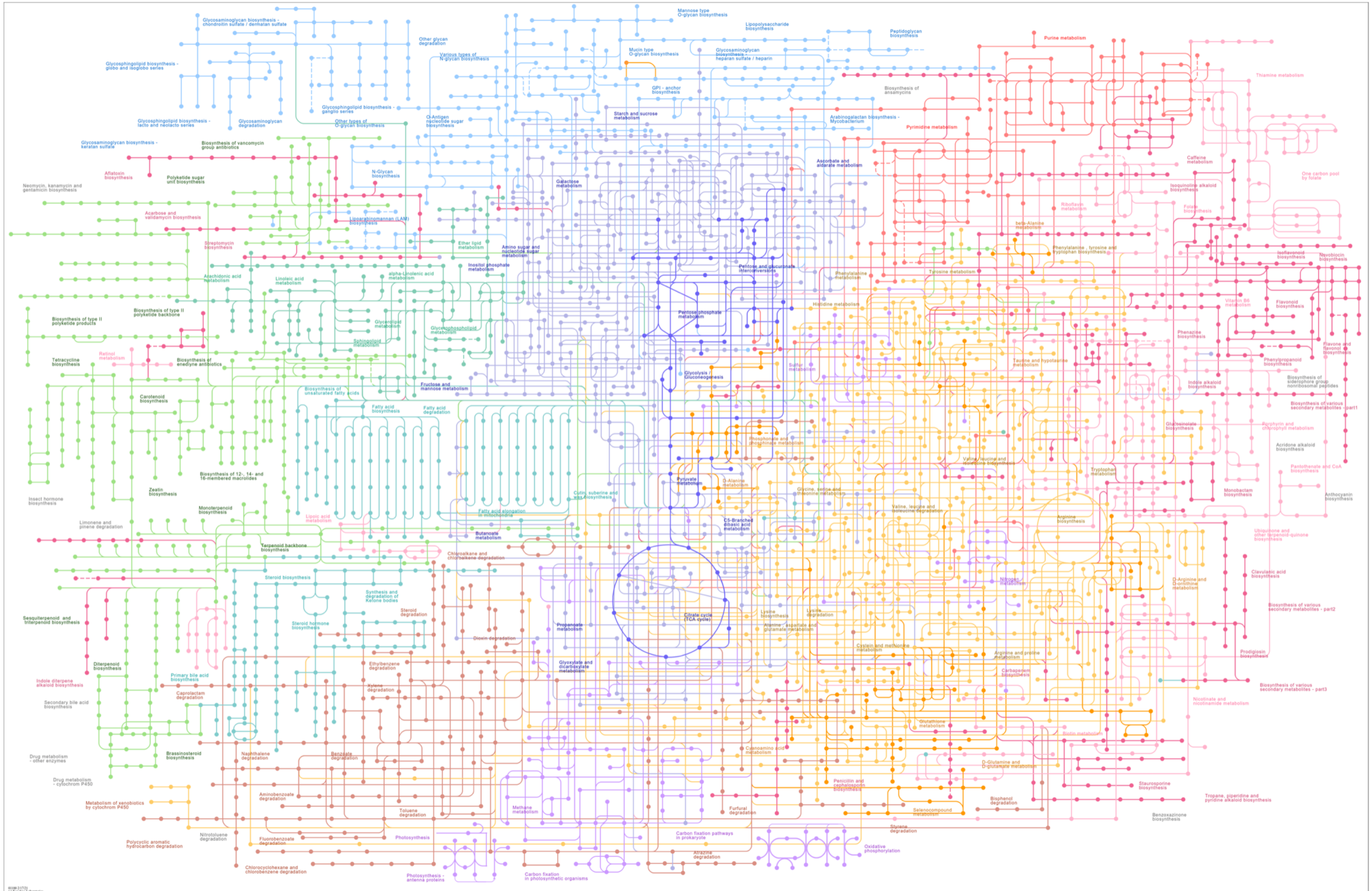
M00549 Nucleotide sugar b

M00554 Nucleotide sugar b

M00892 UDP-N-acetyl-D-gl

M00909 UDP-N-acetyl-D-gl

M00761 Undecaprenylphos



Metabolic pathways - Reference pathway

[Pathway menu | Pathway entry | Image (png) file | Help]

Change pathway type

Option

Scale: 30%

Link: Normal

Search

Go

User data +

Module

Pathway modules

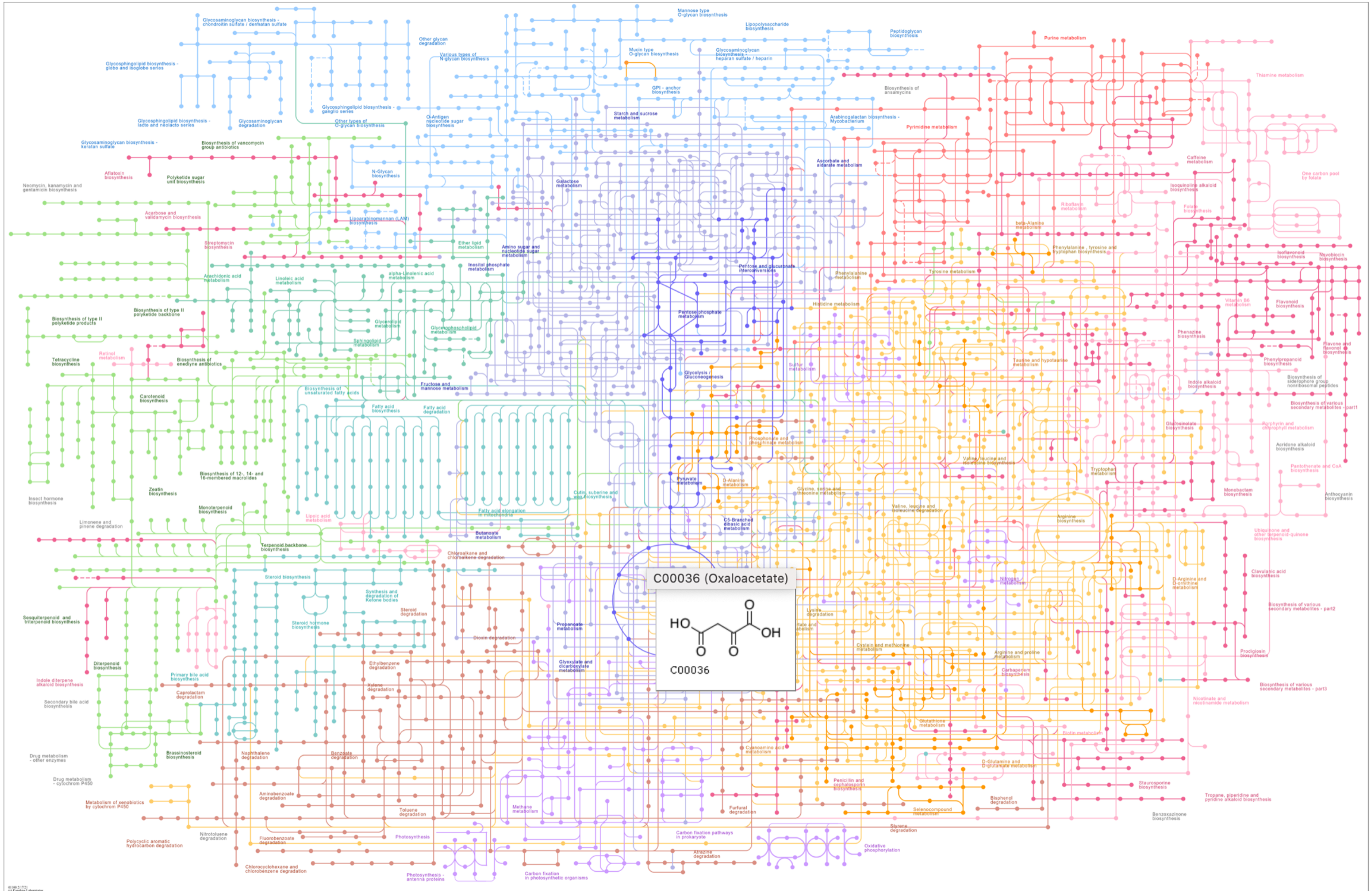
Carbohydrate metabolism

Central carbohydrate metabo

M00001 Glycolysis (Embde
M00002 Glycolysis, core m
M00003 Gluconeogenesis
M00307 Pyruvate oxidation
M00009 Citrate cycle (TCA
M00010 Citrate cycle, first
M00011 Citrate cycle, seco
M00004 Pentose phosphate
M00006 Pentose phosphate
M00007 Pentose phosphate
M00580 Pentose phosphate
M00005 PRPP biosynthesis
M00008 Entner-Doudoroff p
M00308 Semi-phosphorylat
M00633 Semi-phosphorylat
M00309 Non-phosphorylati

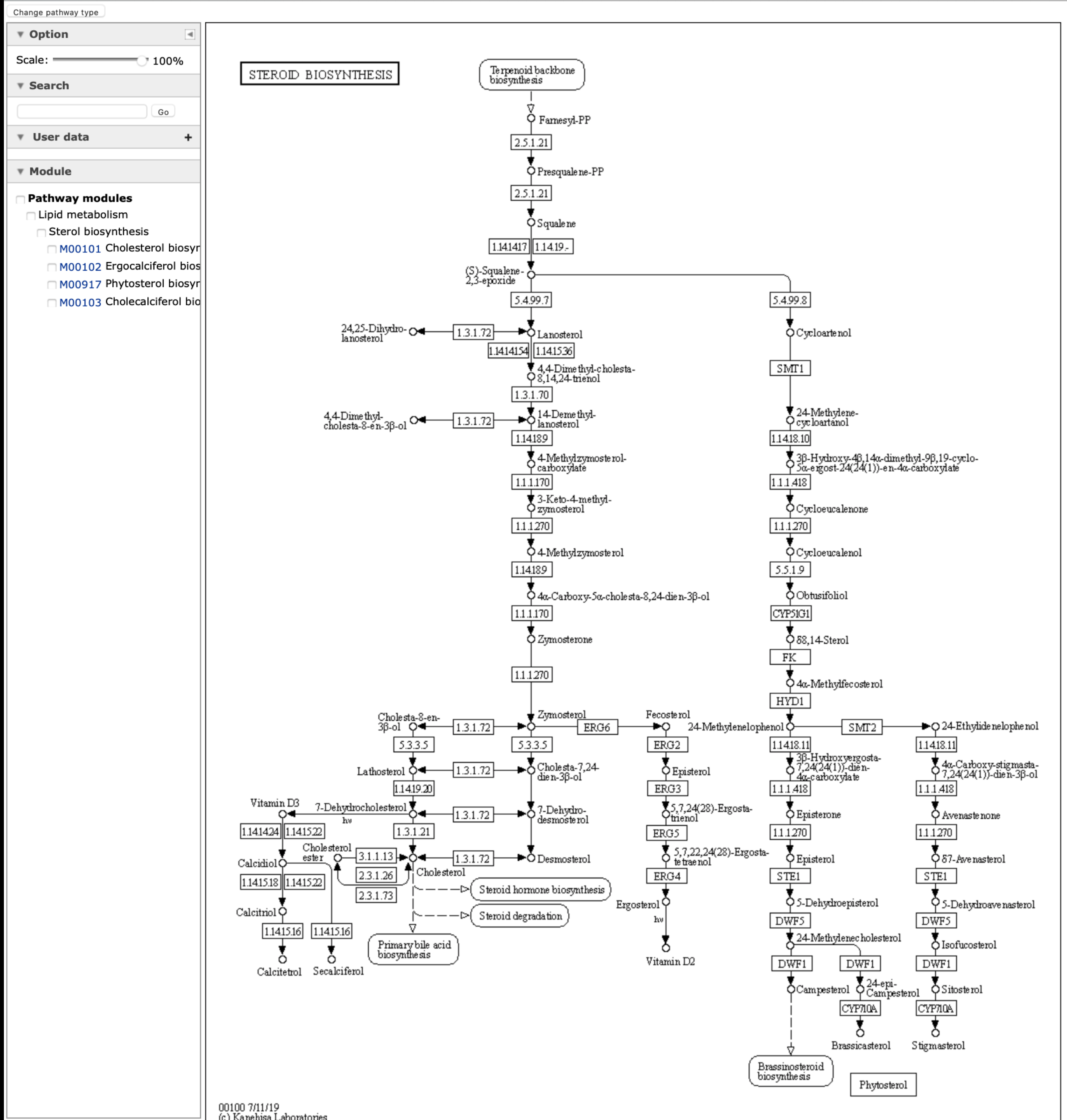
Other carbohydrate metabolis

M00014 Glucuronate pathw
M00630 D-Galacturonate d
M00631 D-Galacturonate d
M00061 D-Glucuronate deg
M00081 Pectin degradation
M00632 Galactose degrada
M00552 D-galactonate deg
M00129 Ascorbate biosynth
M00114 Ascorbate biosynth
M00550 Ascorbate degrada
M00854 Glycogen biosynth
M00855 Glycogen degradat
M00565 Trehalose biosynth
M00549 Nucleotide sugar b
M00554 Nucleotide sugar b
M00892 UDP-N-acetyl-D-gl
M00909 UDP-N-acetyl-D-gl
M00761 Undecaprenylphos



KEGG

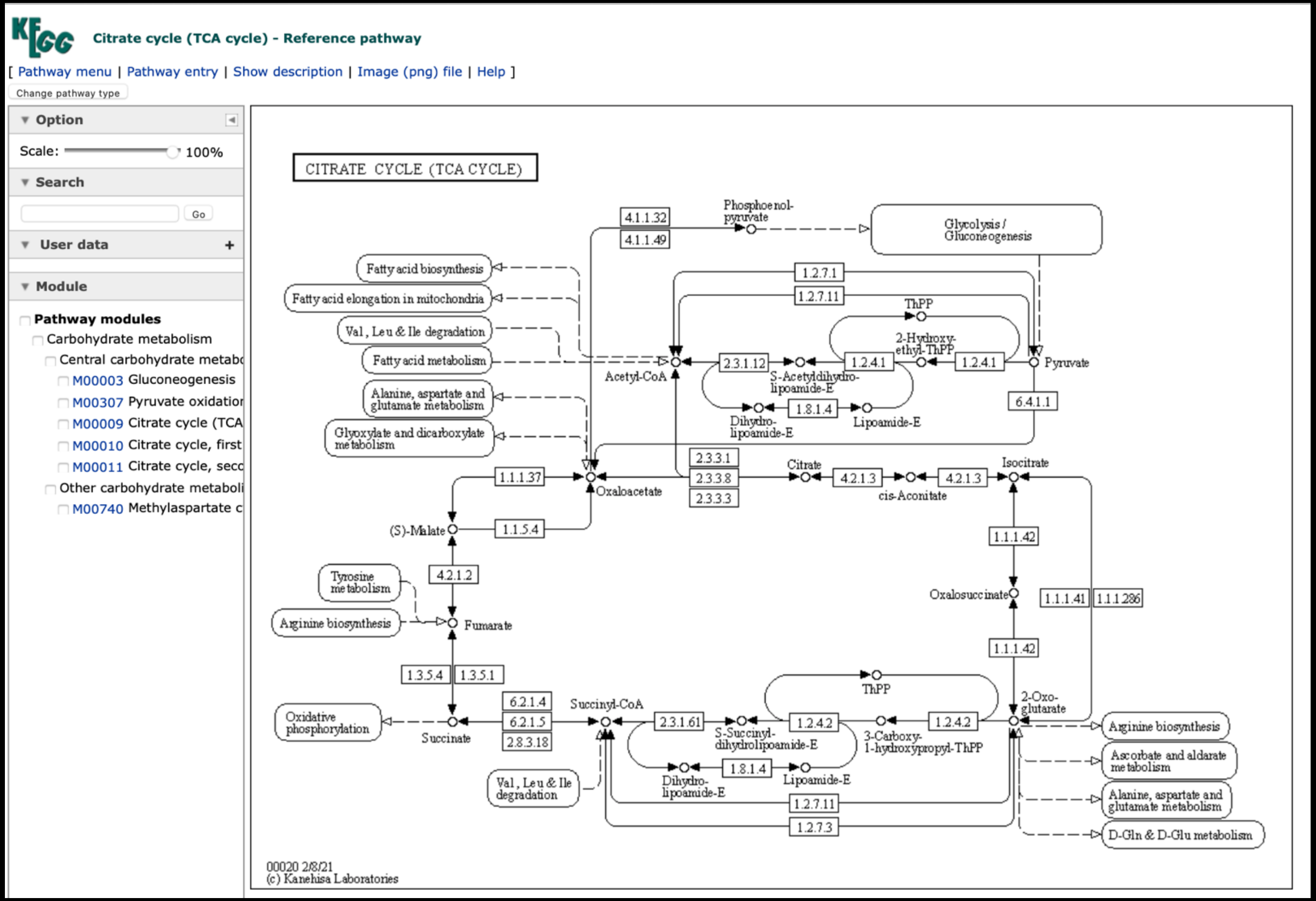
Selecting the “steroid biosynthesis” pathway



KEGG

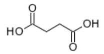
Selecting TCA cycle

Click on a component, e.g., succinate



succinate

COMPOUND: C00042 Help

Entry	C00042 Compound
Name	Succinate; Succinic acid; Butanedionic acid; Ethylenesuccinic acid
Formula	C4H6O4
Exact mass	118.0266
Mol weight	118.088
Structure	 C00042 Mol file KCF file DB search
Reaction	R00402 R00405 R00406 R00407 R00408 R00409 R00410 R00411 R00412 R00432 R00479 R00499 R00713 R00714 R00727 R00999 R01252 R01288 R01412 R01559 R01780 R01867 R01879 R02084 R02160 R02164 R02397 R02407 R02444 R02445 R02486 R02508 R02603 R02734 R02990 R03008 R03009 R03126 R03154 R03218 R03219 R03260 R03376 R03451 R03590 R03640 R03698 R03737 R03806 R03807 R03809 R03812 R04073 R04276 R04946 R05036 R05037 R05039 R05097 R05126 R05127 R05228 R05229 R05301 R05302 R05320 R05419 R05466 R05468 R05469 R05493 R05588 R05722 R05723 R05857 R06322 R06323 R06335 R06336 R06337 » show all
Pathway	map00020 Citrate cycle (TCA cycle) map00190 Oxidative phosphorylation map00250 Alanine, aspartate and glutamate metabolism map00310 Lysine degradation map00350 Tyrosine metabolism map00360 Phenylalanine metabolism map00361 Chlorocyclohexane and chlorobenzene degradation map00620 Pyruvate metabolism map00630 Glyoxylate and dicarboxylate metabolism map00640 Propanoate metabolism map00650 Butanoate metabolism map00720 Carbon fixation pathways in prokaryotes map00760 Nicotinate and nicotinamide metabolism map00920 Sulfur metabolism map01060 Biosynthesis of plant secondary metabolites map01061 Biosynthesis of phenylpropanoids map01062 Biosynthesis of terpenoids and steroids map01063 Biosynthesis of alkaloids derived from shikimate pathway map01064 Biosynthesis of alkaloids derived from ornithine, lysine and nicotinic acid map01065 Biosynthesis of alkaloids derived from histidine and purine map01066 Biosynthesis of alkaloids derived from terpenoid and polyketide map01070 Biosynthesis of plant hormones map01100 Metabolic pathways map01110 Biosynthesis of secondary metabolites map01120 Microbial metabolism in diverse environments map01200 Carbon metabolism map01220 Degradation of aromatic compounds map02020 Two-component system map04024 cAMP signaling pathway map04727 GABAergic synapse map04922 Glucagon signaling pathway map05230 Central carbon metabolism in cancer
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate M00027 GABA (gamma-Aminobutyrate) shunt M00148 Succinate dehydrogenase (ubiquinone) M00149 Succinate dehydrogenase, prokaryotes M00150 Fumarate reductase, prokaryotes M00173 Reductive citrate cycle (Arnon-Buchanan cycle) M00374 Dicarboxylate-hydroxybutyrate cycle M00376 3-Hydroxypropionate bi-cycle M00620 Incomplete reductive citrate cycle, acetyl-CoA => oxoglutarate

All links

Ontology (1)
  KEGG BRITE (1)
Pathway (42)
  KEGG PATHWAY (32)
  KEGG MODULE (10)
Network (1)
  KEGG NETWORK (1)
Drug (9)
  KEGG ENVIRON (8)
  CHEMBL (1)
Chemical substance (25)
  PubChem (1)
  ChEBI (1)
  3DMET (1)
  HMDB (1)
  HSDB (1)
  KNApSAcK (1)
  LIPIDMAPS (1)
  MASSBANK (16)
  NIKKAJI (1)
  PDB-CCD (1)
Chemical reaction (279)
  KEGG ENZYME (104)
  KEGG REACTION (175)
Gene (90353)
  KEGG GENES (90353)
All databases (90710)

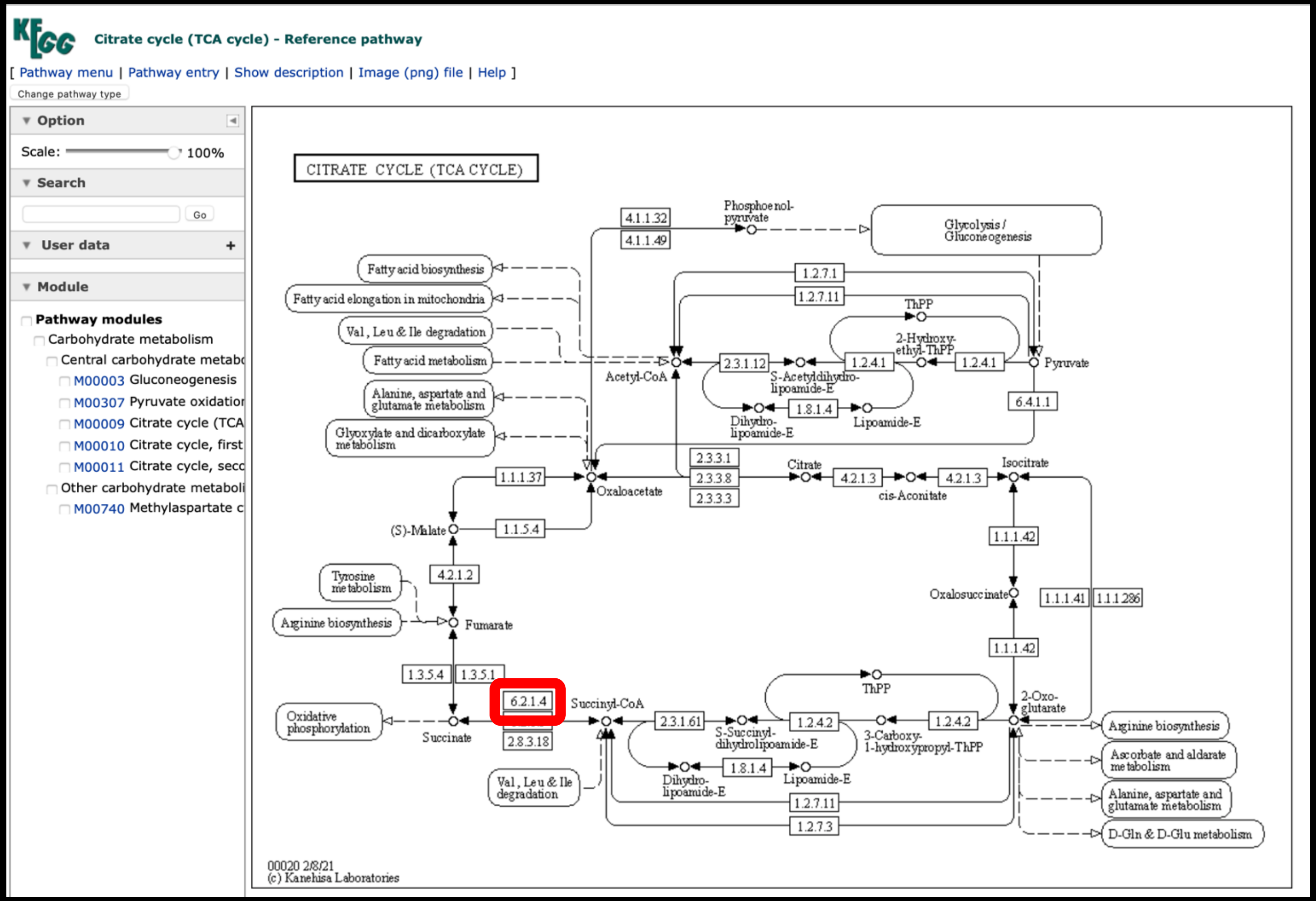
Download RDF

Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate M00027 GABA (gamma-Aminobutyrate) shunt M00148 Succinate dehydrogenase (ubiquinone) M00149 Succinate dehydrogenase, prokaryotes M00150 Fumarate reductase, prokaryotes M00173 Reductive citrate cycle (Arnon-Buchanan cycle) M00374 Dicarboxylate-hydroxybutyrate cycle M00376 3-Hydroxypropionate bi-cycle M00620 Incomplete reductive citrate cycle, acetyl-CoA => oxoglutarate

Enzyme	1.2.1.16 1.2.1.24 1.2.1.79 1.3. 1.3.4.1 1.3.5.1 1.3.5.4 1.3. 1.3.99.- 1.14.11.1 1.14.11.2 1.1 1.14.11.4 1.14.11.6 1.14.11.7 1.1 1.14.11.9 1.14.11.10 1.14.11.11 1.1 1.14.11.13 1.14.11.15 1.14.11.16 1.1 1.14.11.18 1.14.11.20 1.14.11.21 1.1 1.14.11.25 1.14.11.26 1.14.11.27 1.1 1.14.11.31 1.14.11.32 1.14.11.35 1.1 1.14.11.37 1.14.11.38 1.14.11.39 1.1 » show all
Brite	Compounds with biological roles [BR:br08001] Organic acids Carboxylic acids Dicarboxylic acids C00042 Succinate BRITE hierarchy
Other DBs	CAS: 110-15-6 PubChem: 3344 ChEBI: 15741 ChEMBL: CHEMBL576 LIPIDMAPS: LMFA01170043 KNApSAcK: C00001205 PDB-CCD: SIN[PDBj] 3DMET: B00012 NIKKAJI: J2.879G
KCF data	Show

KEGG

Go back to the map and choose and enzyme (6.2.1.4)



KEGG

ORTHOLOGY: K01899 Help

Entry	K01899 KO
Name	LSC1
Definition	succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Disease	H00469 Mitochondrial DNA depletion syndrome
Brite	KEGG Orthology (KO) [BR:ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01899 LSC1; succinyl-CoA synthetase alpha subu 00640 Propanoate metabolism K01899 LSC1; succinyl-CoA synthetase alpha subu Enzymes [BR:ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul 6.2.1.5 succinate---CoA ligase (ADP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul BRITE hierarchy
Other DBs	RN: R00432 R00727 GO: 0004776 0004775
Genes	HSA: 8802(SUCLG1) PTR: 470417(SUCLG1) PPS: 100985425(SUCLG1) GGO: 101143056(SUCLG1) PON: 100173831(SUCLG1) NLE: 100579244(SUCLG1) MCC: 693991(SUCLG1) MCF: 101867049(SUCLG1) CSAB: 103219966(SUCLG1) RRO: 104665951(SUCLG1) » show all Taxonomy UniProt

All links

- Ontology (4)
  - KEGG BRITE (2)
  - GO (2)
- Pathway (14)
  - KEGG PATHWAY (12)
  - KEGG MODULE (2)
- Disease (1)
  - KEGG DISEASE (1)
- Chemical reaction (6)
  - KEGG ENZYME (2)
  - KEGG REACTION (2)
  - KEGG RCLASS (2)
- Gene (10967)
  - KEGG GENES (718)
  - KEGG MGENES (468)
  - RefGene (9743)
  - OC (38)
- All databases (10992)

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ORTHOLOGY: K01900 Help

Entry	K01900 KO
Name	LSC2
Definition	succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Disease	H00469 Mitochondrial DNA depletion syndrome
Brite	KEGG Orthology (KO) [BR:ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01900 LSC2; succinyl-CoA synthetase beta subu 00640 Propanoate metabolism K01900 LSC2; succinyl-CoA synthetase beta subu 09180 Brite Hierarchies 09183 Protein families: signaling and cellular pro 04147 Exosome K01900 LSC2; succinyl-CoA synthetase beta subu Enzymes [BR:ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01900 LSC2; succinyl-CoA synthetase beta subu 6.2.1.5 succinate---CoA ligase (ADP-forming) K01900 LSC2; succinyl-CoA synthetase beta subu Exosome [BR:ko04147] Exosomal proteins Exosomal proteins of other body fluids (saliva and K01900 LSC2; succinyl-CoA synthetase beta subun: BRITE hierarchy
Other DBs	RN: R00432 R00727 GO: 0004776 0004775
Genes	HSA: 8801(SUCLG2) 8803(SUCLA2) PTR: 452711(SUCLA2) 460492(SUCLG2) PPS: 100971297(SUCLA2) 100973485(SUCLG2) GGO: 101127943(SUCLG2) 101128782(SUCLA2) PON: 100171525(SUCLA2) 100443464(SUCLG2) NLE: 100584620(SUCLA2) 100595372(SUCLG2) MCC: 697454(SUCLG2) 704434(SUCLA2) MCF: 101867314(SUCLA2) 102144513(SUCLG2) CSAB: 103214398(SUCLA2) 103227863(SUCLG2) RRO: 104660507 104661533 104662766(SUCLA2) 104678622 » show all Taxonomy UniProt

All links

- Ontology (5)
  - KEGG BRITE (3)
  - GO (2)
- Pathway (14)
  - KEGG PATHWAY (12)
  - KEGG MODULE (2)
- Disease (1)
  - KEGG DISEASE (1)
- Chemical reaction (6)
  - KEGG ENZYME (2)
  - KEGG REACTION (2)
  - KEGG RCLASS (2)
- Gene (12528)
  - KEGG GENES (908)
  - KEGG MGENES (551)
  - RefGene (11044)
  - OC (25)
- All databases (12554)

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ENZYME: 6.2.1.4 Help

Entry	EC 6.2.1.4 Enzyme
Name	succinate---CoA ligase (GDP-forming); succinyl-CoA synthetase (GDP-forming); succinyl coenzyme A synthetase (guanosine diphosphate-forming); succinate thiokinase (ambiguous); succinic thiokinase (ambiguous); succinyl coenzyme A synthetase (ambiguous); succinate-phosphorylating enzyme (ambiguous); P-enzyme; SCS (ambiguous); G-STK; succinyl coenzyme A synthetase (GDP-forming); succinyl CoA synthetase (ambiguous)
Class	Ligases; Forming carbon-sulfur bonds; Acid-thiol ligases BRITE hierarchy
Sysname	succinate:CoA ligase (GDP-forming)
Reaction(IUBMB)	GTP + succinate + CoA = GDP + phosphate + succinyl-CoA [RN:R00432]
Reaction(KEGG)	R00432; (other) R00727 R02405 R02406 Reaction
Substrate	GTP [CPD:C00044]; succinate [CPD:C00042]; CoA [CPD:C00010]
Product	GDP [CPD:C00035]; phosphate [CPD:C00009]; succinyl-CoA [CPD:C00091]
Comment	Itaconate can act instead of succinate, and ITP instead of GTP.
History	EC 6.2.1.4 created 1961
Pathway	ec00020 Citrate cycle (TCA cycle) ec00640 Propanoate metabolism ec01100 Metabolic pathways ec01110 Biosynthesis of secondary metabolites ec01120 Microbial metabolism in diverse environments
Orthology	K01899 succinyl-CoA synthetase alpha subunit K01900 succinyl-CoA synthetase beta subunit
Genes	HSA: 8801(SUCLG2) 8802(SUCLG1) 8803(SUCLA2) PTR: 452711(SUCLA2) 460492(SUCLG2) 470417(SUCLG1) PPS: 100971297(SUCLA2) 100973485(SUCLG2) 100985425(SUCLG1) GGO: 101127943(SUCLG2) 101128782(SUCLA2) 101143056(SUCLG1) PON: 100171525(SUCLA2) 100173831(SUCLG1) 100443464(SUCLG2) NLE: 100579244(SUCLG1) 100584620(SUCLA2) 100595372(SUCLG2) MCC: 693991(SUCLG1) 697454(SUCLG2) 704434(SUCLA2) MCF: 101867049(SUCLG1) 101867314(SUCLA2) 102144513(SUCLG2) CSAB: 103214398(SUCLA2) 103219966(SUCLG1) 103227863(SUCLG2) RRO: 104660507 104661533 104662766(SUCLA2) 104665951(SUCLG1) 104678622 » show all Taxonomy
Reference	1
Authors	Hager, L.P.
Title	Succinyl CoA synthetase.
Journal	In: Boyer, P.D., Lardy, H. and Myrback, K. (Eds.), The Enzymes, 2nd ed., vol. 6, Academic Press, New York, 1962, p. 387-399.
Reference	2 [PMID:13084656]
Authors	KAUFMAN S, GILVARG C, CORI O, OCHOA S.
Title	Enzymatic oxidation of alpha-ketoglutarate and coupled phosphorylation.
Journal	J Biol Chem 203:869-88 (1953)
Reference	3 [PMID:13768680]
Authors	MAZUMDER R, SANADI DR, RODWELL VW.
Title	Purification and properties of hog kidney succinic thiokinase.
Journal	J Biol Chem 235:2546-50 (1960)
Reference	4 [PMID:13181903]
Authors	SANADI DR, GIBSON M, AYENGAR P.
Title	Guanosine triphosphate, the primary product of phosphorylation coupled to the breakdown of succinyl coenzyme A.
Journal	Biochim Biophys Acta 14:434-6 (1954) DOI:10.1016/0006-3002(54)90205-X
Other DBs	ExplorEnz - The Enzyme Database: 6.2.1.4 IUBMB Enzyme Nomenclature: 6.2.1.4 ExPASy - ENZYME nomenclature database: 6.2.1.4 BRENDA, the Enzyme Database: 6.2.1.4 CAS: 9014-36-2

All links

Pathway (12)
- KEGG PATHWAY (10)
- KEGG MODULE (2)

Chemical substance (10)
- KEGG COMPOUND (10)

Chemical reaction (6)
- KEGG REACTION (4)
- KEGG RCLASS (2)

Gene (23434)
- KEGG ORTHOLOGY (2)
- KEGG GENES (1626)
- KEGG MGENES (1019)
- RefGene (20787)

Protein sequence (1749)
- UniProt (1613)
- SWISS-PROT (18)
- RefSeq(pep) (86)
- PDBSTR (32)

DNA sequence (322)
- RefSeq(nuc) (142)
- GenBank (112)
- EMBL (68)

3D Structure (14)
- PDB (14)

Protein domain (14)
- InterPro (13)
- PROSITE(DOC) (1)

All databases (25561)

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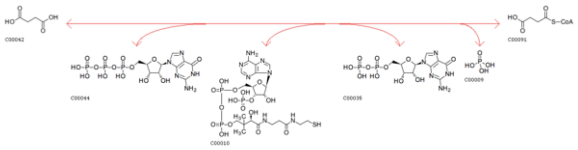
REACTION: R00432 Help

Entry	R00432 Reaction
Name	Succinate:CoA ligase (GDP-forming)
Definition	GTP + Succinate + CoA <=> GDP + Orthophosphate + Succinyl-CoA
Equation	C00044 + C00042 + C00010 <=> C00035 + C00009 + C00091
	
Reaction class	RC00002 C00035_C00044 RC00004 C00010_C00091 RC00014 C00042_C00091
Enzyme	6.2.1.4
Pathway	rn00020 Citrate cycle (TCA cycle) rn00640 Propanoate metabolism rn01100 Metabolic pathways rn01110 Biosynthesis of secondary metabolites rn01200 Carbon metabolism
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Orthology	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5] K01900 succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]
Other DBs	RHEA: 22123

All links

Ontology (1)
- KEGG BRITE (1)

Pathway (12)
- KEGG PATHWAY (10)
- KEGG MODULE (2)

Chemical substance (6)
- KEGG COMPOUND (6)

Chemical reaction (5)
- KEGG ENZYME (1)
- KEGG RCLASS (3)
- SABIO-RK (1)

Gene (1628)
- KEGG ORTHOLOGY (2)
- KEGG GENES (1626)

All databases (1652)

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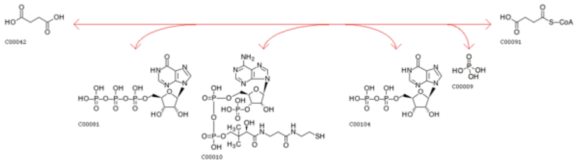
REACTION: R00727 Help

Entry	R00727 Reaction
Name	Succinate:CoA ligase (IDP-forming)
Definition	ITP + Succinate + CoA <=> IDP + Orthophosphate + Succinyl-CoA
Equation	C00081 + C00042 + C00010 <=> C00104 + C00009 + C00091
	
Reaction class	RC00002 C00081_C00104 RC00004 C00010_C00091 RC00014 C00042_C00091
Enzyme	6.2.1.4
Pathway	rn00020 Citrate cycle (TCA cycle) rn01100 Metabolic pathways rn01110 Biosynthesis of secondary metabolites rn01120 Microbial metabolism in diverse environments rn01200 Carbon metabolism
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Orthology	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5] K01900 succinyl-CoA synthetase beta subunit [EC:6.2.1.4 6.2.1.5]

All links

Ontology (1)
- KEGG BRITE (1)

Pathway (12)
- KEGG PATHWAY (10)
- KEGG MODULE (2)

Chemical substance (6)
- KEGG COMPOUND (6)

Chemical reaction (4)
- KEGG ENZYME (1)
- KEGG RCLASS (3)

Gene (1628)
- KEGG ORTHOLOGY (2)
- KEGG GENES (1626)

All databases (1651)

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DBGET integrated database retrieval system

KEGG

Select the human gene

ORTHOLOGY: K01899 Help

Entry	K01899 KO
Name	LSC1
Definition	succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]
Pathway	ko00020 Citrate cycle (TCA cycle) ko00640 Propanoate metabolism ko01100 Metabolic pathways ko01110 Biosynthesis of secondary metabolites ko01120 Microbial metabolism in diverse environments ko01200 Carbon metabolism
Module	M00009 Citrate cycle (TCA cycle, Krebs cycle) M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Disease	H00469 Mitochondrial DNA depletion syndrome
Brite	KEGG Orthology (KO) [BR:ko00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) K01899 LSC1; succinyl-CoA synthetase alpha subu 00640 Propanoate metabolism K01899 LSC1; succinyl-CoA synthetase alpha subu Enzymes [BR:ko01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul 6.2.1.5 succinate---CoA ligase (ADP-forming) K01899 LSC1; succinyl-CoA synthetase alpha sul BRITE hierarchy
Other DBs	RN: R00432 R00727 GO: 0004776 0004775
Genes	HSA: 8802(SUCLG1) PTR: 470417(SUCLG1) PPS: 100985425(SUCLG1) GGO: 101143056(SUCLG1) PON: 100173831(SUCLG1) NLE: 100579244(SUCLG1) MCC: 693991(SUCLG1) MCF: 101867049(SUCLG1) CSAB: 103219966(SUCLG1) RRO: 104665951(SUCLG1) » show all Taxonomy UniProt

All links

- Ontology (4)
  - KEGG BRITE (2)
  - GO (2)
- Pathway (14)
  - KEGG PATHWAY (12)
  - KEGG MODULE (2)
- Disease (1)
  - KEGG DISEASE (1)
- Chemical reaction (6)
  - KEGG ENZYME (2)
  - KEGG REACTION (2)
  - KEGG RCLASS (2)
- Gene (10967)
  - KEGG GENES (718)
  - KEGG MGENES (468)
  - RefGene (9743)
  - OC (38)
- All databases (10992)

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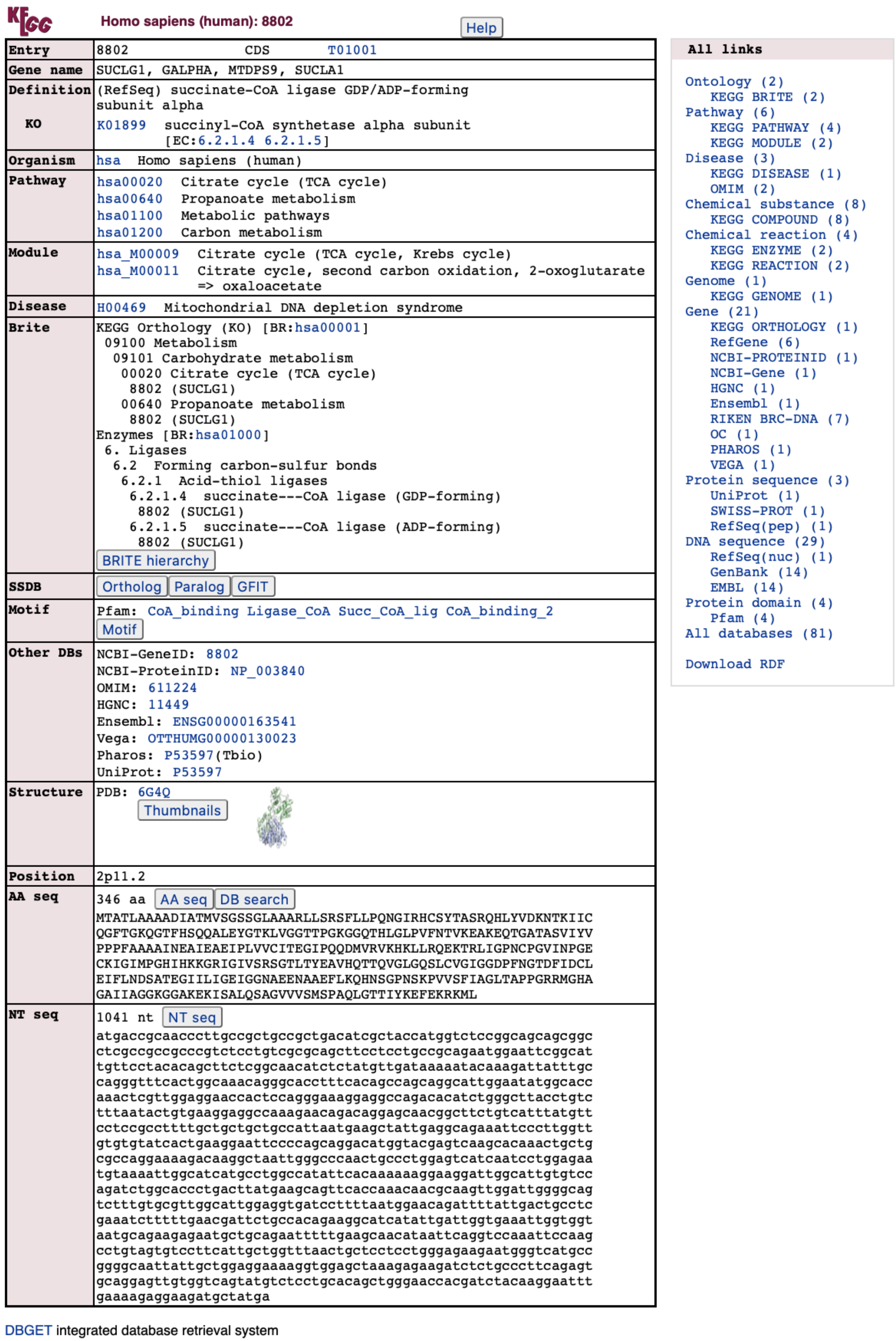
KEGG

Gene information for *SUCLG1.*

Pathways/modules the gene is associated with are listed.

There are links to other databases.



Homo sapiens (human): 8802

Help

Entry	8802 CDS T01001
Gene name	SUCLG1, GALPHA, MTDPS9, SUCLA1
Definition	(RefSeq) succinate-CoA ligase GDP/ADP-forming subunit alpha
KO	K01899 succinyl-CoA synthetase alpha subunit [EC:6.2.1.4 6.2.1.5]
Organism	hsa Homo sapiens (human)
Pathway	hsa00020 Citrate cycle (TCA cycle) hsa00640 Propanoate metabolism hsa01100 Metabolic pathways hsa01200 Carbon metabolism
Module	hsa_M00009 Citrate cycle (TCA cycle, Krebs cycle) hsa_M00011 Citrate cycle, second carbon oxidation, 2-oxoglutarate => oxaloacetate
Disease	H00469 Mitochondrial DNA depletion syndrome
Brite	KEGG Orthology (KO) [BR:hsa00001] 09100 Metabolism 09101 Carbohydrate metabolism 00020 Citrate cycle (TCA cycle) 8802 (SUCLG1) 00640 Propanoate metabolism 8802 (SUCLG1) Enzymes [BR:hsa01000] 6. Ligases 6.2 Forming carbon-sulfur bonds 6.2.1 Acid-thiol ligases 6.2.1.4 succinate---CoA ligase (GDP-forming) 8802 (SUCLG1) 6.2.1.5 succinate---CoA ligase (ADP-forming) 8802 (SUCLG1) BRITE hierarchy
SSDB	Ortholog Paralog GFIT
Motif	Pfam: CoA_binding Ligase_CoA Succ_CoA_lig CoA_binding_2 Motif
Other DBs	NCBI-GeneID: 8802 NCBI-ProteinID: NP_003840 OMIM: 611224 HGNC: 11449 Ensembl: ENSG00000163541 Vega: OTTHUMG00000130023 Pharos: P53597(Tbio) UniProt: P53597
Structure	PDB: 6G4Q Thumbnails
Position	2p11.2
AA seq	346 aa AA seq DB search MTATLAAAADIATMVSGSSGLAAARLLSRSFLLPQNGIRHCSYTASRQHLYVDKNTKIIC QGFTGKQGTFHSQQALEYGTKLVGGTTPGKGGQTHLGLPVFNTVKEAKEQTGATASVIYV PPPFAAAAINEAIEAEIPLVVCITEGIPQQDMVRVKHKLLRQEKTRLIGPNCPGVINPGE CKIGIMPGHIHKKGRIGIVSRSGTLTYEAVHQTTQVGLGQSLCVGIGGDPFNGTDFIDCL EIFLNDSATEGIILIGEIGGNAEENAAEFLKQHNSGPNSKPVVSFIAGLTAPPGRRMGHA GAIIAGGKGGAKEKISALQSAGVVVSMSPAQLGTTIYKEFEKRKML
NT seq	1041 nt NT seq atgaccgcaacccttgccgctgccgctgacatcgctaccatggtctccggcagcagcggc ctcgccgccgcccgtctcctgtcgcgcagcttcctcctgccgcagaatggaattcggcat tgttcctacacagcttctcggcaacatctctatgttgataaaaatacaaagattatttgc cagggtttcactggcaaacagggcacctttcacagccagcaggcattggaatatggcacc aaactcgttggaggaaccactccagggaaaggaggccagacacatctgggcttacctgtc tttaatactgtgaaggaggccaaagaacagacaggagcaacggcttctgtcatttatgtt cctccgccttttgctgctgctgccattaatgaagctattgaggcagaaattcccttggtt gtgtgtatcactgaaggaattccccagcaggacatggtacgagtcaagcacaaactgctg cgccaggaaaagacaaggctaattgggcccaactgccctggagtcatcaatcctggagaa tgtaaaattggcatcatgcctggccatattcacaaaaaaggaaggattggcattgtgtcc agatctggcaccctgacttatgaagcagttcaccaaacaacgcaagttggattggggcag tctttgtgcgttggcattggaggtgatccttttaatggaacagattttattgactgcctc gaaatctttttgaacgattctgccacagaaggcatcatattgattggtgaaattggtggt aatgcagaagagaatgctgcagaattttttgaagcaacataattcaggtccaaattccaag cctgtagtgtccttcattgctggtttaactgctcctcctgggagaagaatgggtcatgcc ggggcaattattgctggaggaaaaggtggagctaaagagaagatctctgcccttcagagt gcaggagttgtggtcagtatgtctcctgcacagctgggaaccacgatctacaaggaattt gaaaagaggaagatgctatga

DBGET integrated database retrieval system

All links

- Ontology (2)
  - KEGG BRITE (2)
- Pathway (6)
  - KEGG PATHWAY (4)
  - KEGG MODULE (2)
- Disease (3)
  - KEGG DISEASE (1)
  - OMIM (2)
- Chemical substance (8)
  - KEGG COMPOUND (8)
- Chemical reaction (4)
  - KEGG ENZYME (2)
  - KEGG REACTION (2)
- Genome (1)
  - KEGG GENOME (1)
- Gene (21)
  - KEGG ORTHOLOGY (1)
  - RefGene (6)
  - NCBI-PROTEINID (1)
  - NCBI-Gene (1)
  - HGNC (1)
  - Ensembl (1)
  - RIKEN BRC-DNA (7)
  - OC (1)
  - PHAROS (1)
  - VEGA (1)
- Protein sequence (3)
  - UniProt (1)
  - SWISS-PROT (1)
  - RefSeq(pep) (1)
- DNA sequence (29)
  - RefSeq(nuc) (1)
  - GenBank (14)
  - EMBL (14)
- Protein domain (4)
  - Pfam (4)
- All databases (81)

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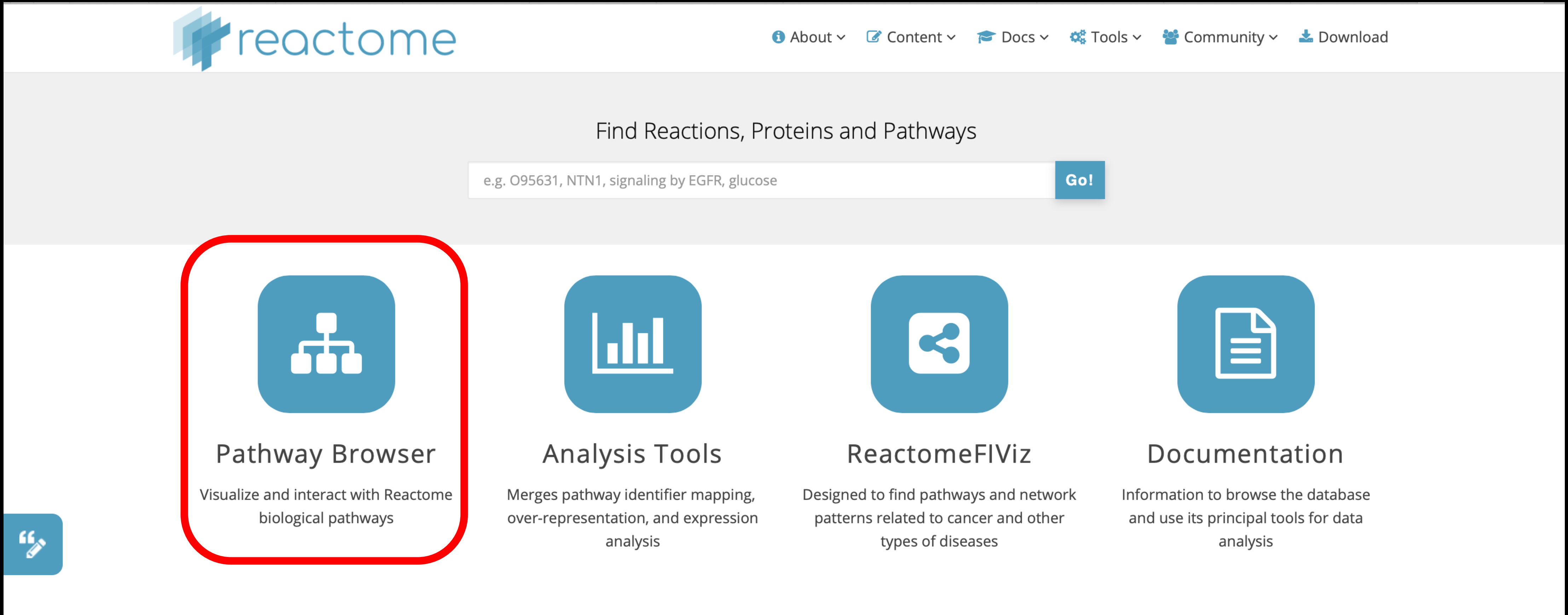
Reactome

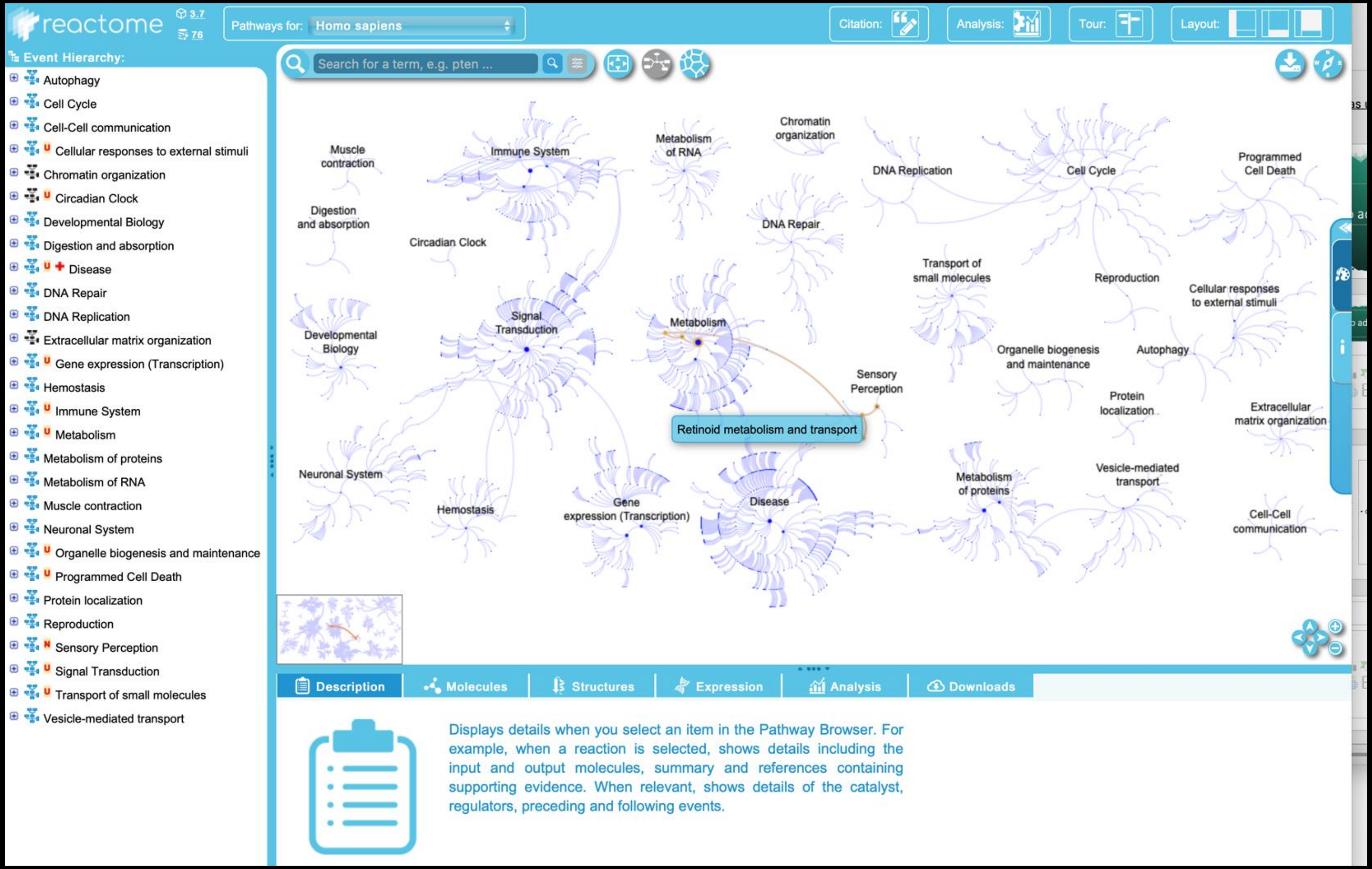
reactome.org

Reactome is an online pathway database, with a focus on reactions.

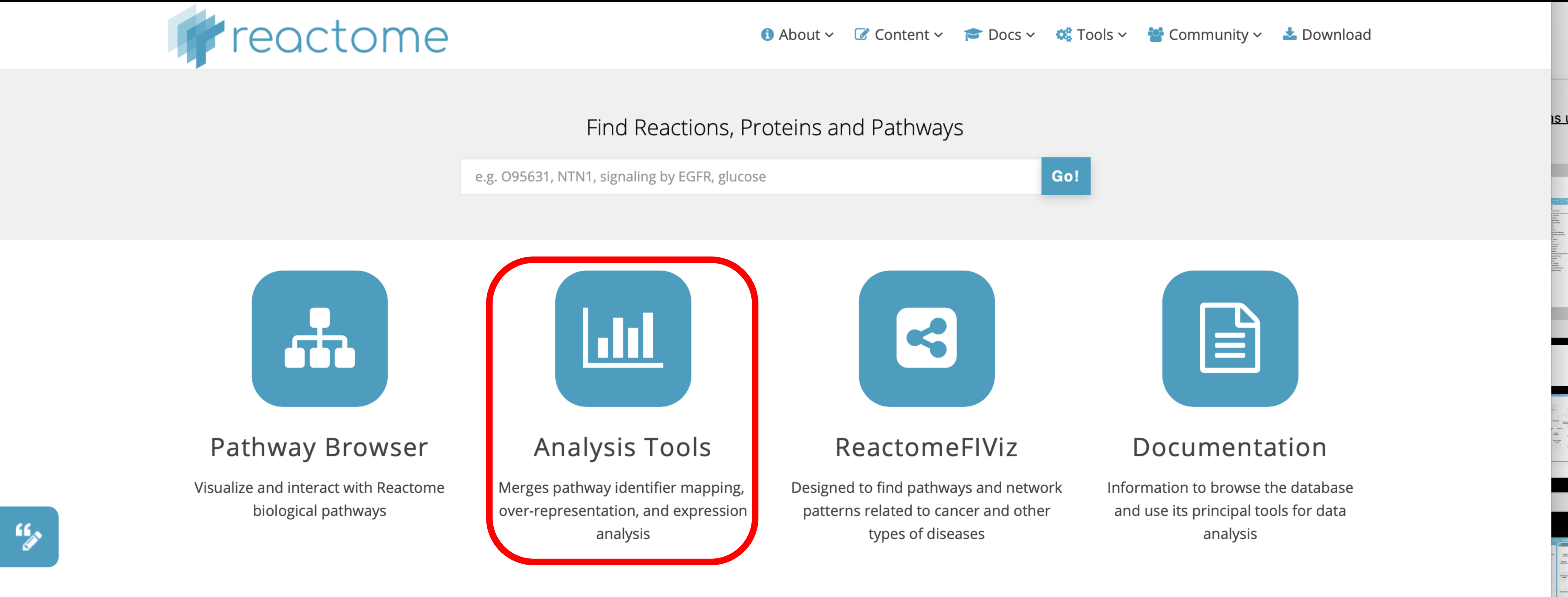


Reactome

https://reactome.org/PathwayBrowser/



Reactome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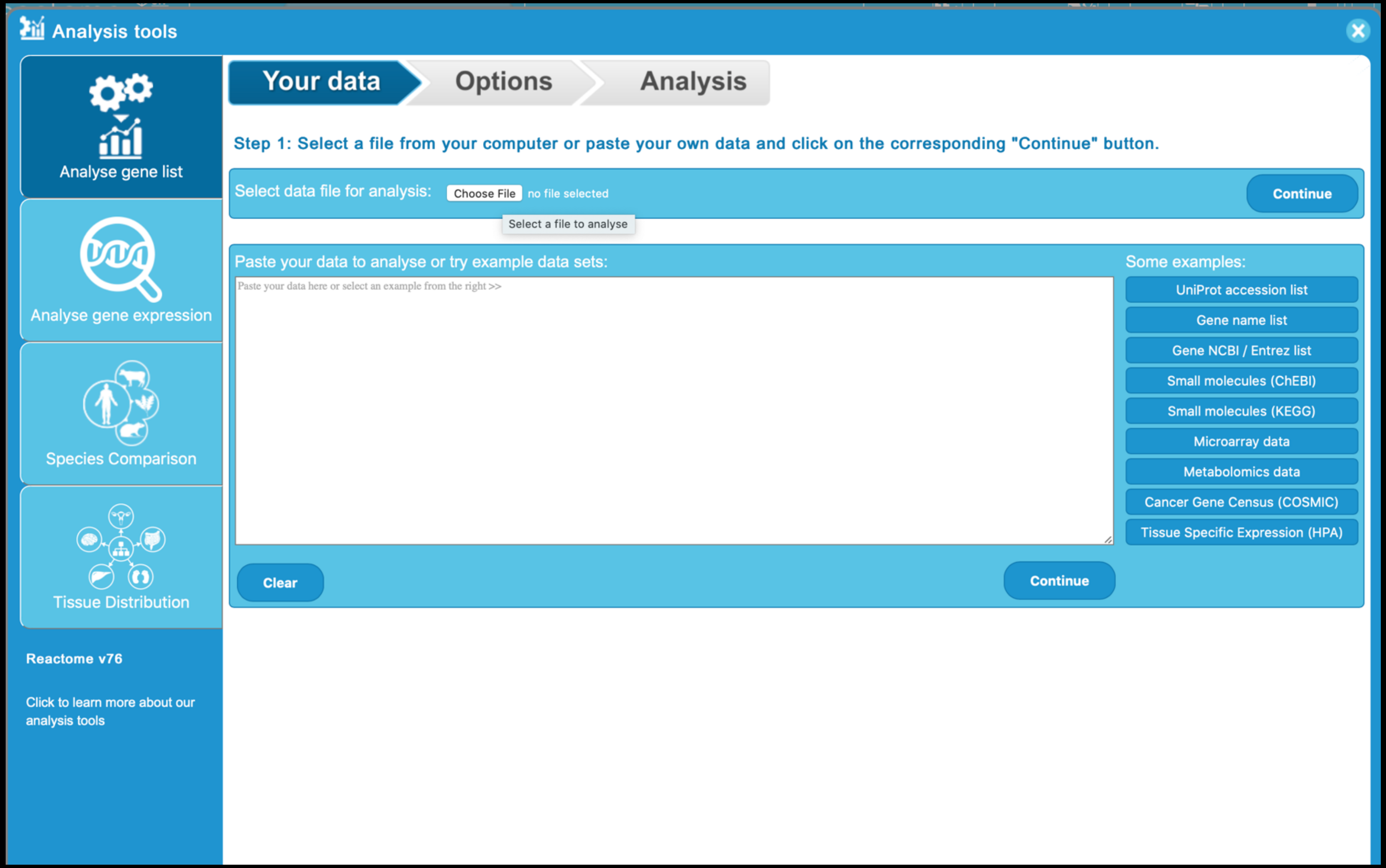
Visualize and interact with Reactome biological pathways
Analysis Tools
Merges pathway identifier mapping, over-representation, and expression analysis
ReactomeFIViz
Designed to find pathways and network patterns related to cancer and other types of diseases
Documentation
Information to browse the database and use its principal tools for data analysis

Reactome

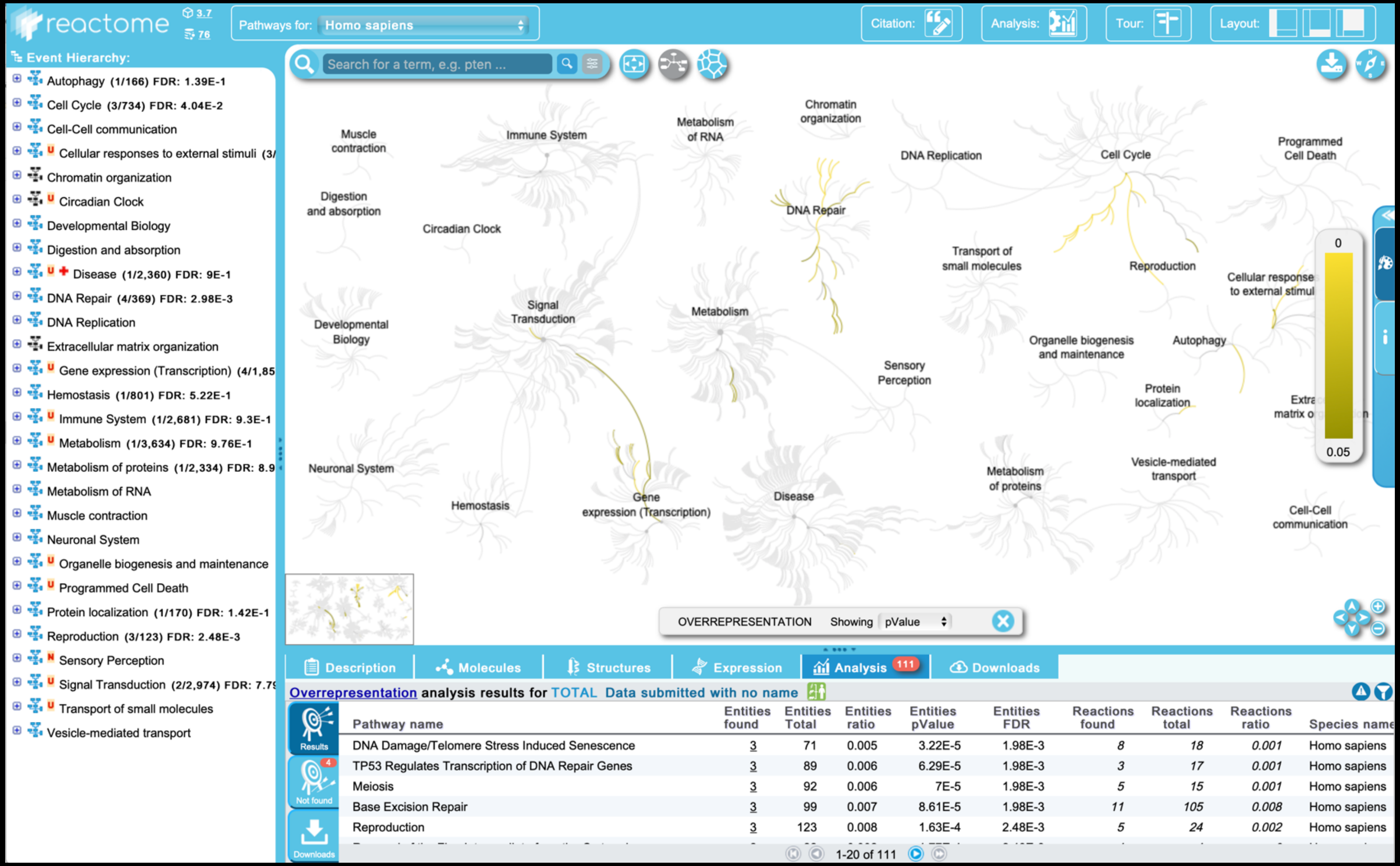
Input:

SENP7
MOB1B
CARMIL1
PRRC2A
TERF2
RFWD3
PARP1
POT1
ATM
MPHOSPH6



Reactome

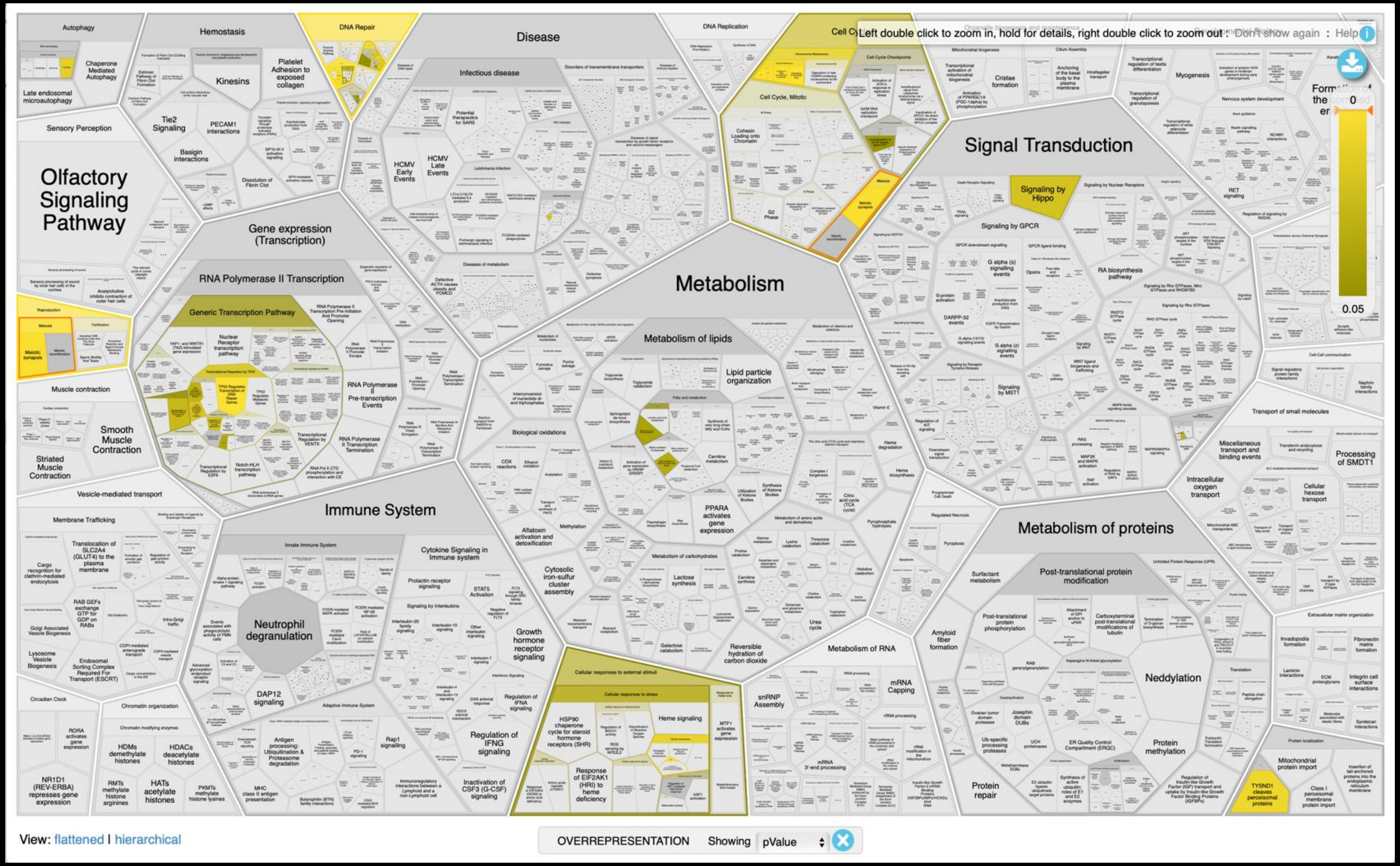
Results



Pathway name	Entities found	Entities Total	Entities ratio	Entities pValue	Entities FDR	Reactions found	Reactions total	Reactions ratio	Species name
DNA Damage/Telomere Stress Induced Senescence	3	71	0.005	3.22E-5	1.98E-3	8	18	0.001	Homo sapiens
TP53 Regulates Transcription of DNA Repair Genes	3	89	0.006	6.29E-5	1.98E-3	3	17	0.001	Homo sapiens
Meiosis	3	92	0.006	7E-5	1.98E-3	5	15	0.001	Homo sapiens
Base Excision Repair	3	99	0.007	8.61E-5	1.98E-3	11	105	0.008	Homo sapiens
Reproduction	3	123	0.008	1.63E-4	2.48E-3	5	24	0.002	Homo sapiens

Reactome

Results

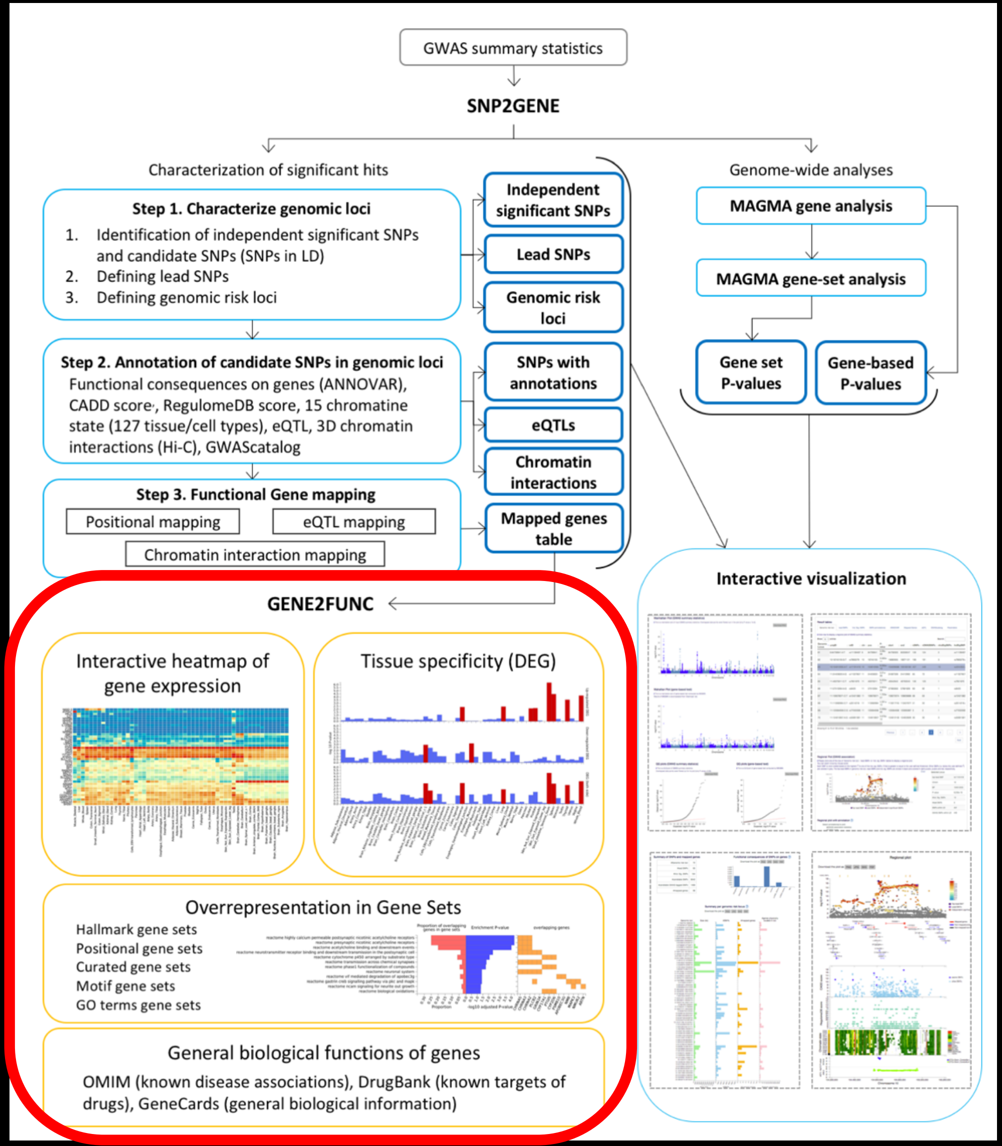


Gene set/pathway analysis in FUMA

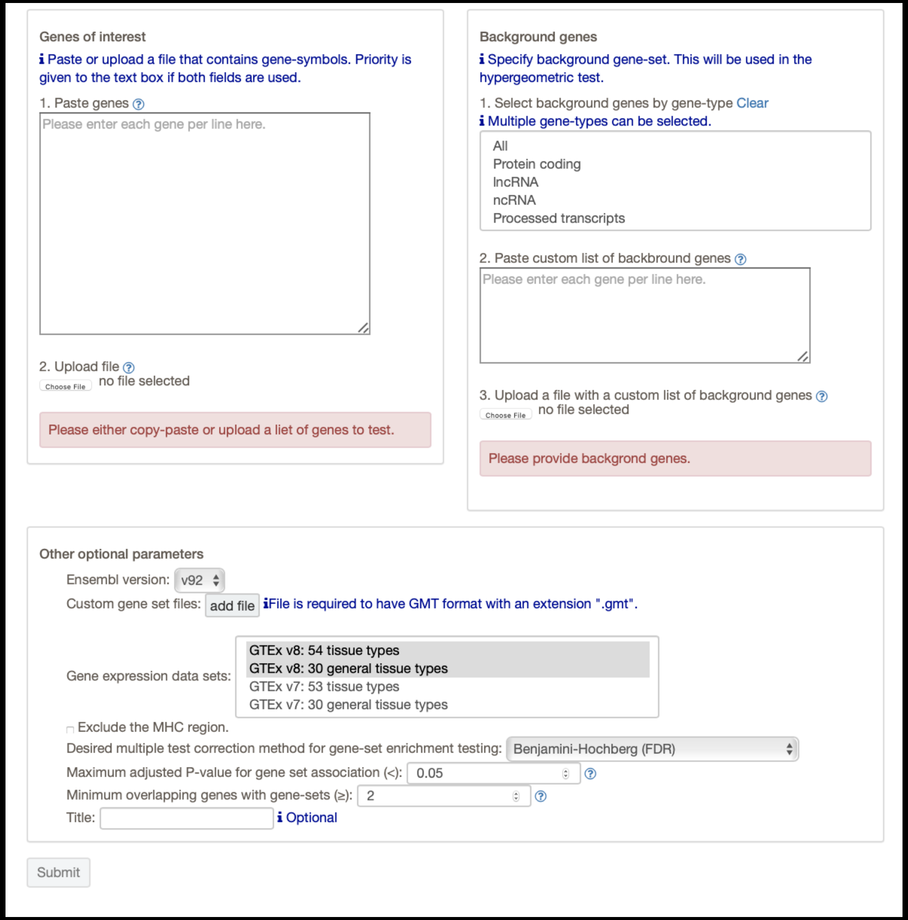
Recall that FUMA's SNP2GENE feature takes GWAS summary statistics and outputs a list of mapped genes.

The GENE2FUNC feature takes a set of genes as input and can perform overrepresentation analyses.

Gene sets include: Gene Ontologies, KEGG pathways, Reactome pathways, and DE genes across various tissues/cell types



Gene set analysis in FUMA – GENE2FUNC

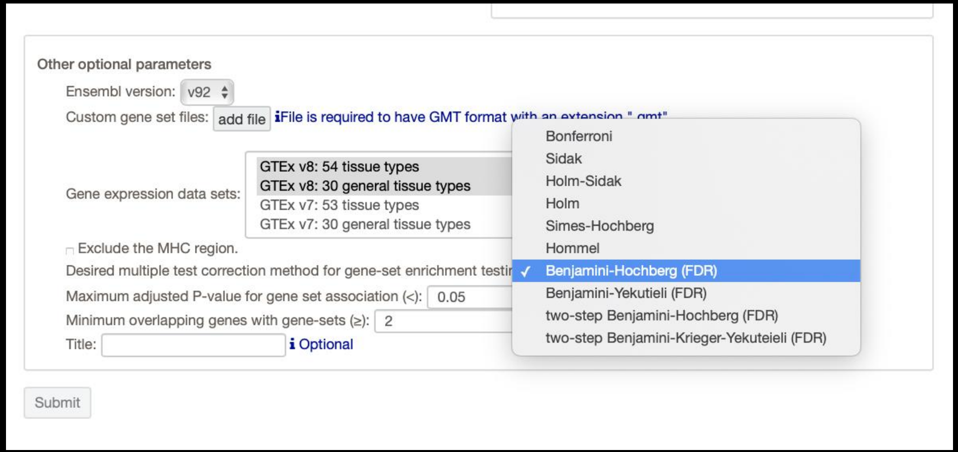


Provide a list of genes (copy and paste or upload a file).

You need to specify a “background” set of genes to compare your list with.

FUMA lets you select several different gene sets defined by gene expression profiles of various tissues.

You can also choose one of several multiple-testing correction methods and a “significance” threshold.

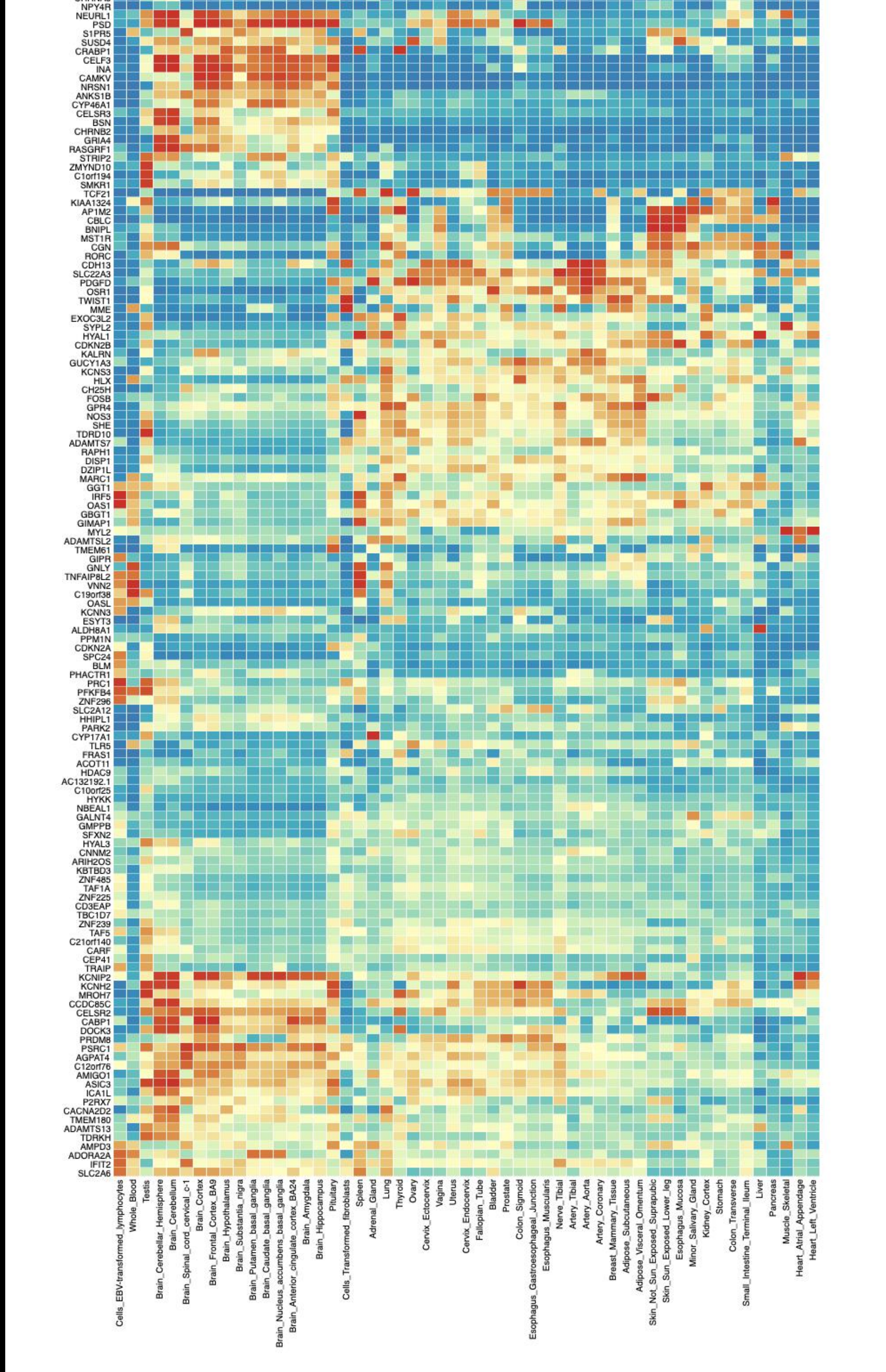


GENE2FUNC

Input: 644 genes from the hw4 data.

Output (first part):
gene expression heatmap

https://fuma.ctglab.nl/browse/179#g2f_summaryPanel



GENE2FUNC

Output (second part)
Differential expression of genes across tissues

(Are the genes you supplied enriched for
DE genes in particular tissues?)

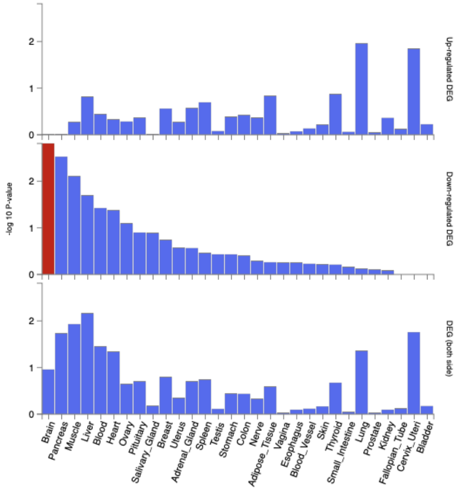
Differentially expressed genes

ℹ Significantly enriched DEG sets (P_{bon} < 0.05) are highlighted in red.

Order tissue by : down-regulated DEG P-value

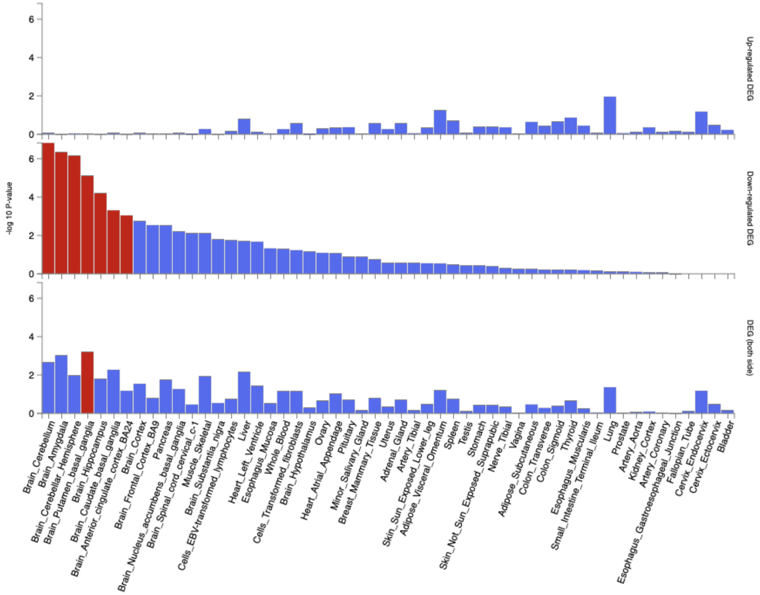
GTEx v7 30 general tissue types

Download the plot as PNG JPG SVG PDF



GTEx v7 53 tissue types

Download the plot as PNG JPG SVG PDF



GENE2FUNC

Overrepresentation of your genes in pre-defined gene sets

Enrichment of input genes in Gene Sets

ℹ Plots and tables only display gene sets with adjusted P-value < 0.05. When adjusted P-value threshold is set to > 0.05, all results passed threshold are included in the GS.txt field downloadable from "Summary" tab.

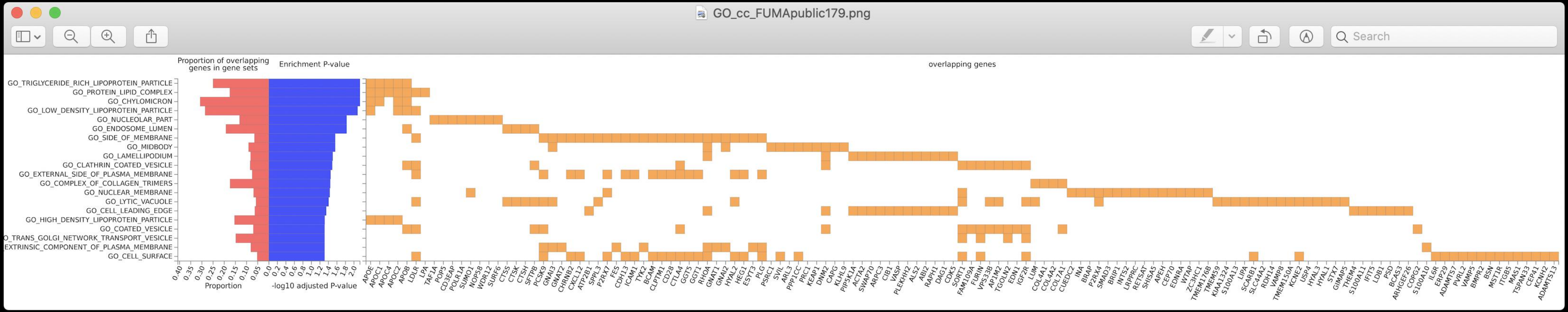
ℹ If there is no significant gene sets (adjusted P-value < 0.05) in user provided custom gene sets, they are not displayed in this page, but all results passed threshold are included in the GS.txt field downloadable from "Summary" tab.

- Hallmark gene sets (MsigDB h) (3)
- Positional gene sets (MsigDB c1) (28)
- Curated_gene_sets (39)
- Chemical and Genetic pertubation gene sets (MsigDB c2) (17)
- All Canonical Pathways (MsigDB c2) (21)
- BioCarta (MsigDB c2) (1)
- KEGG (MsigDB c2) (2)
- Reactome (MsigDB c2) (16)
- microRNA targets (MsigDB c3) (0)
- TF targets (MsigDB c3) (3)
- All computational gene sets (MsigDB c4) (11)
- Cancer gene neighborhoods (MsigDB c4) (3)
- Cancer gene modules (MsigDB c4) (9)
- GO biological processes (MsigDB c5) (124)
- GO cellular components (MsigDB c5) (20)
- GO molecular functions (MsigDB c5) (17)
- Oncogenic signatures (MsigDB c6) (1)
- Immunologic signatures (MsigDB c7) (4)
- WikiPathways (12)
- GWAS catalog reported genes (152)

GENE2FUNC

For example, choose “GO cellular components”

You can see exactly which genes in our list overlapped with these 20 GO terms, and how significant the overlap/enrichment is.



GENE2FUNC

Results are also viewable as a table.

GO cellular components (MsigDB c5) (20)

Plot / Table

GeneSet	N	n	P-value	adjusted P	genes
GO_TRIGLYCERIDE_RICH_LIPOPROTEIN_PARTICLE	20	5	3.06e-5	7.23e-3	APOE, APOC1, APOC4, APOC2, APOB
GO_PROTEIN_LIPID_COMPLEX	40	7	3.71e-5	7.23e-3	LDLR, APOE, APOC1, APOC4, APOC2, APOB, LPA
GO_CHYLOMICRON	13	4	3.74e-5	7.23e-3	APOE, APOC1, APOC2, APOB
GO_LOW_DENSITY_LIPOPROTEIN_PARTICLE	14	4	5.66e-5	8.21e-3	LDLR, APOE, APOC2, APOB
GO_NUCLEOLAR_PART	61	8	1.49e-4	1.49e-2	TAF1A, POP5, CD3EAP, POLR1A, SUMO1, NOP58, WDR12, SURF6
GO_ENDOSOME_LUMEN	26	5	1.54e-4	1.49e-2	CTSS, CTSK, CTSH, APOB, SFTPB
GO_SIDE_OF_MEMBRANE	402	26	3.35e-4	2.76e-2	PCSK9, GNAI3, GNAT2, CHRNB2, CXCL12, ATP2B1, SPPL3, P2RX7, FES, CDH13, ICAM1, TYK2, LDLR, BCAM, CLPTM1, CD28, CTLA4, GGT5, GGT1, RHOA, GNAT1, GNAI2, HYAL2, HEG1, ESYT3, PLG
GO_MIDBODY	132	12	3.81e-4	2.76e-2	PSRC1, GNAI3, SVIL, ARL3, PPP1CC, PRC1, KEAP1, DNM2, CAPG, RHOA, GNAI2, KLHL9
GO_LAMELLIPODIUM	171	14	4.90e-4	3.16e-2	PIP5K1A, ACTA2, SWAP70, ARPC3, CIB1, DNM2, VASP, PLEKHH2, ALS2, ABI2, RAPH1, RHOA, DAG1, CDK5
GO_CLATHRIN_COATED_VESICLE	155	13	5.58e-4	3.24e-2	SORT1, FAM109A, FURIN, VPS33B, AP1M2, DNM2, LDLR, APOB, TGOLN2, SFTPB, CTLA4, EDN1, IGF2R
GO_EXTERNAL_SIDE_OF_PLASMA_MEMBRANE	214	16	6.70e-4	3.53e-2	PCSK9, CHRNB2, CXCL12, P2RX7, CDH13, ICAM1, LDLR, BCAM, CLPTM1, CD28, CTLA4, GGT5, GGT1, HYAL2, HEG1, PLG
GO_COMPLEX_OF_COLLAGEN_TRIMERS	23	4	7.46e-4	3.61e-2	LUM, COL4A1, COL4A2, COL7A1
GO_NUCLEAR_MEMBRANE	278	19	8.29e-4	3.70e-2	SORT1, CUEDC2, INA, BRAP, P2RX7, P2RX4, SMAD3, BRIP1, INTS2, LRPPRC, RETSAT, SUMO1, SHISA5, APEH, CEP70, EDNRA, WTAP, ZC3HC1, TMEM176B
GO_LYTIC_VACUOLE	520	30	9.54e-4	3.95e-2	TMEM59, PCSK9, KIAA1324, SORT1, GNAI3, CTSS, CTSK, S100A13, LIPA, LUM, SPPL3, P2RX4, SCARB1, CTSH,

One other FUMA function

Recall that FUMA uses GWAS summary statistics to perform gene-based testing in MAGMA.

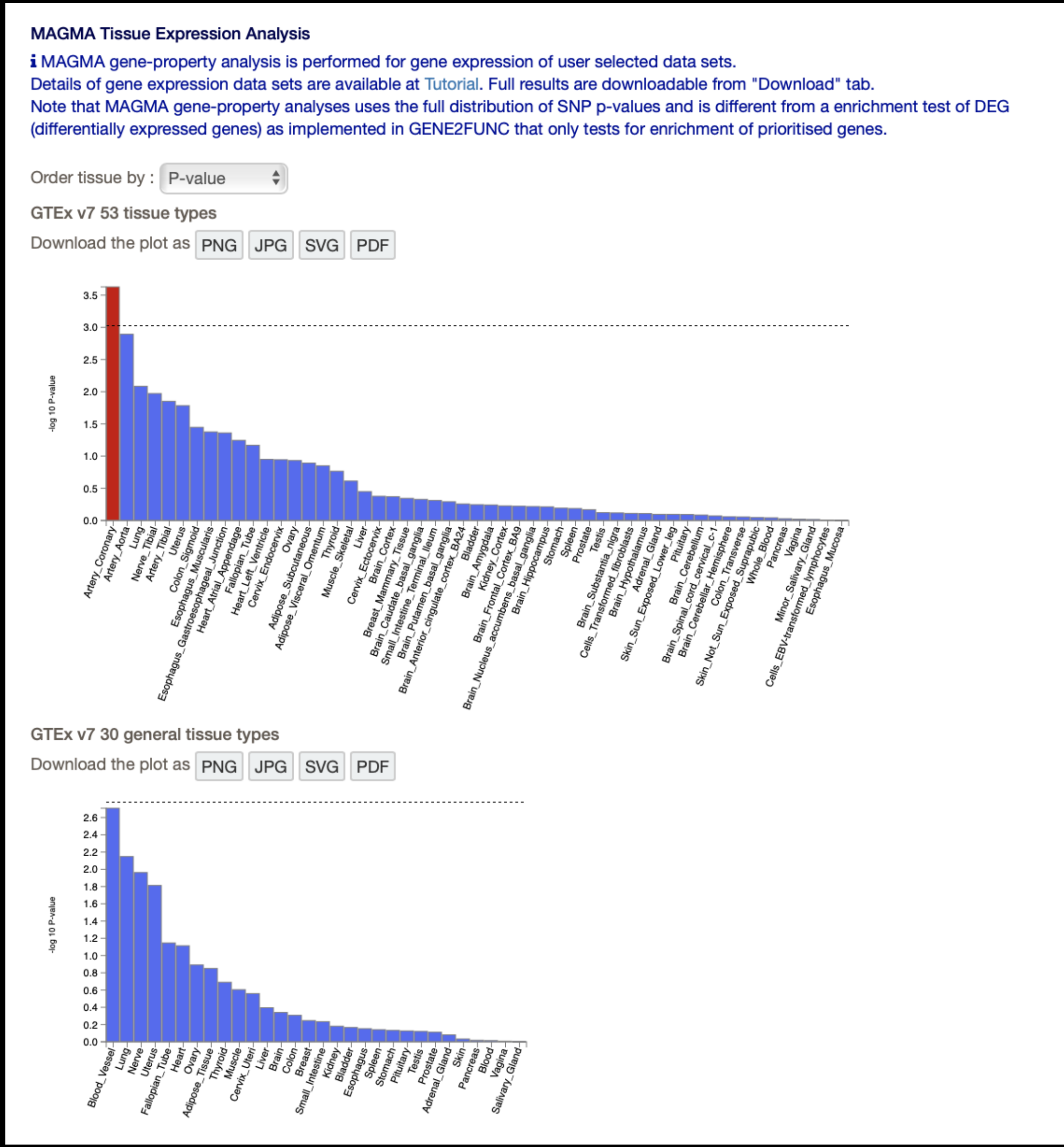
MAGMA can integrate the gene-based p-values into a gene set testing framework.

Essentially this tests whether genes in a gene set are more strongly associated with the phenotype (compared to other genes).

Recall that this uses ALL genes, not just a set you chose.

MAGMA results

Gene sets defined by differential expression in various tissues.



GREAT

Genomic Regions Enrichment of Annotations

Publication
http://bejerano.stanford.edu/papers/GREAT.pdf

Website
http://great.stanford.edu/public/html/index.php

More documentation

https://great-help.atlassian.net/wiki/spaces/GREAT/overview?homepageId=655450

GREAT

Focus on “b”

Scenario: you want to know if some <u>regions</u> identified by your study are enriched for an annotation.

Arrows = gene transcription start sites
Black arrows = for genes without a specific annotation of interest
Green arrows = for genes without the annotation

Black lines = regulatory regions for gene without the annotation
Green arrows = regulatory regions for genes with the annotation

Calculate the fraction of the genome overlapping the green lines.

How many of your genomic regions overlap the green lines?

Perform a binomial test.

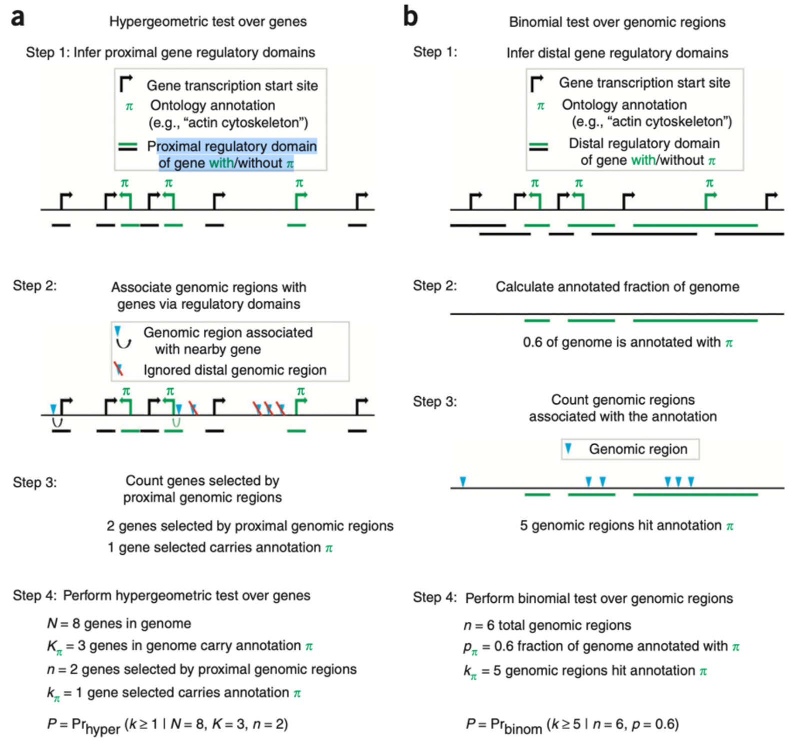
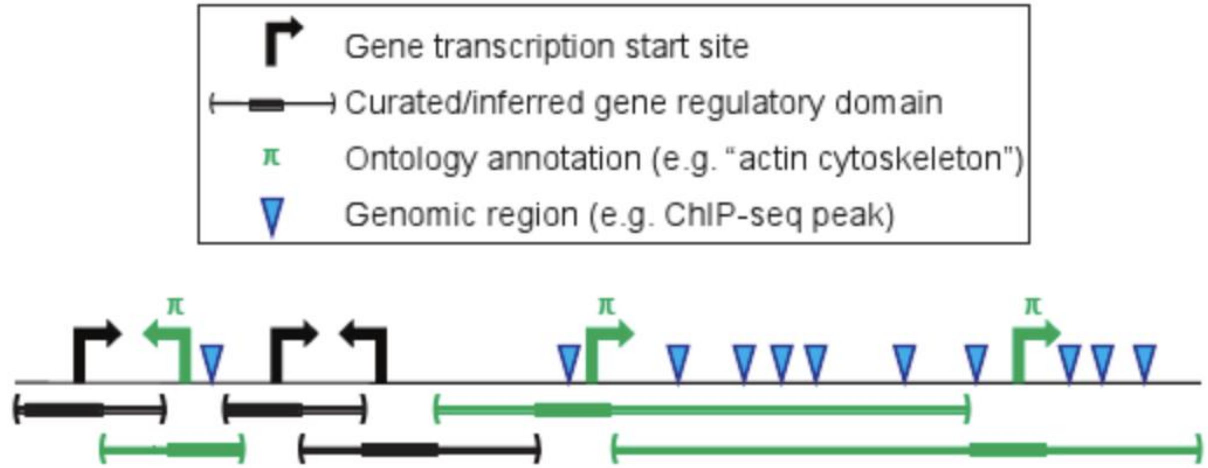


Figure 1 Enrichment analysis of a set of *cis*-regulatory regions. (**a**) The current prevailing methodology associates only proximal binding events with genes and performs a gene-list test of functional enrichments using tools originally designed for microarray analysis. (**b**) GREAT’s binomial approach over genomic regions uses the total fraction of the genome associated with a given ontology term (green bar) as the expected fraction of input regions associated with the term by chance.

GREAT

Example



Binomial test over genomic regions

n = 11 total genomic regions

p_π = 0.75 fraction of genome annotated with π

k_π = 11 genomic regions annotated with π

P-value = 0.04

Hypergeometric test over genes

N = 6 total genes

K_π = 3 genes annotated with π

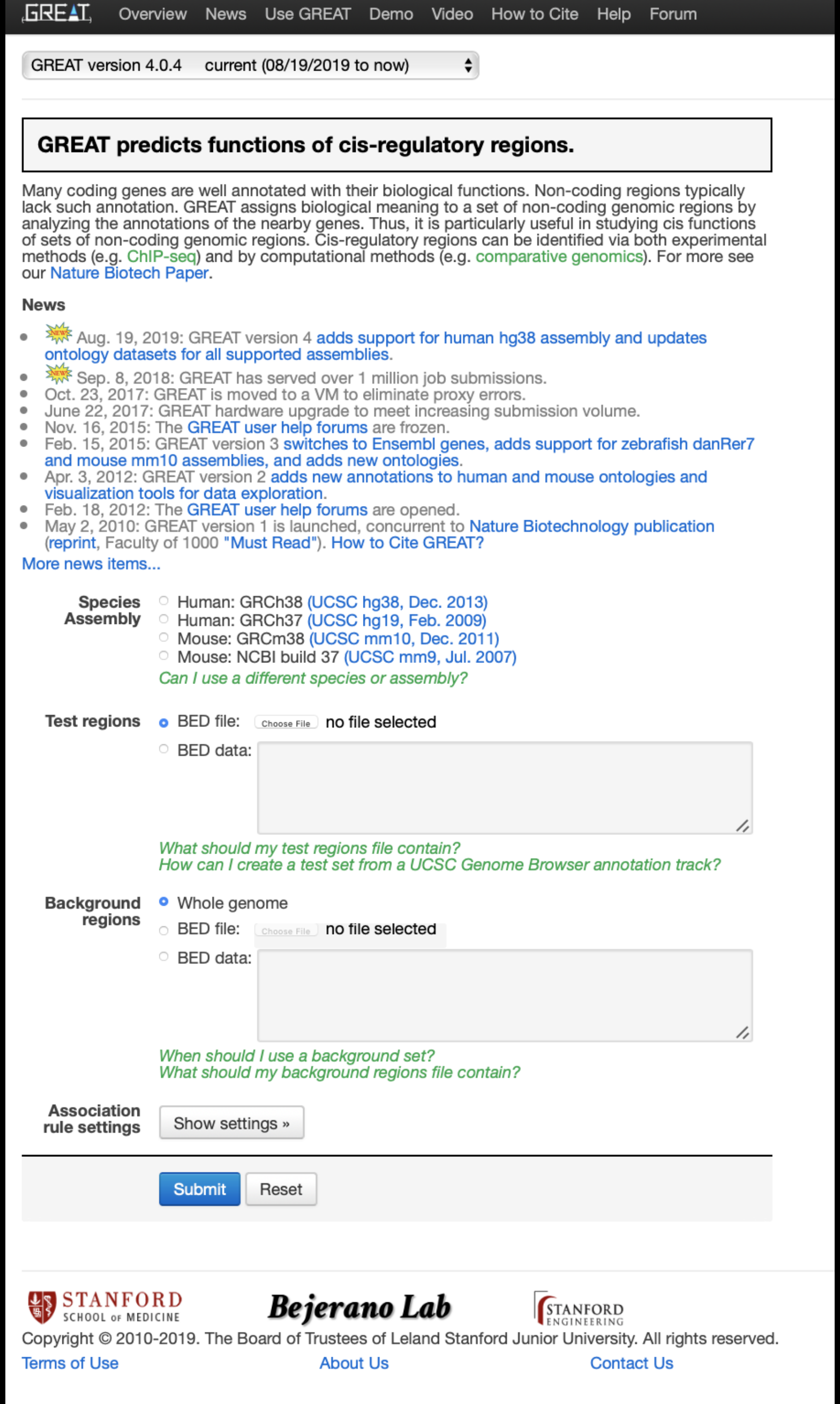
n = 3 genes with an associated genomic region

k_π = 3 genes annotated and with a genomic region

P-value = 0.05

GREAT

Select a species and provide input in BED format



GREAT

Example: plug in a list of 235 SNPs from a GWAS of height (http://great.stanford.edu/public/html/demo.php)

GREAT Overview News Use GREAT Demo Video How to Cite Help Forum

GREAT version 4.0.4 current (08/19/2019 to now)

Choose one of the data sets below to run. In each case, the data set is run against a whole genome background, using the *basal plus extension* genomic association rule with its default values.

Set	Experiment	Set size	Species	Cell/tissue type	Reference	GREAT Enrichment Highlights
SRF (activator)	ChIP-seq	556	Human (hg19)	Jurkat	Valouev et al. 2008	Actin-related terms
NRSF/REST (repressor)	ChIP-seq	1,712	Human (hg19)	Jurkat	Valouev et al. 2008	Neurotransmitters and ion channels
p300 transcriptional co-activator	ChIP-seq	2,105	Mouse (mm10)	Embryonic limb E11.5	Visel et al. 2009	Embryonic limb development
Ultraconserved elements	Comparative genomics	481	Human (hg19)	N/A	Bejerano et al. 2004	Transcription regulation and developmental defects
Height-associated unlinked SNPs	GWAS	235	Human (hg19)	N/A	NHGRI compendium	Bone growth terms

Submit

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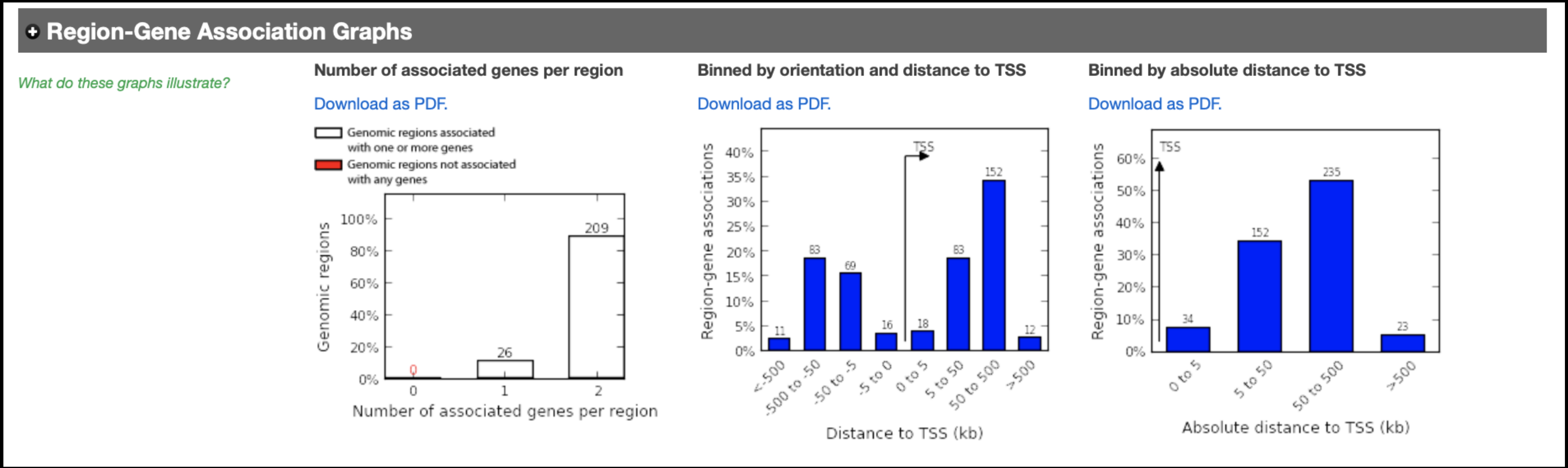
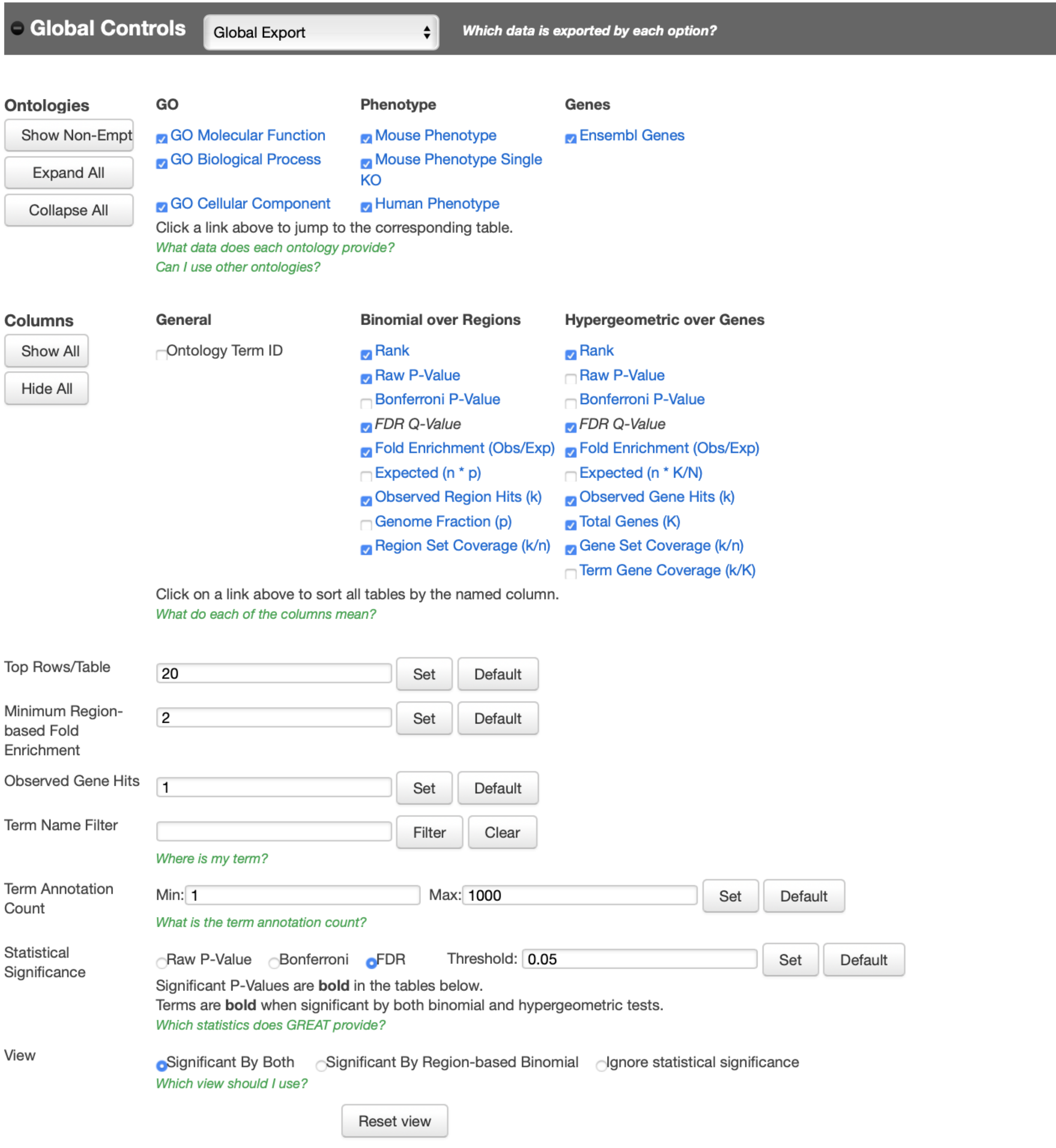


Region-Gene Association Graphs
What do these graphs illustrate?
Number of associated genes per region
Download as PDF.
Genomic regions associated with one or more genes
Genomic regions not associated with any genes
Genomic regions
Number of associated genes per region
Binned by orientation and distance to TSS
Download as PDF.
Region-gene associations
TSS
Distance to TSS (kb)
Binned by absolute distance to TSS
Download as PDF.
Region-gene associations
TSS
Absolute distance to TSS (kb)

GREAT

Output controls



GREAT

Example output for GO biological process

GO Biological Process (20+ terms) Global controls

Table controls: Export Shown top rows in this table: 20 Set Term annotation count: Min: 1 Max: Inf Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	*Binom FDR Q-Val*	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	*Hyper FDR Q-Val*	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
oncogene-induced cell senescence	3	**1.7272e-18**	**7.5682e-15**	128.5039	10	4.26%	75	**4.1815e-2**	64.6307	2	2	0.70%
skeletal system development	28	**1.7143e-11**	**8.0479e-9**	3.0246	46	19.57%	1	**5.5453e-7**	4.1697	30	465	10.45%
negative regulation of nitrogen compound metabolic process	36	**5.2151e-11**	**1.9042e-8**	2.0717	79	33.62%	12	**3.3690e-3**	1.9925	48	1,557	16.72%
chondrocyte differentiation	40	**8.0475e-11**	**2.6446e-8**	6.4496	20	8.51%	3	**1.8272e-4**	10.2588	10	63	3.48%
negative regulation of nucleobase-containing compound metabolic process	55	**2.8821e-10**	**6.8883e-8**	2.0720	74	31.49%	33	**6.1454e-3**	1.9640	43	1,415	14.98%
negative regulation of gene expression	61	**4.8352e-10**	**1.0419e-7**	2.0340	75	31.91%	17	**3.4228e-3**	2.0034	46	1,484	16.03%
skeletal system morphogenesis	62	**4.8529e-10**	**1.0289e-7**	4.0645	28	11.91%	2	**1.9009e-4**	5.2570	17	209	5.92%
cartilage development	66	**6.2737e-10**	**1.2495e-7**	4.4829	25	10.64%	5	**2.7834e-4**	5.9528	14	152	4.88%
negative regulation of RNA metabolic process	68	**7.7255e-10**	**1.4934e-7**	2.1220	68	28.94%	41	**8.0168e-3**	2.0068	39	1,256	13.59%
chondrocyte differentiation involved in endochondral bone morphogenesis	83	**3.9875e-9**	**6.3152e-7**	17.4059	9	3.83%	7	**6.8854e-4**	29.3776	5	11	1.74%
cartilage development involved in endochondral bone morphogenesis	97	**1.3825e-8**	**1.8735e-6**	12.3285	10	4.26%	10	**1.3848e-3**	16.8602	6	23	2.09%
endochondral bone morphogenesis	98	**1.3953e-8**	**1.8715e-6**	7.2616	14	5.96%	4	**2.5376e-4**	11.4054	9	51	3.14%
negative regulation of transcription, DNA-templated	103	**1.7062e-8**	**2.1775e-6**	2.0619	62	26.38%	36	**6.7704e-3**	2.0830	37	1,148	12.89%
connective tissue development	105	**1.8601e-8**	**2.3287e-6**	3.6482	26	11.06%	8	**7.8921e-4**	4.8963	15	198	5.23%
negative regulation of nucleic acid-templated transcription	117	**3.4671e-8**	**3.8953e-6**	2.0229	62	26.38%	45	**1.1910e-2**	2.0078	37	1,191	12.89%
negative regulation of RNA biosynthetic process	119	**4.4820e-8**	**4.9509e-6**	2.0088	62	26.38%	52	**1.3639e-2**	1.9812	37	1,207	12.89%
response to growth hormone	120	**4.9684e-8**	**5.4425e-6**	10.7355	10	4.26%	65	**3.3361e-2**	9.5045	5	34	1.74%
positive regulation of fibroblast proliferation	131	**1.2924e-7**	**1.2968e-5**	7.3800	12	5.11%	64	**3.3341e-2**	7.3167	6	53	2.09%
embryonic morphogenesis	151	**4.1544e-7**	**3.6165e-5**	2.2891	42	17.87%	63	**3.0900e-2**	2.4588	21	552	7.32%
morphogenesis of a branching epithelium	183	**3.4599e-6**	**2.4853e-4**	3.2088	21	8.94%	22	**4.2208e-3**	4.8778	12	159	4.18%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
GO Biological Process has 13,145 terms covering 16,621 (90%) of all 18,549 genes, and 1,251,831 term - gene associations.
13,145 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Other enrichment categories

GO Cellular Component (1 term)

Global controls

Table controls: Export | Shown top rows in this table: 20 Set | Term annotation count: Min: 1 Max: Inf Set | Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	*Binom FDR Q-Val*	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	*Hyper FDR Q-Val*	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
proteinaceous extracellular matrix	9	**3.0842e-8**	**5.9148e-6**	2.9578	33	14.04%	1	**2.3838e-2**	3.2315	18	360	6.27%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
GO Cellular Component has 1,726 terms covering 17,726 (96%) of all 18,549 genes, and 380,581 term - gene associations.
1,726 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GO Molecular Function (no terms)

Global controls

Human Phenotype (20+ terms)

Global controls

Table controls: Export | Shown top rows in this table: 20 Set | Term annotation count: Min: 1 Max: Inf Set | Visualize this table: [select one]

Click to sort ascending

Term Name	Binom Rank	Raw P-Value	*FDR Q-Val*	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	*Hyper FDR Q-Val*	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
Growth delay	2	**1.8159e-10**	**6.0578e-7**	2.5886	52	22.13%	4	**1.8731e-4**	2.6802	35	844	12.20%
Aplasia/hypoplasia affecting bones of the axial skeleton	6	**9.8631e-9**	**1.0968e-5**	3.1774	32	13.62%	25	**5.5121e-3**	3.1357	18	371	6.27%
Growth abnormality	7	**1.1452e-8**	**1.0915e-5**	2.1202	60	25.53%	14	**7.1577e-4**	2.2064	41	1,201	14.29%
Short stature	8	**1.6792e-8**	**1.4004e-5**	2.7019	39	16.60%	1	**6.2158e-5**	3.3201	30	584	10.45%
Abnormality of the rib cage	10	**4.1760e-8**	**2.7863e-5**	4.4357	20	8.51%	10	**5.2829e-4**	5.0549	14	179	4.88%
Aplasia/hypoplasia involving the skeleton	15	**9.0336e-8**	**4.0181e-5**	2.5752	38	16.17%	11	**5.6075e-4**	3.0315	25	533	8.71%
Abnormality of body height	16	**1.0135e-7**	**4.2264e-5**	2.5228	39	16.60%	2	**4.0652e-5**	3.2006	31	626	10.80%
Hypoplastic iliac wing	29	**6.3629e-7**	**1.4639e-4**	11.5678	8	3.40%	75	**4.1556e-2**	10.7718	4	24	1.39%
Abnormality of lower limb epiphysis morphology	31	**8.4191e-7**	**1.8120e-4**	11.1381	8	3.40%	62	**2.3452e-2**	8.9765	5	36	1.74%
Short ribs	33	**8.9185e-7**	**1.8032e-4**	9.1222	9	3.83%	79	**4.3026e-2**	7.5152	5	43	1.74%
Abnormality of limb epiphysis morphology	34	**9.0304e-7**	**1.7721e-4**	6.8411	11	4.68%	41	**1.5263e-2**	6.5567	7	69	2.44%
Aplasia/Hypoplasia of the middle phalanges of the toes	39	**1.4074e-6**	**2.4077e-4**	17.8770	6	2.55%	35	**1.3501e-2**	32.3153	3	6	1.05%
Abnormal enchondral ossification	44	**1.9662e-6**	**2.9814e-4**	6.3055	11	4.68%	53	**2.1822e-2**	5.9528	7	76	2.44%
Cone-shaped epiphyses of the phalanges of the hand	50	**2.9126e-6**	**3.8865e-4**	11.7523	7	2.98%	63	**2.3683e-2**	12.9261	4	20	1.39%
Abnormality of the wing of the ilium	53	**3.2709e-6**	**4.1176e-4**	7.7668	9	3.83%	82	**4.6182e-2**	7.3444	5	44	1.74%
Abnormality of the radius	55	**3.5733e-6**	**4.3347e-4**	4.2519	15	6.38%	16	**1.2989e-3**	6.5283	10	99	3.48%
Abnormality of the ribs	56	**3.6811e-6**	**4.3857e-4**	4.2413	15	6.38%	33	**1.4125e-2**	4.5837	10	141	3.48%
Deviation of the hand or of fingers of the hand	57	**3.9991e-6**	**4.6811e-4**	3.5865	18	7.66%	64	**2.6620e-2**	3.9170	10	165	3.48%
Radial deviation of the hand or of fingers of the hand	62	**4.8814e-6**	**5.2530e-4**	5.7246	11	4.68%	31	**1.2413e-2**	7.0690	7	64	2.44%
Abnormality of the epiphyses of the phalanges of the hand	64	**6.5559e-6**	**6.8345e-4**	8.4058	8	3.40%	40	**1.4823e-2**	10.7718	5	30	1.74%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
Human Phenotype has 6,672 terms covering 3,411 (18%) of all 18,549 genes, and 256,883 term - gene associations.
6,672 ontology terms (100%) were tested using an annotation count range of [1, Inf].

GREAT

Mouse-based categories

Mouse Phenotype (20+ terms) Global controls

Table controls: Export Shown top rows in this table: Set Term annotation count: Min: Max: Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	*Binom FDR Q-Val*	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	*Hyper FDR Q-Val*	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
abnormal long bone morphology	8	**2.8089e-8**	**3.3546e-5**	2.3932	46	19.57%	8	**9.1817e-4**	2.9035	27	601	9.41%
abnormal uterus morphology	10	**4.2296e-8**	**4.0410e-5**	3.7437	24	10.21%	101	**2.9091e-2**	3.5370	11	201	3.83%
disproportionate dwarf	11	**6.2005e-8**	**5.3854e-5**	6.4261	14	5.96%	37	**2.8400e-3**	7.4934	8	69	2.79%
abnormal tail length	15	**1.1933e-7**	**7.6008e-5**	4.3609	19	8.09%	122	**3.5014e-2**	4.4573	8	116	2.79%
delayed endochondral bone ossification	20	**2.8930e-7**	**1.3820e-4**	5.6497	14	5.96%	24	**1.9441e-3**	8.3394	8	62	2.79%
increased embryo size	21	**3.5922e-7**	**1.6343e-4**	16.1615	7	2.98%	99	**2.7732e-2**	21.5435	3	9	1.05%
abnormal limb morphology	24	**4.0997e-7**	**1.6320e-4**	2.0502	52	22.13%	18	**1.1652e-3**	2.4798	32	834	11.15%
abnormal long bone epiphyseal plate proliferative zone	26	**4.6680e-7**	**1.7153e-4**	5.0273	15	6.38%	57	**6.0422e-3**	6.3833	8	81	2.79%
small uterus	27	**4.7403e-7**	**1.6774e-4**	5.4178	14	5.96%	29	**2.5908e-3**	7.8340	8	66	2.79%
abnormal appendicular skeleton morphology	28	**5.1735e-7**	**1.7653e-4**	2.0950	49	20.85%	7	**8.2587e-4**	2.6857	31	746	10.80%
abnormal humerus morphology	29	**5.2714e-7**	**1.7367e-4**	3.9491	19	8.09%	17	**1.1468e-3**	6.0764	11	117	3.83%
short limbs	31	**6.6151e-7**	**2.0387e-4**	3.7188	20	8.51%	32	**2.4579e-3**	5.2662	11	135	3.83%
abnormal uterus size	34	**7.3734e-7**	**2.0719e-4**	4.5353	16	6.81%	50	**4.3922e-3**	5.8755	9	99	3.14%
abnormal stomach glandular epithelium morphology	37	**9.7430e-7**	**2.5158e-4**	10.9199	8	3.40%	128	**3.4867e-2**	10.7718	4	24	1.39%
abnormal thoracic cage morphology	38	**1.0304e-6**	**2.5907e-4**	2.4052	36	15.32%	10	**9.8105e-4**	3.1023	24	500	8.36%
abnormal forelimb stylopod morphology	40	**1.0906e-6**	**2.6050e-4**	3.7581	19	8.09%	20	**1.4725e-3**	5.8274	11	122	3.83%
abnormal cartilage development	46	**1.4597e-6**	**3.0317e-4**	2.9797	25	10.64%	12	**1.0252e-3**	4.5302	15	214	5.23%
female infertility	47	**1.6991e-6**	**3.4538e-4**	2.8754	26	11.06%	35	**2.3819e-3**	3.3430	18	348	6.27%
abnormal cartilage morphology	49	**1.9670e-6**	**3.8352e-4**	2.3751	35	14.89%	1	**6.4078e-4**	3.7349	23	398	8.01%
abnormal basicranium morphology	52	**2.4972e-6**	**4.5882e-4**	4.1214	16	6.81%	16	**1.0465e-3**	6.9495	10	93	3.48%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
Mouse Phenotype has 9,554 terms covering 9,592 (52%) of all 18,549 genes, and 709,451 term - gene associations.
9,554 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Mouse Phenotype Single KO (20+ terms) Global controls

Table controls: Export Shown top rows in this table: Set Term annotation count: Min: Max: Set Visualize this table: [select one]

Term Name	Binom Rank	Binom Raw P-Value	*Binom FDR Q-Val*	Binom Fold Enrichment	Binom Observed Region Hits	Binom Region Set Coverage	Hyper Rank	*Hyper FDR Q-Val*	Hyper Fold Enrichment	Hyper Observed Gene Hits	Hyper Total Genes	Hyper Gene Set Coverage
abnormal long bone epiphyseal plate proliferative zone	4	**6.8640e-10**	**1.5700e-6**	8.3213	15	6.38%	13	**2.0555e-3**	8.9146	8	58	2.79%
disproportionate dwarf	8	**2.4537e-9**	**2.8062e-6**	8.3522	14	5.96%	9	**1.9632e-3**	9.4008	8	55	2.79%
abnormal long bone morphology	14	**5.8168e-8**	**3.8013e-5**	2.5775	39	16.60%	20	**9.5033e-3**	2.8217	21	481	7.32%
abnormal thoracic cage morphology	15	**8.2821e-8**	**5.0515e-5**	2.7775	34	14.47%	5	**1.5564e-3**	3.3456	22	425	7.67%
abnormal cartilage development	19	**1.3192e-7**	**6.3524e-5**	3.9455	21	8.94%	11	**2.2577e-3**	5.3487	12	145	4.18%
abnormal long bone epiphyseal plate morphology	21	**1.5758e-7**	**6.8654e-5**	3.9027	21	8.94%	12	**2.2242e-3**	5.3121	12	146	4.18%
abnormal chondrocyte morphology	25	**3.1048e-7**	**1.1362e-4**	5.1950	15	6.38%	8	**1.8600e-3**	8.0788	9	72	3.14%
short limbs	26	**3.1104e-7**	**1.0945e-4**	4.8465	16	6.81%	36	**2.1187e-2**	5.6818	8	91	2.79%
abnormal appendicular skeleton morphology	27	**3.1544e-7**	**1.0689e-4**	2.2548	44	18.72%	6	**1.6094e-3**	2.8560	27	611	9.41%
abnormal skeleton physiology	29	**3.5088e-7**	**1.1070e-4**	2.4777	37	15.74%	7	**1.6367e-3**	2.9811	25	542	8.71%
abnormal bone ossification	33	**5.1574e-7**	**1.4299e-4**	2.7783	30	12.77%	2	**1.7090e-3**	3.8018	20	340	6.97%
abnormal epiphyseal plate morphology	37	**6.8280e-7**	**1.6884e-4**	3.5620	21	8.94%	15	**4.3087e-3**	4.8778	12	159	4.18%
abnormal limb morphology	39	**7.4482e-7**	**1.7473e-4**	2.1843	44	18.72%	34	**1.9492e-2**	2.3485	25	688	8.71%
small uterus	41	**9.9329e-7**	**2.2165e-4**	6.0696	12	5.11%	43	**2.7804e-2**	7.6036	6	51	2.09%
abnormal long bone hypertrophic chondrocyte zone	42	**1.0042e-6**	**2.1875e-4**	4.4277	16	6.81%	45	**3.2926e-2**	5.1705	8	100	2.79%
increased bone marrow adipose tissue amount	45	**1.1861e-6**	**2.4114e-4**	154.5616	3	1.28%	51	**4.2799e-2**	64.6307	2	2	0.70%
abnormal uterus size	48	**1.4928e-6**	**2.8454e-4**	4.9085	14	5.96%	50	**3.7337e-2**	5.8002	7	78	2.44%
abnormal cartilage morphology	49	**1.5706e-6**	**2.9325e-4**	2.6870	29	12.34%	3	**1.5025e-3**	4.0963	18	284	6.27%
abnormal skeleton development	57	**2.4260e-6**	**3.8940e-4**	2.4761	32	13.62%	10	**2.4481e-3**	3.3487	20	386	6.97%
respiratory failure	62	**3.3191e-6**	**4.8977e-4**	3.3399	20	8.51%	22	**1.0502e-2**	4.6772	11	152	3.83%

The test set of 235 genomic regions picked 287 (2%) of all 18,549 genes.
Mouse Phenotype Single KO has 9,149 terms covering 9,408 (51%) of all 18,549 genes, and 554,948 term - gene associations.
9,149 ontology terms (100%) were tested using an annotation count range of [1, Inf].

Some limitations

For “over-representation”-based methods we’ve discussed

- Genes are effectively all treated equally
  (expression fold-change, effect sizes of each gene is not considered)
- Tends to discard genes not meeting a threshold
  (a somewhat arbitrary significance or fold-change cutoff is applied)
- Genes are assumed independent
  (e.g., types/strengths of genes’ biological interactions are not considered)
- Pathways/gene sets are assumed independent
  (though, e.g., GO terms are hierarchically ordered)

Some extensions

"Functional class scoring" (FCS) methods

- Rather than using an arbitrary cutoff for including genes and weighing all genes equally, try to use, e.g., full matrix of expression data
- One basic idea: calculate a score for each gene, then aggregate genes' scores into pathway-level scores, and test for significance
- Example: GSEA (gene set enrichment analysis)

Example: GSEA

Rank genes by effect on/correlation with phenotype.

GSEA calculates the ES by walking down the ranked list of genes, increasing a running-sum statistic when a gene is in the gene set and decreasing it when it is not.

The magnitude of the increment depends on the correlation of the gene with the phenotype. The ES is the maximum deviation from zero encountered in walking the list. A positive ES indicates gene set enrichment at the top of the ranked list; a negative ES indicates gene set enrichment at the bottom of the ranked list.

The top portion of the plot shows the running ES for the gene set as the analysis walks down the ranked list. The score at the peak of the plot (the score furthest from 0.0) is the ES for the gene set. Gene sets with a distinct peak at the beginning (such as the one shown here) or end of the ranked list are generally the most interesting.

Use a permutation procedure to normalize the ES and assign each gene set an enrichment p-value. Adjust for multiple testing/try to control FDR.

(Numerous implementations exist.)

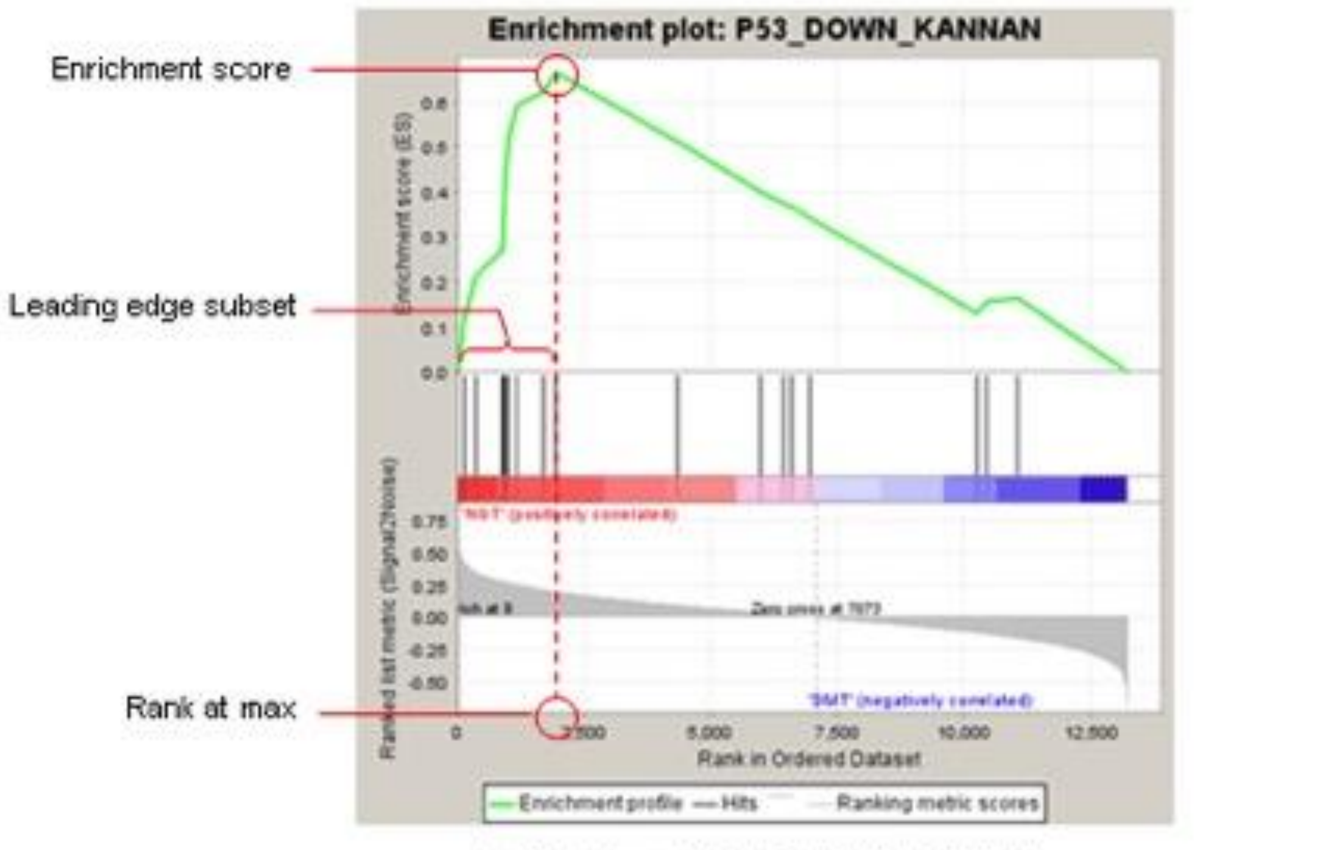


Fig 1: Enrichment plot: P53_DOWN_KANNAN
Profile of the Running ES Score & Positions of GeneSet Members on the Rank Ordered List

Other extensions

Pathway topology (PT)-based approaches

- ORA/FCS methods don't use "structure" of gene sets
- PT-based methods do

  Review/comparison: https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3146-1
- Some approaches
  - Computing similarities for gene pairs in a pathway and adjusting for their "distance" in the pathway → gene score → pathway score
  - "Impact factor" approach: treat signaling pathway as a graph, using its connectivity structure to model a pathway score as a weighted sum involving differential expression of each gene in the pathway
- Some limitations: pathway connectivity could vary across cell types or with experimental conditions

Some other challenges

- Need for "higher-resolution" databases
  - Lots of alternative splicing → the "same gene" (as labeled in a pathway or gene set) could have various different meanings/contexts in a given pathway, though the pathway may have only one representation in databases
  - In an association study, how does a SNP affect associated with a phenotype affect a *pathway*?
- Missing/biased data
  - How should non protein-coding elements like pseudogenes be included? (They may be transcribed/affect pathways)
  - > 95% of GO term assertions (as of 2007) are inferred from electronic annotations (and more likely lower in quality or incorrect)
  - Lots more "manual" annotation needed

Some other challenges (con't.)

- Condition- and cell-specificity
  - Databases are built from experiments on certain cell types in certain tissues and may not generalize; this context may be unavailable.
  - Different "variations" of the "same" pathway from different contexts may erroneously lead to two genes that are actually independent to be grouped together in a gene set; this can inflate the number of genes in a pathway and lead to spurious GSA associations
  - "Canonical" pathways way differ from disease- or phenotypic-specific pathways

Some other challenges (con't.)

- Benchmarks
  - Lack of a "gold standard" since "the full biology" is generally not known; simulations can't answer everything.

When to use each approach

- If you have GWAS results
  - Often converted to a gene list for ORA
  - Use enrichment tools that test many gene sets at once
    - FUMA
    - Enrichr
    - clusterProfiler (ORA)
    - DAVID
  - If working with genomic regions (not just genes), use GREAT
  - If you have gene-level stats, use rank-based methods (e.g., MAGMA in FUMA)
- If you have differential expression results
  - Usually you have a ranked gene list with effect sizes → use GSEA
  - Use GSEA or R packages (clusterProfiler [GSEA] or fgsea)

A note on interpreting results

Multiple testing problem: we usually test many, many gene sets

- FDR = false discovery rate = proportion of significance results that are false positives
- E.g., 20 significant pathways at FDR=0.05 → 1/20 expected to be a false positive
- FDR < 0.05 somewhat stringent; FDR < 0.1 often used for GSEA

q-value = minimum FDR at which a test is significant

- "If I call this significant, what fraction of my discoveries would be false?"
- E.g.: pathway has p=0.0001, q=0.03 → Including it among significance results means you expect 3% of significant findings are false positives
- Using FDR < 0.05 → select all results with q < 0.05
- Think of q-values as adjusted p-values, controlling for FDR